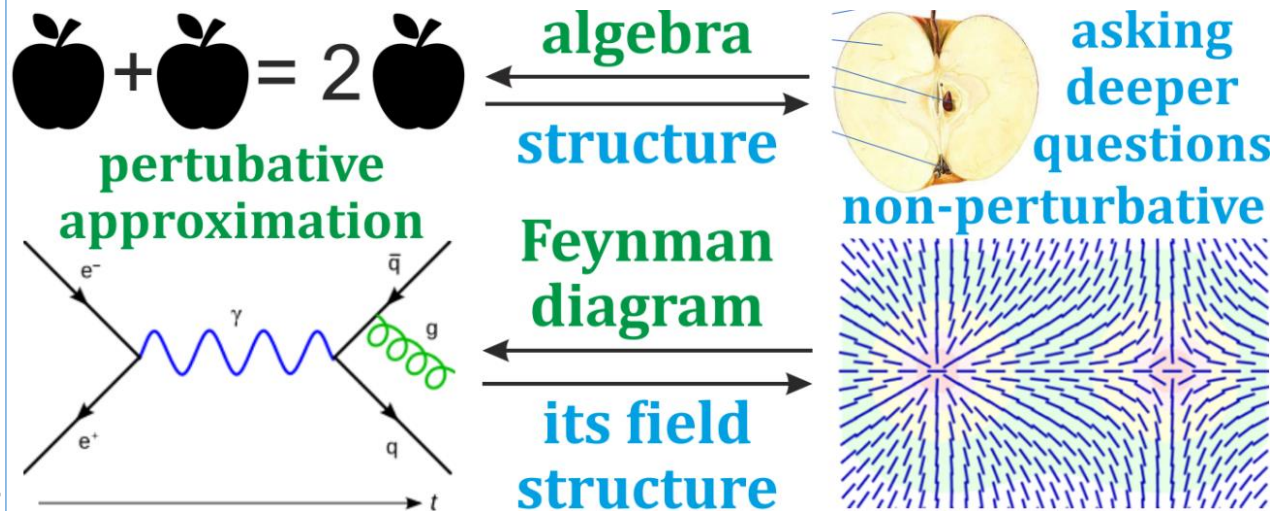


Non-perturbative Standard Model field configurations as liquid crystal-like topological excitations

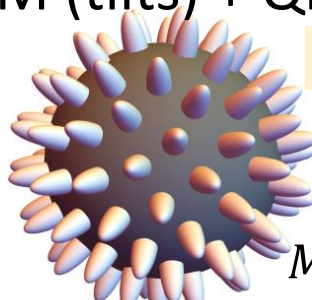
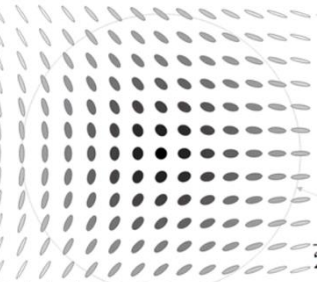
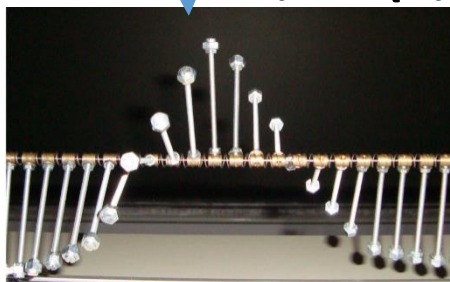


Electron: effectively described as perfect point in **perturbative approximation**
in non-perturbative: at least $E \propto r^{-2}$ field of **quantized(!)** charge

Electric charge = $\oint_S E \cdot dA \stackrel{\text{Gauss law}}{=} \frac{e_0}{4\pi} \oint_{S(u,v)} \stackrel{\text{Gauss-Bonnet}}{du dv} (\partial_u \vec{n} \times \partial_v \vec{n}) \cdot \vec{n} = \text{topological}$

Skyrme/EM-like: $\mathcal{L} = -F_{\mu\nu\alpha\beta} F^{\mu\nu\alpha\beta} + V_{\text{Higgs}}(M)$ for $R \sim F_{\mu\nu\alpha\beta}^{\text{dual}} = [\partial_\mu M, \partial_\nu M]_{\alpha\beta}$
 SO(1,3) vacuum V_{min} : $M = O \text{diag}(g, 1, \delta, 0) \cdot O^T$, weights: $g_{\text{grav.}} \gg 1_{\text{EM}} \gg \delta \sim \hbar > 0$
 Rotations + boosts of ellipsoid field M of (Higgs) preferred ax lengths $(g, 1, \delta, 0)$

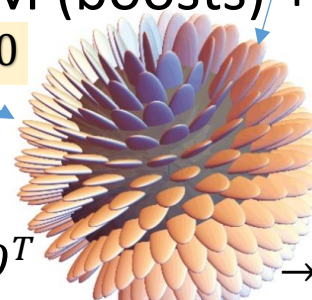
1. 1D sine-Gordon $\rightarrow$ 3D Faber's model: **Coulomb** long-range interaction
2. $\sim$ Superfluid ($\sim$ QGPlasma) liquid crystal agreeing with **Standard Model**
3. 4D, SO(1,3) vacuum: EM (tilts) + QM (twists) + GEM (boosts) + $\perp$ gluons



$\delta = 0$

1

$\delta > 0$



Landau-de Gennes

$$M = O \text{diag}(1, \delta, 0) O^T$$

Jarek Duda
 (github, video, intro, article)

$\rightarrow$ **teleparallelism**

Teleparallelism in liquid crystal view with GEM+EM+QM unification and particles as topological excitations

Teleparallelism – “attempt by Albert Einstein to base a **unified theory of electromagnetism and gravity**” – **let’s make it work!**

Tetrad: 4 orthogonal vectors, like liquid crystal Landau-de Gennes model

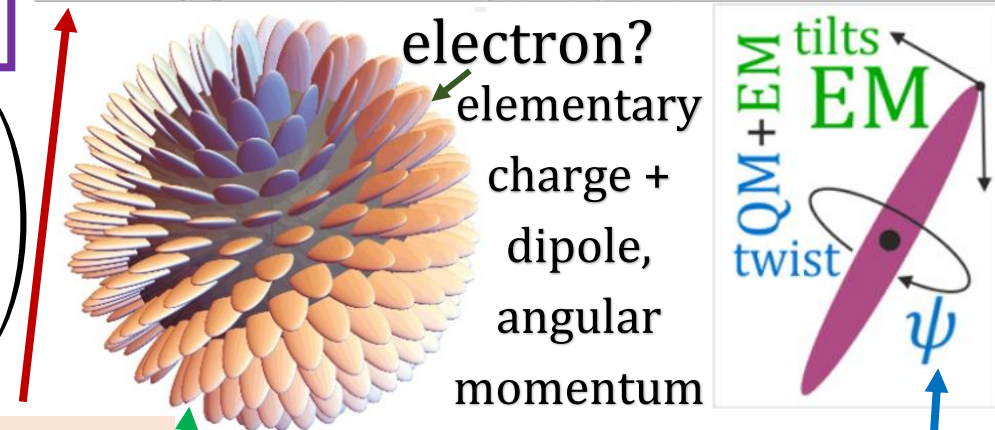
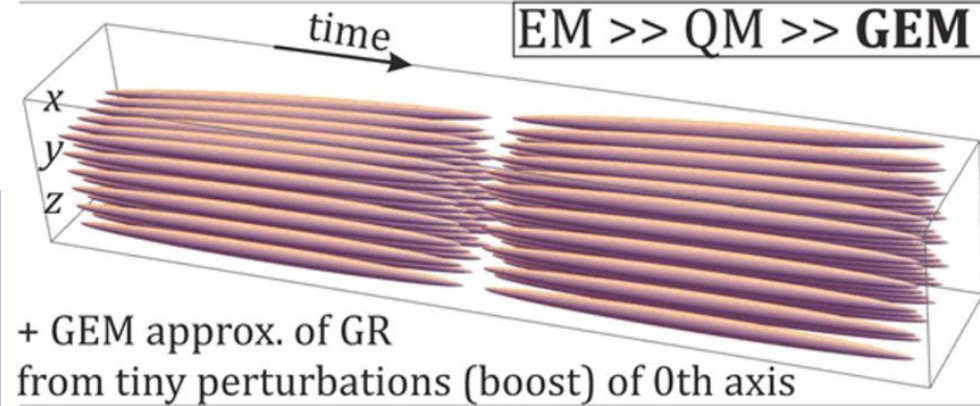
combine 3 (space) or 4 (+time) vectors $\vec{n}$ into real tensor field:

$$M = \sum_i \lambda_i \vec{n}_i n_i^T \approx O(g_{\gg 1}, 1, \delta_{\sim \hbar}, 0) O^T$$

dynamics:

$$O^T \partial_\mu O = \begin{pmatrix} 0 & \tilde{\Gamma}^1 \text{ boost} & \tilde{\Gamma}^2 & \tilde{\Gamma}^3 \\ \tilde{\Gamma}^1 & 0 & -\Gamma^3 & \Gamma^2 \\ \tilde{\Gamma}^2 & \Gamma^3 \text{ tilt} & 0 & -\Gamma^1 \\ \tilde{\Gamma}^3 & -\Gamma^2 \text{ tilt} & \Gamma^1 \text{ twist} & 0 \end{pmatrix}$$

$O \in SO(1,3)$



- tiny **boosts** of longest **time axis** give **GEM** (gravitoelectromagnetism)
- **tilts** of 2nd axis give **EM with topologically quantized charge**
- low energy **twists** around 2nd: **quantum phase** with $\sim$ Klein-Gordon

Various energy: $L_{QED} = -\hbar c \bar{\psi} \gamma^\mu D_\mu \psi - mc^2 \bar{\psi} \psi - F_{\mu\nu} F^{\mu\nu} / 4$

Liquid crystal topological defects – particle physics correspondence

Repair classical EM (Faber):

$$1) \int_0^\infty E^2 r^2 dr \propto \int_0^\infty r^{-4} r^2 dr = \infty$$

Deform/regularize charge to finite energy (Higgs-like potential allowing deformation)

2) Gauss law charge $q \in \mathbb{R}$, nature $q \in \mathbb{Z}e$

Interpret curvature as electric field to count (quantized) topological charge with Gauss law.

~Landau-de Gennes model, skyrmions

3 distinguishable axes like biaxial nematic

Use real symmetric tensor field $M = ODO^T$

~Higgs e.g. $V = \sum_i (\lambda_i - \Lambda_i)^2$ for D shape

Getting 3 leptons, baryons, nuclei ...

→ 4 axes Einstein's teleparallelism to unify:

EM >> quantum phase >> GEM

1st axis ~ Klein-Gordon/Dirac? 0th axis in 4D
tilts rotat. 1st axis twists (~Berry) tiny boosts

$$\mathcal{L}_{QED} = \bar{\psi}(i\gamma^\mu \partial_\mu - e\gamma^\mu A_\mu - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

spin/particle + phase $\partial_\mu?$ Lagrangian/Hamiltonian

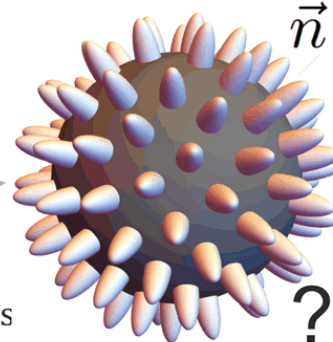
Ahar. Bohm
Maxwell:
 $\square A_\mu \propto J_\mu$
extended quantum phase for topological charge quantization
(dual)
 $A_\mu = [M, M_\mu]$
 $\cong$ affine connection
 $F^* \cong$ curvature
 $\psi \xrightarrow{2D} M = ODO^T \xrightarrow{3D} \hat{P} = -i\hbar\nabla - qA$
QM+EM EM

2D charge (+1) as director hedgehog
regularized $\vec{n} \rightarrow 0$ by Higgs pot.
 $V(\vec{n}) = (\|\vec{n}\|^2 - 1)^2$

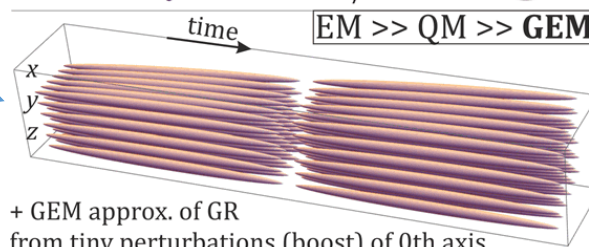
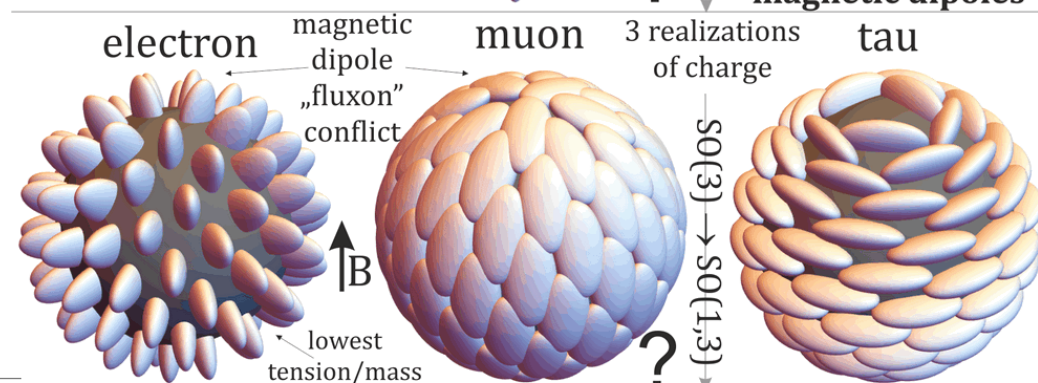
electric/topological (3D winding number) = $\oint_S E \cdot dA = \frac{e_0}{4\pi} \oint_{S(u,v)} du dv (\partial_u \vec{n} \times \partial_v \vec{n}) \cdot \vec{n}$
3D charge in 3D

uniaxial nematic unitary vector in 3D of director field $\vec{n}(x)$
1 distinguished axis

charge + Coulomb Maxwell equations from director dynamics



biaxial nematic 1 → 3 in 3D distinguished axes „uniaxial + quantum phase“ (for „pilot wave“) field of real symmetric M
3 charges with magnetic dipoles



+ GEM approx. of GR from tiny perturbations (boost) of 0th axis

energy density: EM quantized charge 3 uniaxial (Higgs potential) curvature (regularization) biaxial/general, M -matrix curvature (real, symmetric)

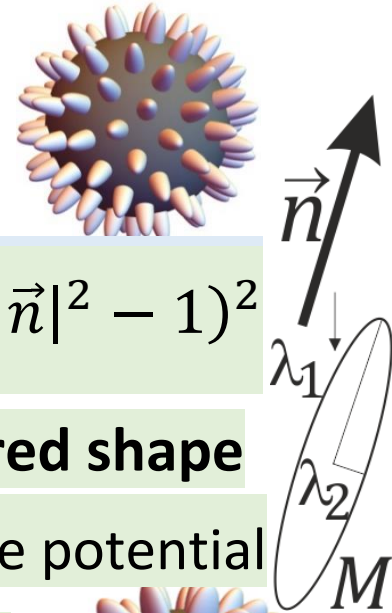
$$\mathcal{H} \sim \sum_{\mu,\nu=0}^3 (\partial_\mu A_\nu - \partial_\nu A_\mu)^2 \rightarrow \sum_{\mu,\nu=0}^3 \|\partial_\mu \vec{n} \times \partial_\nu \vec{n}\|^2 + (\|\vec{n}\|^2 - 1)^2 \rightarrow \sum_{\mu,\nu=0}^3 \|\partial_\mu A_\nu - \partial_\nu A_\mu\|^2 + V(M)$$

final 4D Lagrangian? $\mathcal{L} = -F_{\mu\nu\alpha\beta} F^{\mu\nu\alpha\beta} - V_{\text{Higgs}}(M)$ $(R \sim) F_{\mu\nu\alpha\beta} = [\partial_\mu M, \partial_\nu M]_{\alpha\beta}$

Standard liquid crystal models ([review1](#), [review2](#))

Uniaxial: $\vec{n} \in S^2$ (Higgs potential?), **Oseen-Frank** model (1958) :

$$E[\vec{n}] = \int_{\Omega} K_1 |\nabla \cdot \vec{n}|^2 + K_2 |\vec{n} \cdot (\nabla \times \vec{n})|^2 + K_3 |\vec{n} \times (\nabla \times \vec{n})|^2 + (|\vec{n}|^2 - 1)^2$$

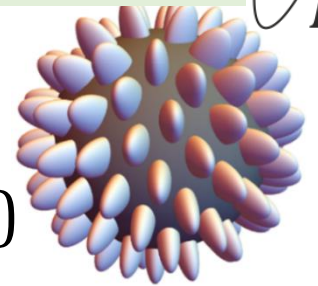


Biaxial: Landau-de Gennes (1974, Nobel Prize in 1991) of **preferred shape**

($Q \equiv$) $M = \sum_{i=1}^3 \lambda_i \vec{n}_i \vec{n}_i^T$ **preferred** $(\lambda_1, \lambda_2, \lambda_3)$: $V(M)$ Higgs-like potential

$$E[M] = \int_{\Omega} |\nabla M|^2 + \frac{A}{2} \text{Tr}(M^2) - \frac{B}{3} \text{Tr}(M^3) + \frac{C}{4} (\text{Tr}(M^2))^2$$

(or $V(M) = \sum_i (\lambda_i - \Lambda_i)^2$ or $\sum_i (\text{Tr}(M^i) - c_i)^2$ tough choice!)



→ **particles?**: Higgs-like potential, e.g. as above, with **Coulomb, EM-like**:

$$\mathcal{L}_{EM} = -F_{\mu\nu} F^{\mu\nu} \quad \text{for} \quad F_{\mu\nu} \propto R_{\mu\nu} = \Gamma_{\mu} \times \Gamma_{\nu} \quad \Gamma_{\nu} = (\partial_{\nu} \vec{n}) \times \vec{n}$$

$R_{\mu\nu}$ **curvature**: Gauss law counts (quantized) topological charge

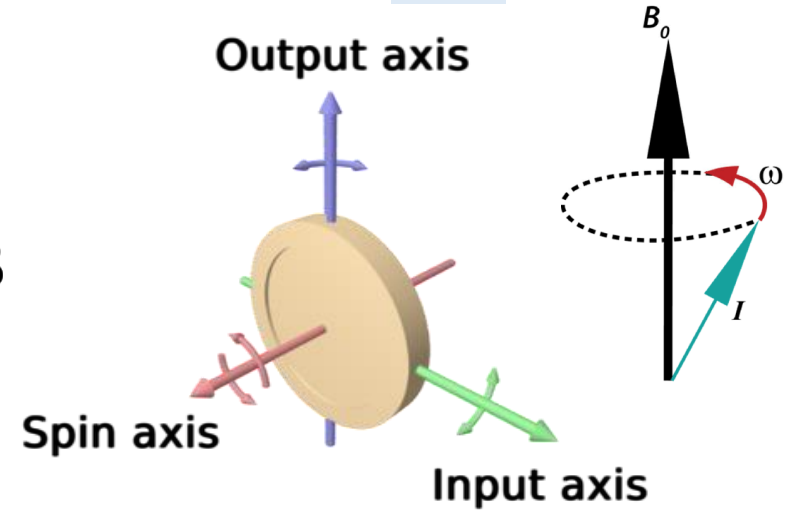
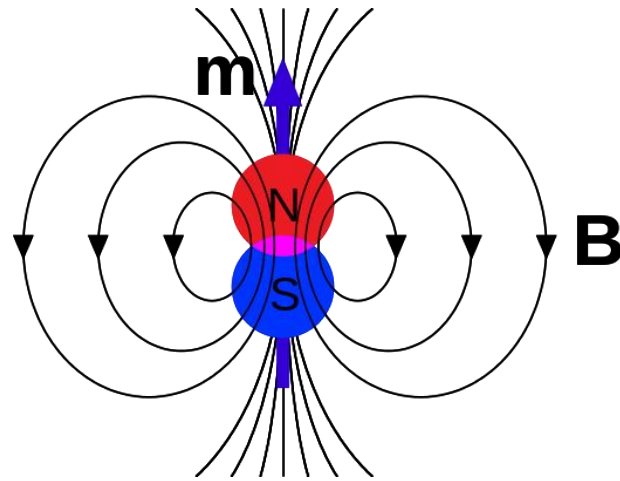
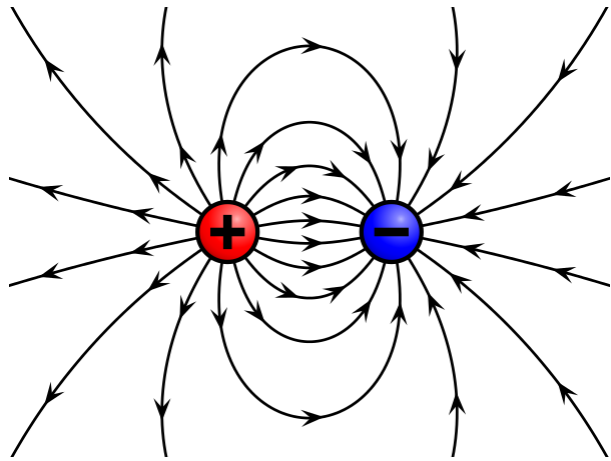
$$\text{with kinetic: } \mathcal{L} \sim \sum_{i=1}^3 ||\partial_i \vec{n} \times \partial_0 \vec{n}||^2 - \sum_{1 \leq i < j \leq 3} ||\partial_i \vec{n} \times \partial_j \vec{n}||^2 - V(\vec{n})$$

$$\text{Biaxial: } F_{\mu\nu\alpha\beta} = [\partial_{\mu} M, \partial_{\nu} M]_{\alpha\beta} \quad \mathcal{L} = -F_{\mu\nu\alpha\beta} F^{\mu\nu\alpha\beta} - V(M)$$

Lorentz-invariant Lagrangian mechanics leading to **electromagnetism**
+ $\sim$ **Klein-Gordon** for **twists** + **gravito-electromagnetism** for 0th axis **boosts**

Einstein-Hilbert: $\mathcal{L} \sim R \sqrt{-g}$ spacetime curvature, here field curvature ... Newton?

non-perturbative
Electron – (at least) a complex configuration of electromagnetic field ... Larmor



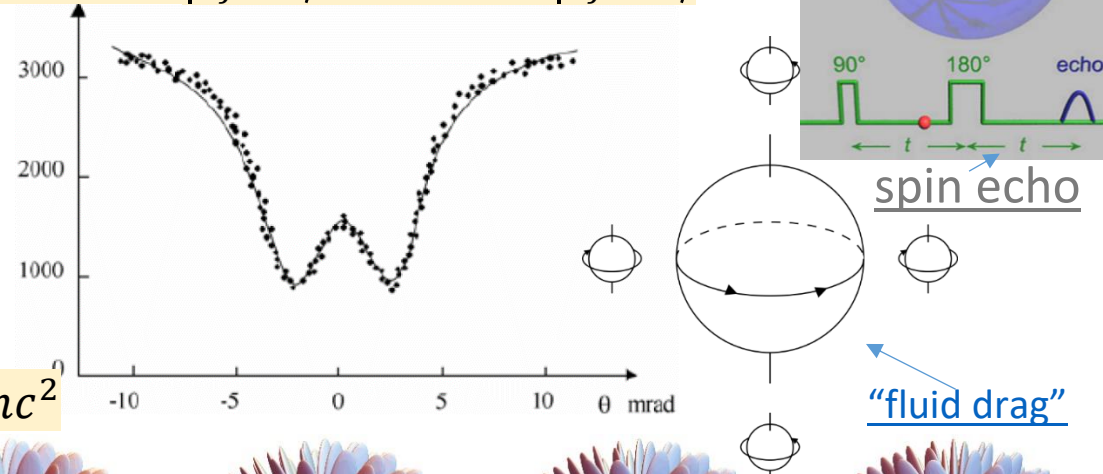
quantized!

Electric charge ($E \propto \frac{1}{r^2}$) + magnetic dipole ($B \propto \frac{1}{r^3}$, magnets) + “gyroscope” ($L = \frac{\hbar}{2}$ of field?)
 + $\approx 10^{21} \text{ Hz}$ zitterbewegung (observ.)/de Broglie’s clock ($E = mc^2 = \hbar\omega$):

some internal periodic process $\sim$ neutrino oscillations: $|v_j(t)\rangle = e^{-iE_j t/\hbar} |v_j(0)\rangle$

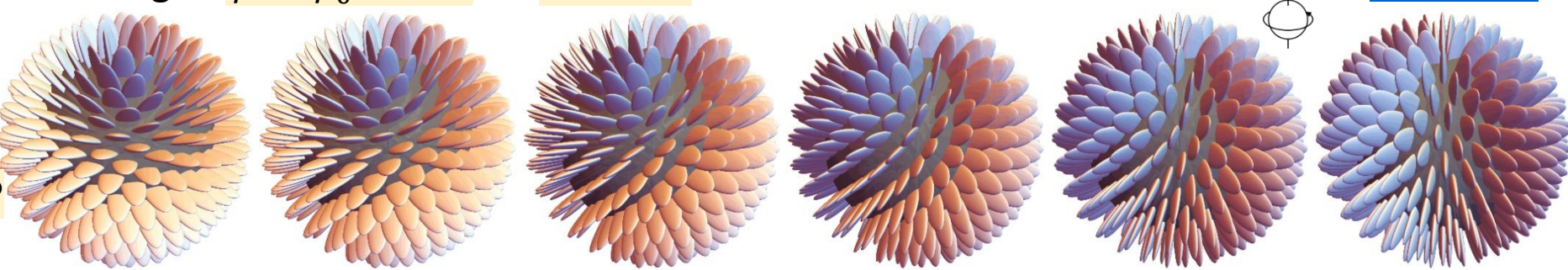
Along <110> axis of silicon crystal atomic spacing
 correspond to ‘internal clock’ period for $E \approx 81 \text{ MeV}$
 electrons – observed resonance:

P. Catillon, N. Cue, M.J. Gaillard, R. Genre, M. Gouanère, R.G. Kirsch, J.-C. Poizat, J. Remillieux, L. Roussel, M. Spighel, *A Search for the de Broglie Particle Internal Clock by Means of Electron Channeling*, Found Phys (2008) 38: 659–664



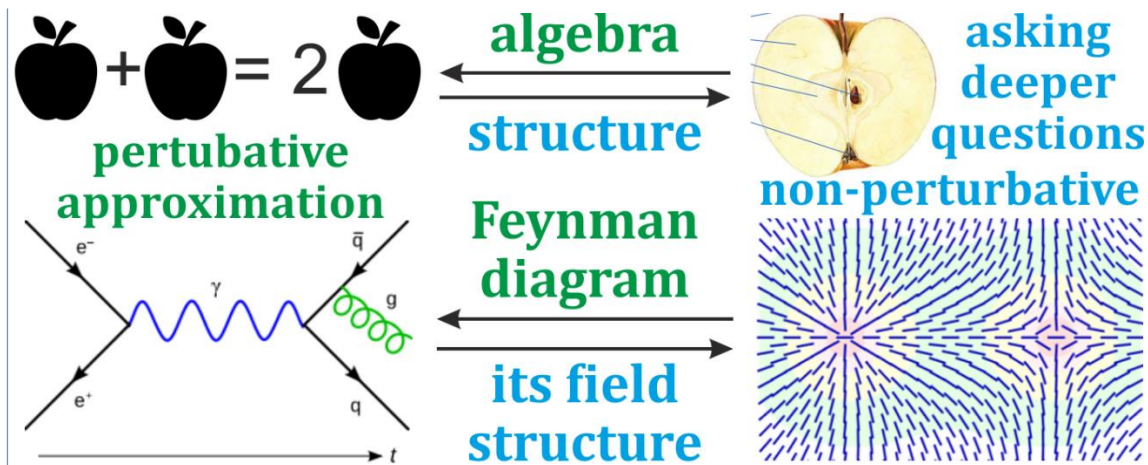
stationary **Schrödinger**: $\psi = \psi_0 e^{-iEt/\hbar}$ for $E = mc^2$

twist as
 phase ψ
 spinning/
 evolution?



In **perturbative approximation** electron is imagined as **perfect point**.
Non-perturbative field configuration: electron needs $E \propto r^{-2}$ for Coulomb

What is **non-perturbative**



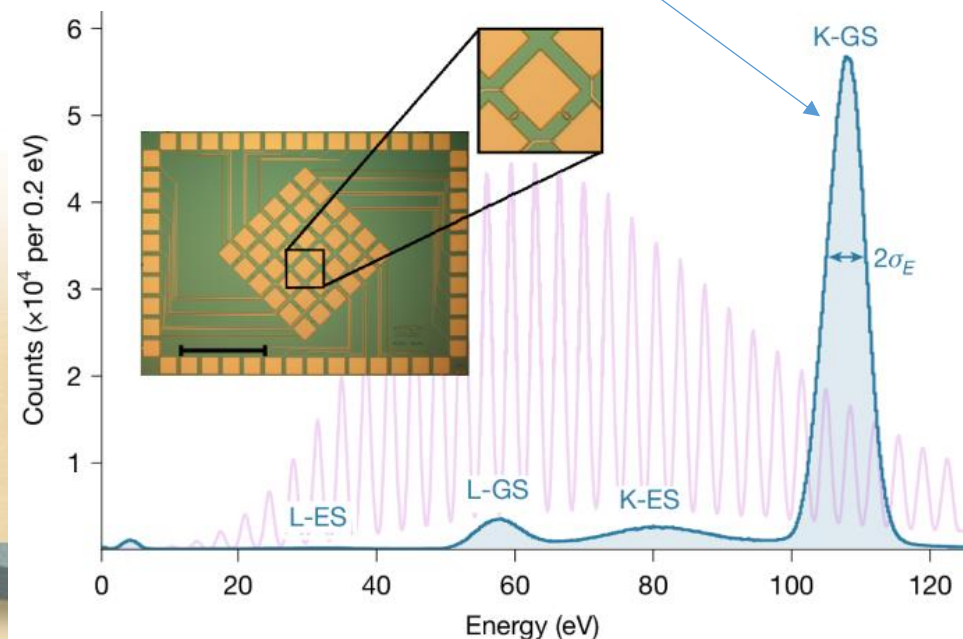
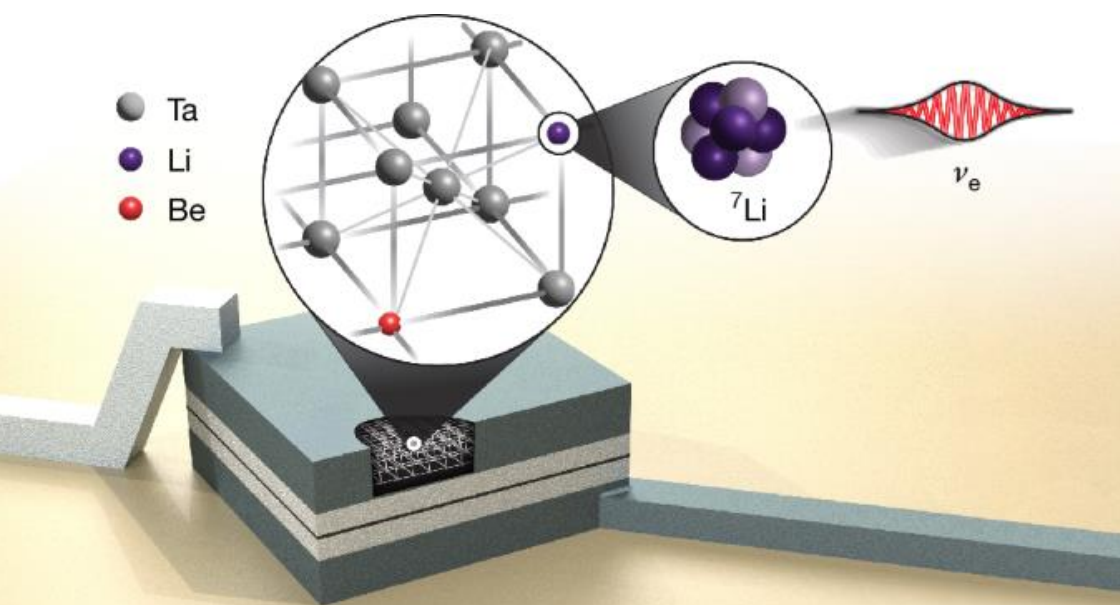
field configuration of e.g. neutrino for its weak interactions???

More stable than EM? 3 types? Resting oscillates ("**time crystal**")?

[nature.com/articles/s41586-024-08479-6](https://www.nature.com/articles/s41586-024-08479-6): **neutrino >1000x larger than nucleus**

${}^7\text{Be} + e^- \rightarrow {}^7\text{Li} + \nu_e$ measure ${}^7\text{Li}$ recoil energy due to emitted neutrino

From Heisenberg ($\Delta x \Delta p \geq \hbar/2$) getting wavefunction $\Delta x > 6.2 \text{ pm}$

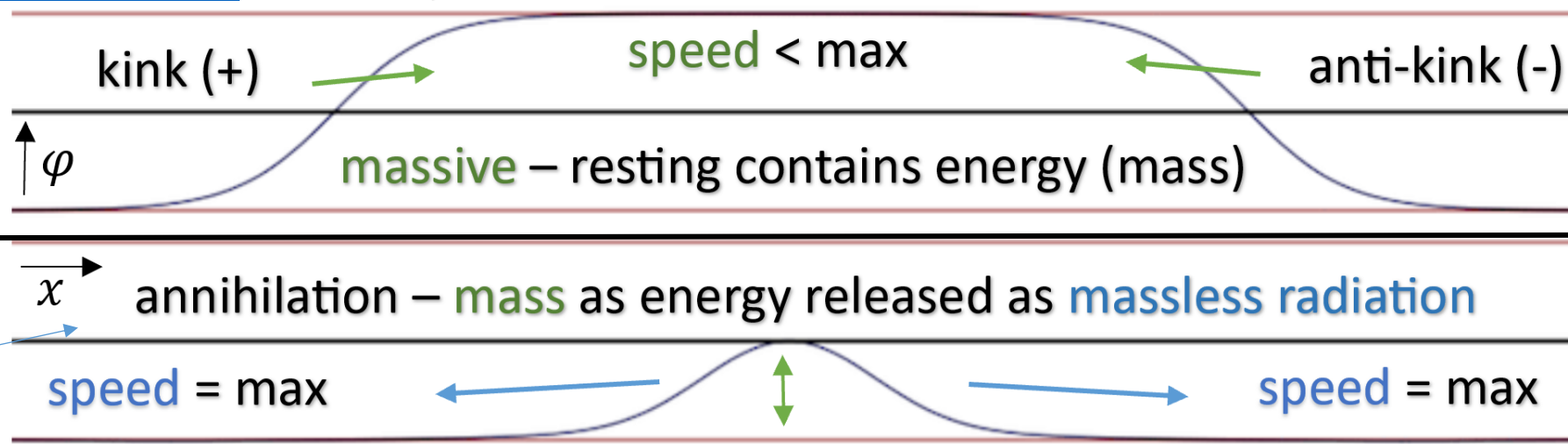


1D sine-Gordon – e.g. lattice of coupled pendula, just $\varphi_{tt} - \varphi_{xx} = -\sin \varphi$
angle wave in gravity: pendula prefer to be down, but “down” is every $2\pi k$

Pair creation/annihilation of massive particles: $m = \int_{-\infty}^{+\infty} \frac{1}{2} (\varphi_x^2 + \varphi_t^2) + V dx$

special relativity: velocity $\leq$ max, Lorentz contraction, time dilation

(anti)kink:
massive
particle,
quantized
[vid.](#), [anim.](#)
[img](#)



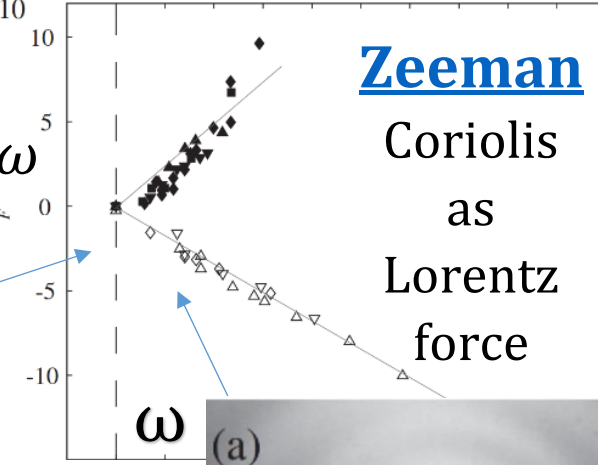
electromagnetism(c) $\leftrightarrow$ **Hydrodynamics(c_s)** analogy

for superfluid: $\nu = 0$ viscosity ... **Barnett effect**: B from ω

$B = \nabla \times A \leftrightarrow \omega = \nabla \times u$ for $A \leftrightarrow u$ **velocity, vorticity** ω

Lorentz force $\leftrightarrow$ **Coriolis force** e.g. for **Zeeman effect**

current/flow e.g. in superfluid: $\vec{j} \sim \vec{v}_s m = \hbar \nabla \varphi - q \vec{A}$

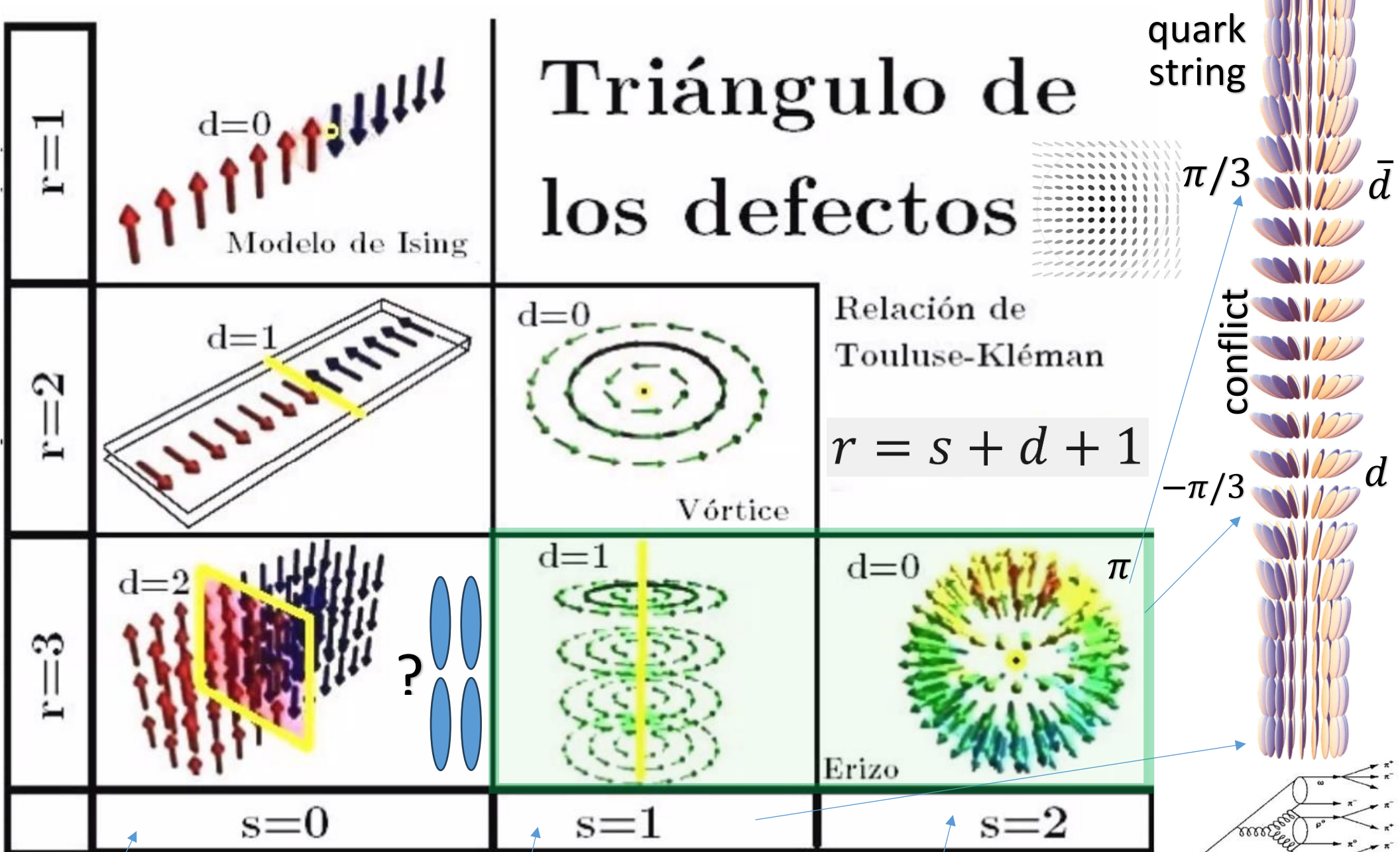


Hydrodynamics	superfluid $\nu = 0$	Electromagnetism	analogs
Navier-Stokes equation	$\frac{\partial u}{\partial t} = -l - \nabla \phi_p + \nu \nabla^2 u$ $\phi_p = p/\rho + u^2/2$	$\frac{\partial A}{\partial t} = -E - \nabla \phi$ E - electric field	vector and scalar potential
vorticity	$\omega = \nabla \times u$	$B = \nabla \times A$	u l
Lamb vector	$l \equiv \omega \times u$	$\nabla \cdot B = 0$	ω B
vorticity tendency	$\frac{\partial \omega}{\partial t} = -\nabla \times l + \nu \nabla^2 \omega$	Coulomb Thomson	
Lamb vector tendency	$\frac{\partial l}{\partial t} = \nabla \times \eta - j$	Faraday's law	
and turbulent current (j)		$\frac{\partial B}{\partial t} = -\nabla \times E$	
vorticity field strength (η)	$\eta = u ^2 \omega - M$	Amper's law	η H
and magnetization	$M = u(u \cdot \omega) + \nu \nabla^2 u$	$\frac{\partial(\epsilon_0 E)}{\partial t} = c^2 \nabla \times H - \mu_0 J$	
turbulent charge density	$\nabla \cdot l = u \cdot \nabla \times \omega - \omega ^2 = \rho_n$	$H = \mu_0^{-1} B - M$	
		strength and magnetization	
		$\nabla \cdot (\epsilon_0 E) = \rho_e$	ρ_n ρ_e
		electric charge density	

lk

Theory	Gauge fields	Circulation	Gauge condition	Matter field
Electro-dynamics	$\varphi, \vec{A}$ four-potential	$\vec{B} = \vec{\nabla} \times \vec{A}$ magnetic f.	$\vec{\nabla} \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$	$\vec{E}_e = -\frac{\partial \vec{A}}{\partial t} - \vec{\nabla} \varphi$
Hydro-dynamics	$\chi = v^2/2, \vec{v}$ flow velocity	$\vec{\omega} = \vec{\nabla} \times \vec{v}$ vorticity	$\vec{\nabla} \cdot \vec{v} + \frac{1}{c_s^2} \frac{\partial \chi}{\partial t} = 0$	$\vec{E}_h = -\frac{\partial \vec{v}}{\partial t} - \vec{\nabla} \chi$

[link](#)



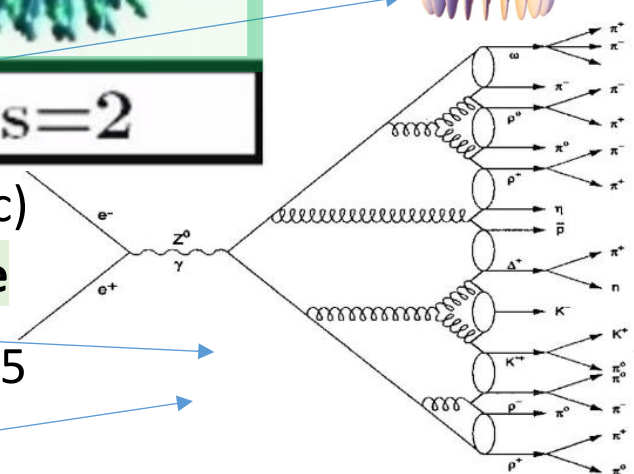
Particle physics in 3D:

domains?, **quark string/gluon flux tube**, **quantized electric charge**

dual $B \leftrightarrow E$ (not magnetic)

Gareth P. Alexander et al., **Liquid crystal** reviews, 10, 1, 91-97, 2005

String hadronization – its decay into particles in LHC collisions

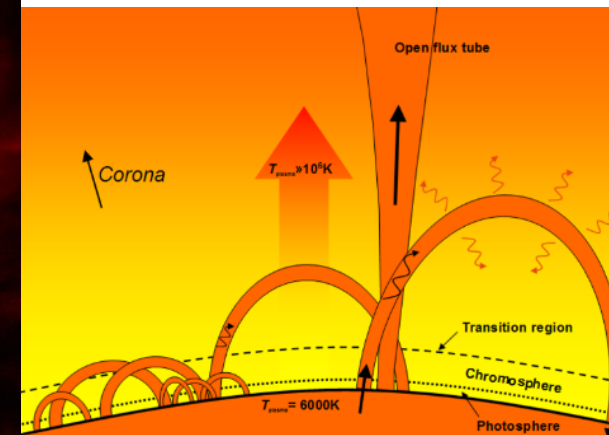
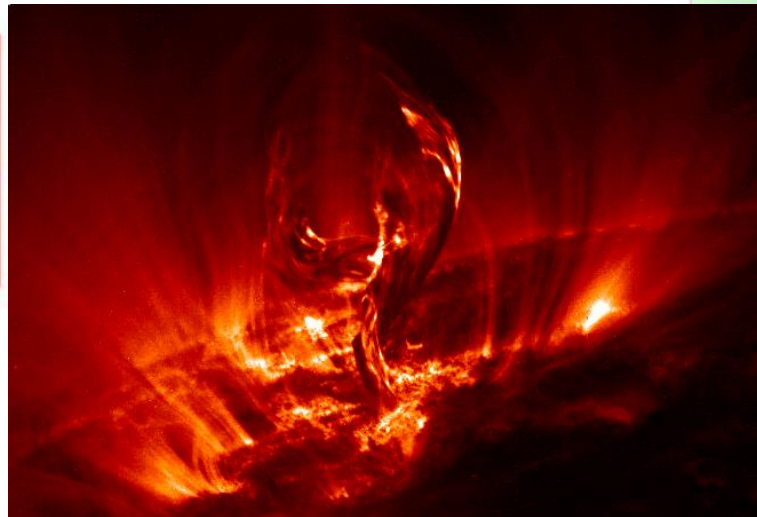
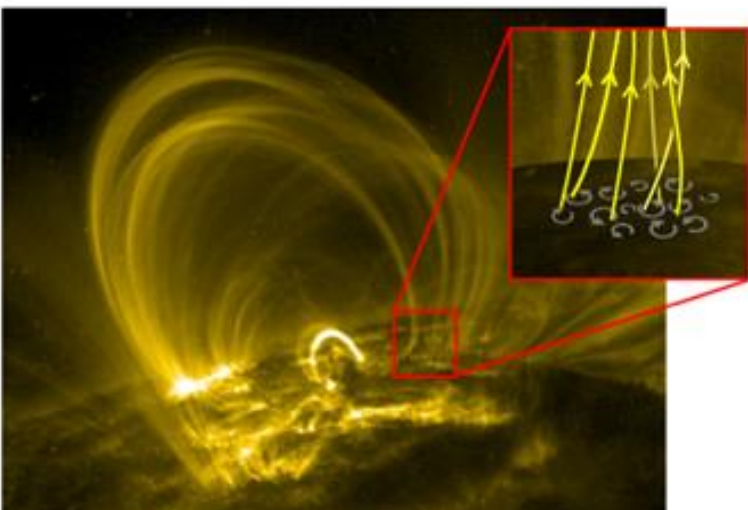
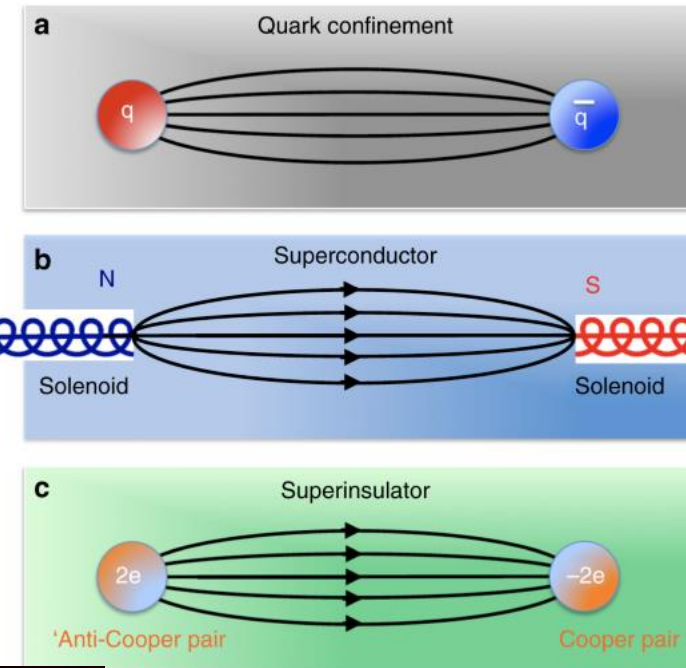
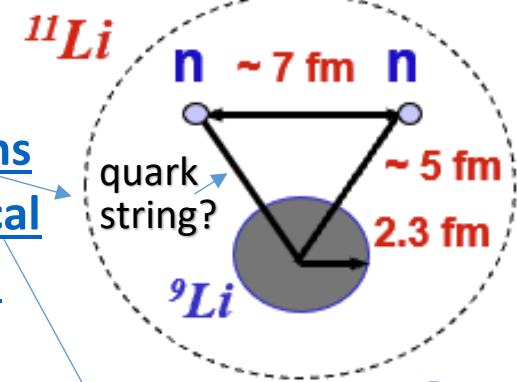


Required: **vacuum quark strings/flux tubes** – with energy density:

- Holding nucleus together against Coulomb repulsion (?), [halo neutrons](#)
- popular [quark string model](#), [Nature article suggesting being topological](#)
- [coronal heating problem](#) (surface 6000K, corona 10^7 K), [reconnections](#)
- [electrons in anti-parallel alignment](#) – in **100-1000nm** for **Cooper pairs**
- Brawley et. al., “[Electron-like scattering of positronium](#)”, Science 2010
- [Cosmic strings](#) – 1D topological defects of $\arg(\psi)$, [evidence?](#)

["Physics of Magnetic Flux Tubes" book](#) by Ryutova:

“Vortices in superfluid Helium and superconductors, [magnetic flux tubes](#) in solar atmosphere and space, filamentation process in biology and chemistry have probably a common ground, which is to be yet established. One conclusion can be made for sure: **formation of filamentary structures in nature is energetically favorable and fundamental process.**”



[wiki/Macroscopic quantum phenomena](#) ([v1](#), [v2](#), [photo](#))



Superfluid: quantized angular momentum (of field alone)

"Perfect liquid" quark-gluon plasma is the most vortical fluid"

Superconductor: quantized magnetic field

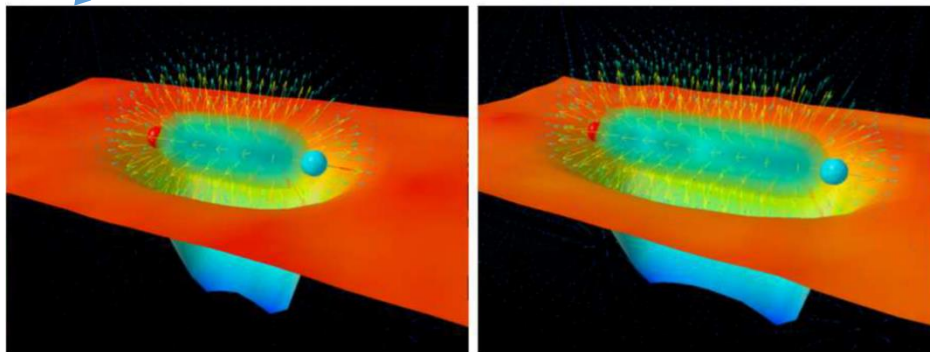
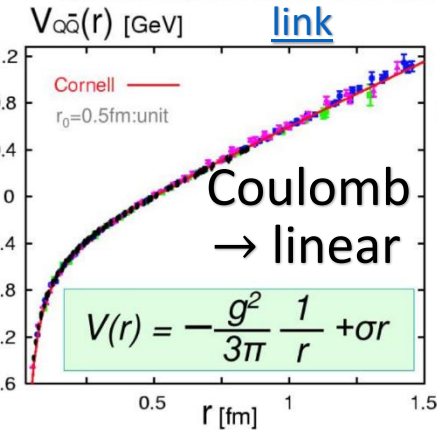
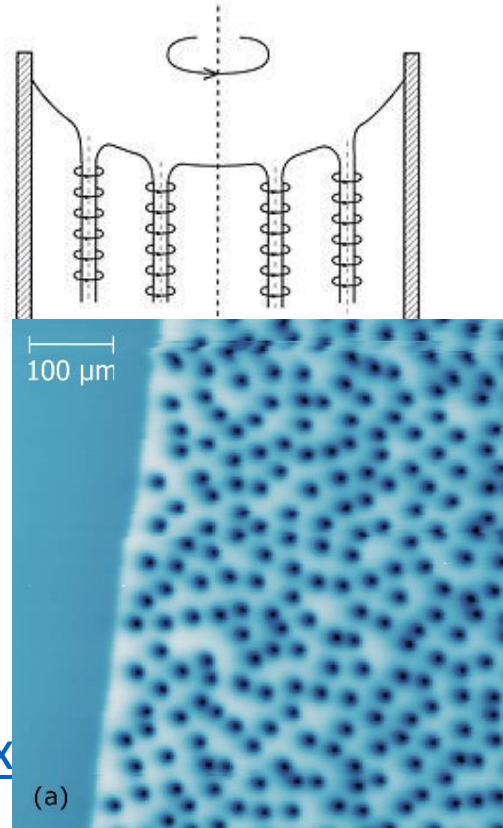
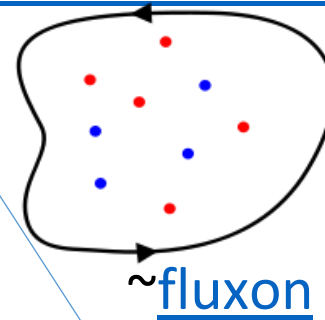
$$2\pi n = \Delta\varphi = \frac{q}{\hbar} \oint_{\partial S} A \cdot dl = \frac{q}{\hbar} \int_S B \cdot dS$$

current/velocity: $\vec{J} \sim \vec{v}_s = (\hbar \vec{\nabla} \varphi - q \vec{A})/m$

2D: [argument principle](#) in complex analysis:

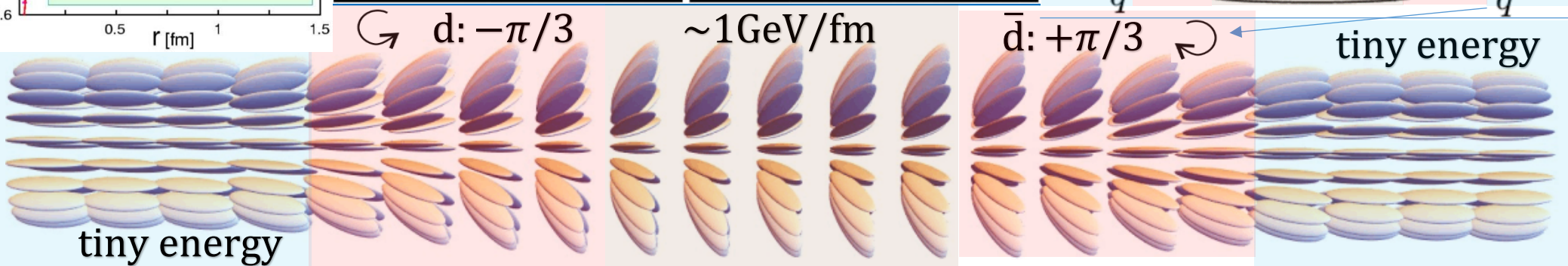
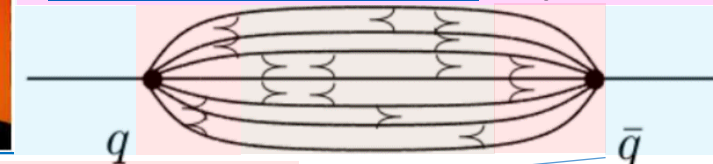
$$\oint_C \frac{f'(z)}{f(z)} dz = \oint_C (\ln f)' dz = 2\pi i (\# \text{zeros} - \# \text{poles})$$

QCD quark strings/flux tubes often [modeled as Abrikosov vortex](#)

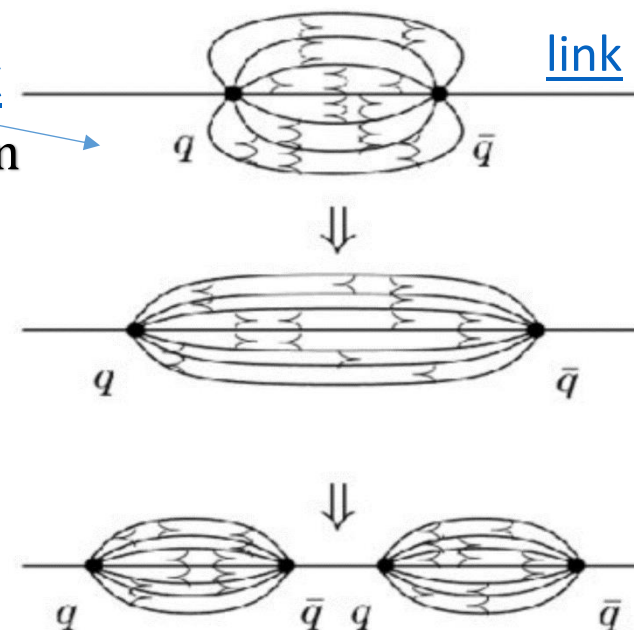
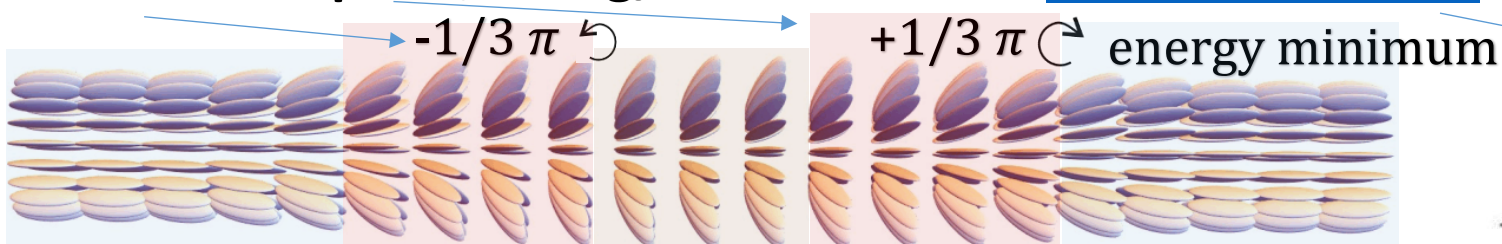


String hadronization?

~QCD color strings...quarks?



Quark-antiquark, string/QCD flux tube: [Abrikosov vortex](#) [link](#)



Binding of [halo nuclei](#) like [Boron-8](#) with [proton](#) for 770ms

[Ne-17](#): 2 halo [protons](#) 109ms, [1p P-26](#), [2p S-27](#) 44,16ms

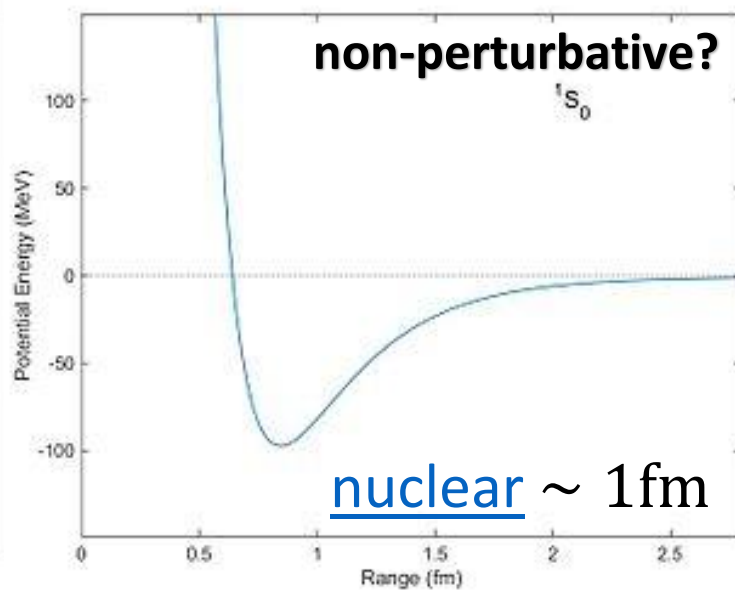
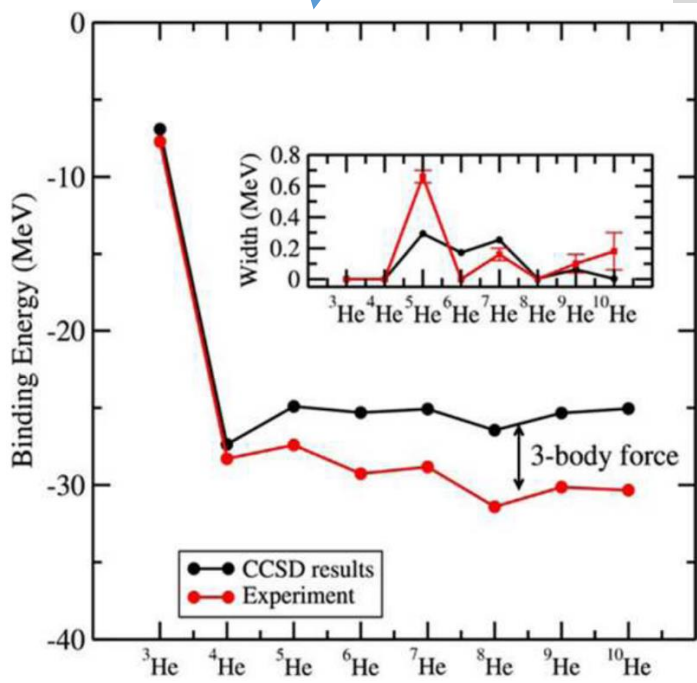
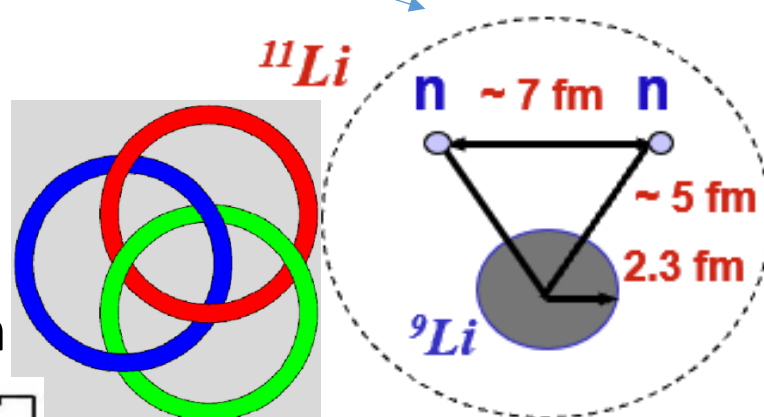
[Li-11](#): 8.6 ms lifetime in ~5 fm distance

[Efimov state](#) field conf.?

[Borromean halo nuclei](#)

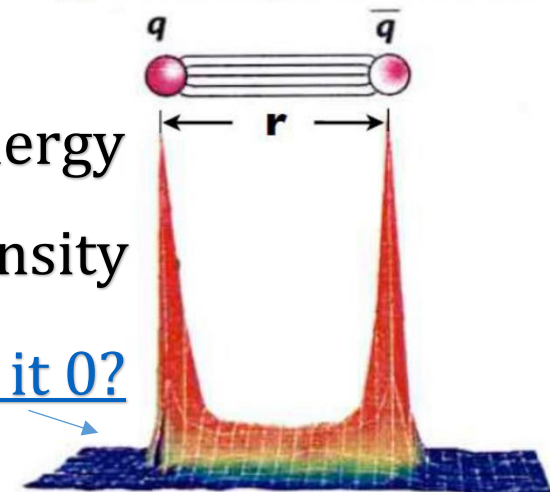
requiring [3-body forces](#)

[diagrams](#) - He6: He4+nn



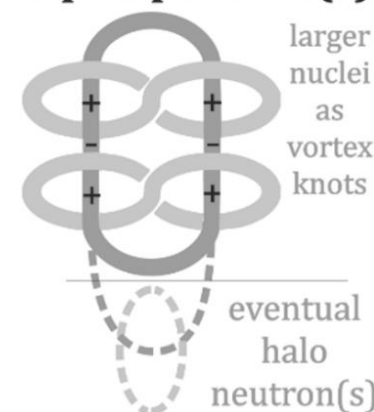
Energy density

[is it 0?](#)



alpha particle(?)

binding
larger
nuclei?



Different perspectives:

"Classical"

vs

"Quantum"

Coupled oscillators



$$m\ddot{u}_1 = -Cu_1 + C(u_2 - u_1)$$

$$m\ddot{u}_2 = -Cu_2 + C(u_1 - u_2)$$

Crystal:

1D lattice of oscillators

$$m\ddot{u}_x = C(u_{x+1} - 2u_x + u_{x-1})$$

"unitary evolution" of normal modes

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \text{Re} \left(c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{i\omega_1 t} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega_2 t} \right)$$

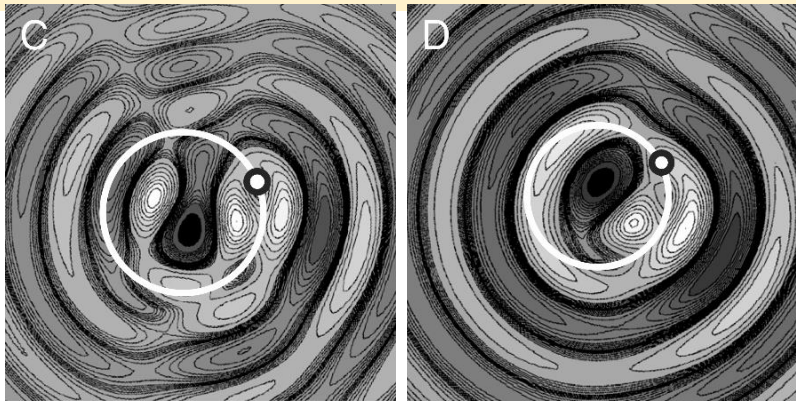
Phonons – quasiparticles in QFT

$$u_x = \text{Re}(\sum_k c_k e^{i(kx - \omega_k t)})$$

Infinitesimal lattice limit → field, like in **hydrodynamics**

e.g. Couder's

Quantized orbits

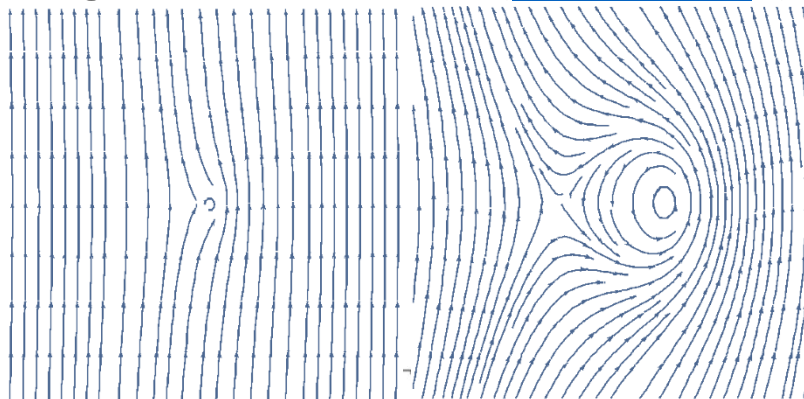


Schrödinger equation:

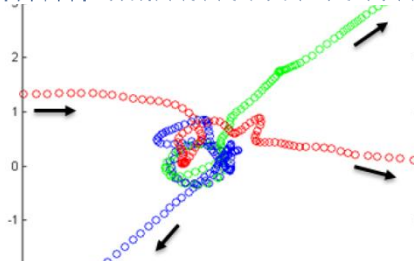
- 1) coupled **standing wave** $\psi_0 e^{iEt/\hbar}$, resonant with electron's clock,
- 2) $\rho = |\psi|^2$ **statistics**, e.g. [MERW?](#), $\arg \psi$: expected $\Delta\psi$?

Stable field configurations: **solitons** ([topological?](#))

Varying numbers of particles



[soliton Feynman diagrams](#):
for incomplete knowledge
non-perturbative

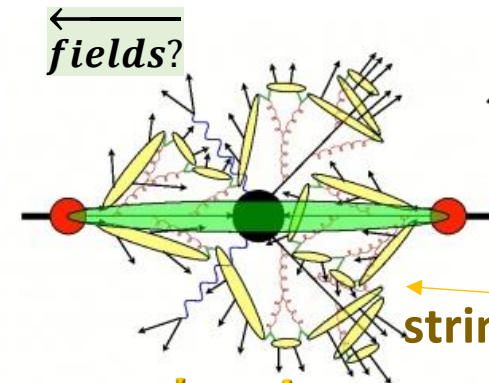
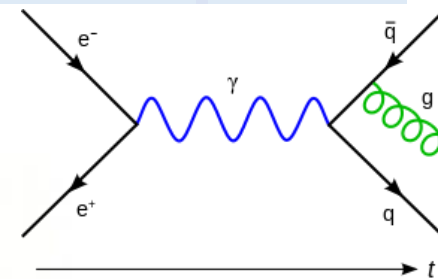


perturbative QFT "algebra of particles"

ensemble of scenarios: Feynman diagrams

description?

fields?

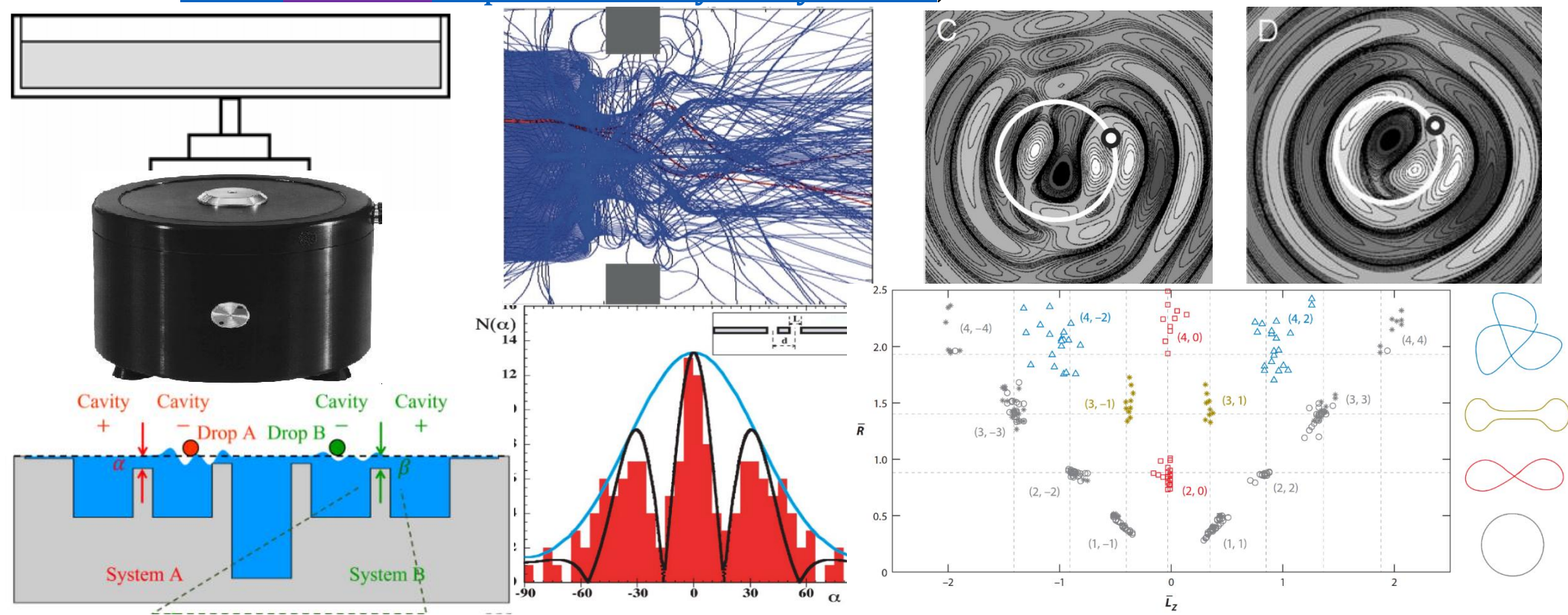


[parton shower](#)

strings hadronization

perturbative approximation

What about quantum phenomena for (topological) solitons? E.g. fluxons:
Experimental demonstration of Aharonov-Casher interference in a Josephson junction circuit, PRB 2012 – for **fluxons**, Aharonov-Casher: for magnetic dipole in electric field
Tunneling and resonant tunneling of **fluxons** in a long Josephson junction, PRB 1997
Aharonov-Bohm type forces between magnetic fluxons, PRA 1997 **short-range**
Hydrodynamical classical wave-particle duality object “walking droplets”:
Single-Particle Diffraction and Interference at a Macroscopic Scale, PRL 2006
Unpredictable Tunneling of a Classical Wave-Particle Association, PRL 2009
Path-memory induced quantization of classical orbits, PNAS 2010
Level Splitting at Macroscopic Scale, PRL 2012 – **Zeeman splitting**
Self-organization into quantized eigenstates of a classical wave-driven particle, Nature 2014
Wavelike statistics from pilot-wave dynamics in a circular corral, PRE 2013
Static Bell test in pilot-wave hydrodynamics, PRF 2024 - **violate CHSH**



2D topological charge

fluxon – quant of magnetic field

resembles spin ($\rightarrow \mu$) also $\frac{1}{2}$

$$2\pi k = \Delta\varphi = \frac{q}{\hbar} \oint_{\partial S} A \cdot dl = \frac{q}{\hbar} \oint_S B \cdot dS$$

quantum rotation operator

spin s particle by θ angle rotates

quantum phase: $\psi \rightarrow \psi e^{-is\theta}$

spin, charge: structural, dynamic:

$\mu_z \sim \ell_z$ twisting  lepton $g \approx 2$

g-factor: $\mu_z = g \frac{1}{2} \left(\frac{\hbar e}{2m} \right) \approx (\mu_B)$

($\frac{1}{2}$ spin) bispinor rotation by ϕ :

$$S[\Lambda_{\text{rot}}] = \begin{pmatrix} e^{+i\phi \cdot \sigma/2} & 0 \\ 0 & e^{+i\phi \cdot \sigma/2} \end{pmatrix}$$

3D topological

charge: electric

$V(r)$ distance dep.

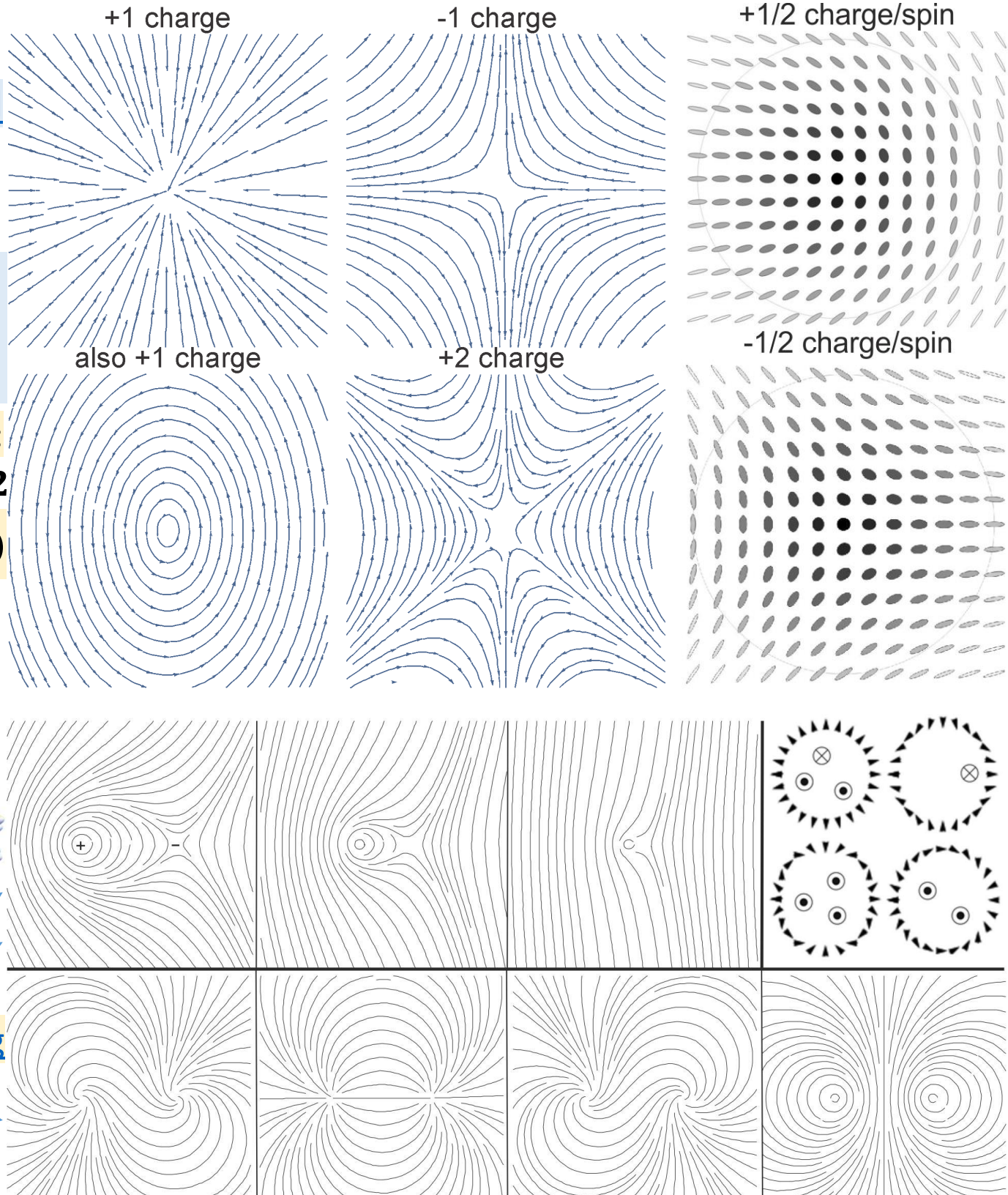
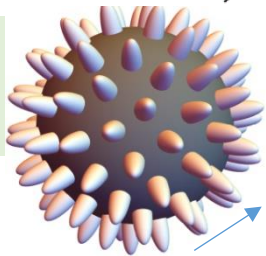
long-range attraction/repulsion:

virtual particle: perturbation toward

de Broglie clock/zitterbewegung

twisting  of quantum phase

around - forming pilot wave

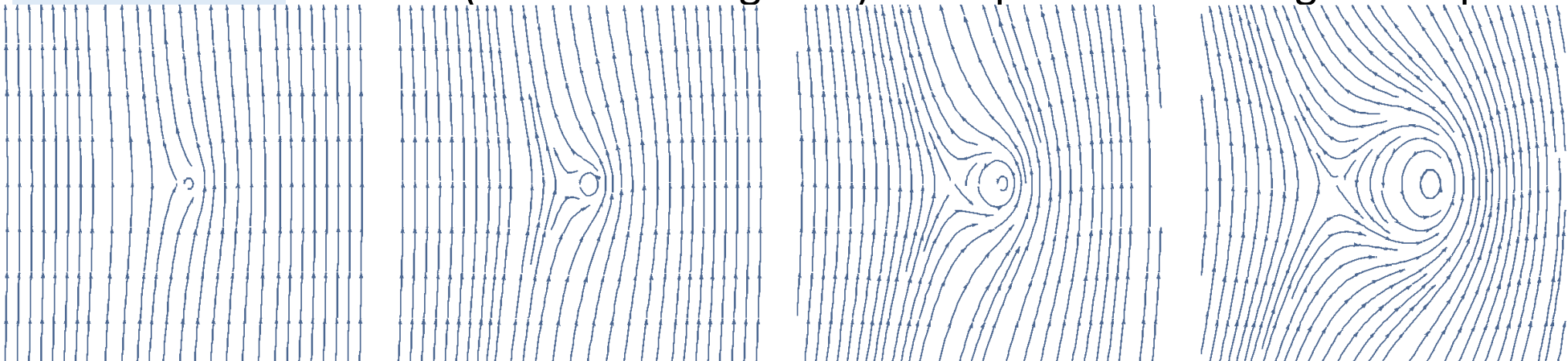


electromagnetism	topological charge/index/ <u>degree</u>
charge conservation	the number of covering of $f: S^{D-1} \rightarrow S^{D-1}$ <u>Hopf thm</u> : f, g are homotopic $\Leftrightarrow \deg f = \deg g$ <u>Poincare-Hopf thm.</u> : $\sum_i \deg_{x_i} = \chi(S)$ <u>Riemann-Hurwitz thm.</u> : $\chi(n \text{ covering of } S) = n \chi(S)$
Gauss law (in 3D) : charge inside $S = \oint_S E \cdot dS$ charge is a real number	<u>argument principle</u> in 2D: $\oint_C (\ln f)' dz = 2\pi i (N - P)$ general - quantized <u>Gauss-Bonnet theorem</u> : $\oint_S R \cdot dS = 2\pi n \chi(S)$ (R : curvature/Jacobian)
Coulomb : same charges repel, opposite attract $F \propto 1/r^2$	to reduce stress of the field <u>entire field causes attraction/repulsion</u>

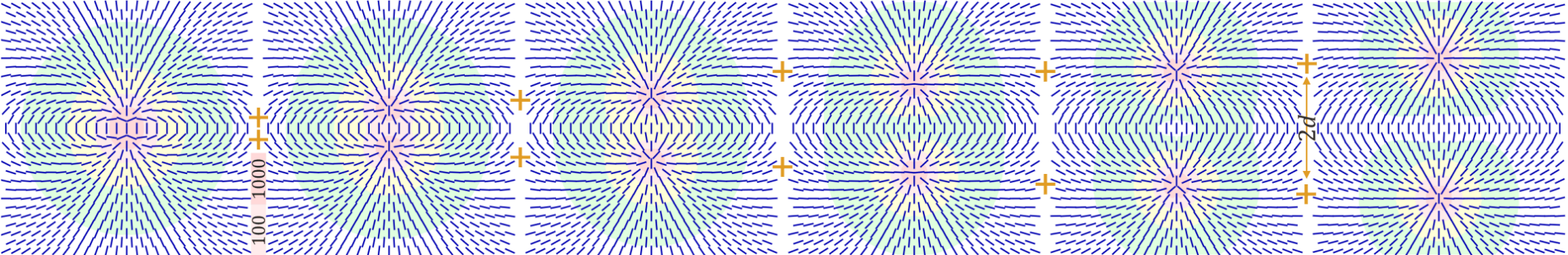
$$\vec{\Gamma}_i = (\partial_i \vec{n}) \times \vec{n} \quad \vec{R}_{\mu\nu} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu \quad \mathcal{L}_{EM} = -\frac{\alpha \hbar c}{16\pi} \vec{R}_{\mu\nu} \vec{R}^{\mu\nu} \quad \text{Faber's quantized EM}$$

$|\vec{n}| = 1$, **connection** $\Gamma \sim A$, **EM Lagrangian** with **curvature** $F_{\mu\nu}^* \sim R_{\mu\nu}$ (+Higgs)

dual: $E \leftrightarrow^* B$: electric (instead of magnetic) monopoles ... and magnetic dipoles



Virtual pair creation: start $< 2 \times 511$ keV and return, perturbation toward pair creation

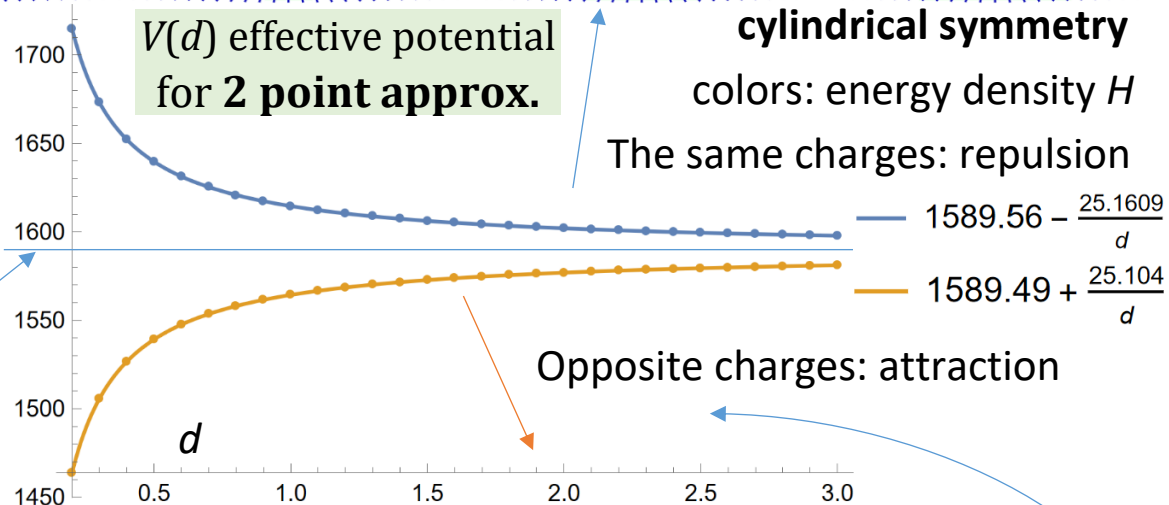


Field energy: ~Coulomb potent.

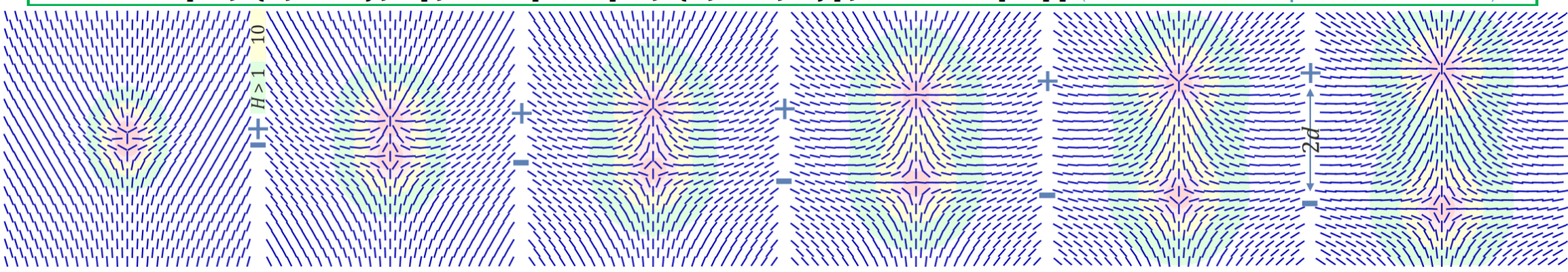
$$E \approx \frac{m_0}{\sqrt{1 - v_1^2}} + \frac{m_0}{\sqrt{1 - v_2^2}} + V(d)$$

cutoff ϵ around singularities:
to be **regularized** to rest masses

Lorentz inv.: SRT scaling, magnetism



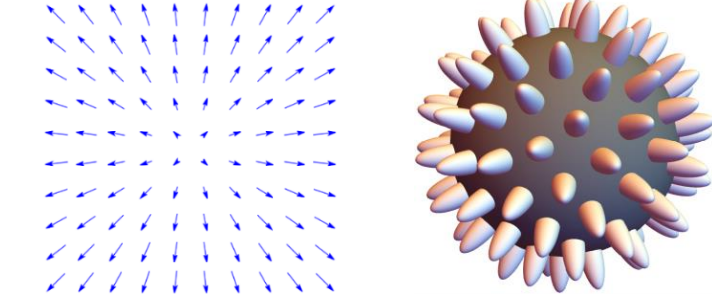
```
cos = 1 + (z - d) / Sqrt[(z - d)^2 + r^2] - (z + d) / Sqrt[(z + d)^2 + r^2]; (*Manfried Faber dipole ansatz*)
n = {Sqrt[1 - cos^2] x / r, Sqrt[1 - cos^2] y / r, cos} /. r -> Sqrt[x^2 + y^2]; (*cylindrical symmetry*)
M = KroneckerProduct[n, n]; dM = {D[M, x], D[M, y], D[M, z]}; (*vector n -> matrix M field*)
H = Simplify[Sum[Total[(dM[[i]].dM[[j]] - dM[[j]].dM[[i]])^2, 2], {i, 2}, {j, i + 1, 3}]]; (*Hamiltonian*)
Es = Table[{d, NIntegrate[4 Pi * x (H /. {y -> 0}) * Boole[x^2 + (z - d)^2 > 0.001], (*0.001 cutoff*)
  {x, 0, Infinity}, {z, 0, Infinity}]}], {d, 0.1, 3, 0.1}]; (*integrate H: total field energy*)
ft = Fit[Es, {1, 1/d}, d]; Show[Plot[ft, {d, 0.2, 3}], ListPlot[Es]] (*fit Coulomb potential*)
```



Regularization to finite energy e.g. for “hedgehog”:

Asymptotically (vacuum) $||\vec{n}|| \approx 1$, but $\vec{n}(0) = 0$

thanks to **Higgs-like potential**, e.g. $V(\vec{n}) = (||\vec{n}||^2 - 1)^2$



$$C \Gamma \propto r^{-1}$$

$$R \propto r^{-2}$$

$$\vec{C}: \vec{\Gamma}_i = (\partial_i \vec{n}) \times \vec{n} \quad \rightarrow \quad \frac{1}{r} ||\vec{n}||^2 \rightarrow \frac{1}{r}$$

$$\vec{R}_{\mu\nu} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu \quad \rightarrow \quad \frac{1}{r^2}$$

electric field in 3D, dual:

$$*_F_{\mu\nu} = \frac{-e_0}{4\pi\epsilon_0 c} \vec{R}_{\mu\nu} \sim \begin{pmatrix} 0 & B_1 & B_2 & B_3 \\ -B_1 & 0 & E_3 & -E_2 \\ -B_2 & -E_3 & 0 & E_1 \\ -B_3 & E_2 & -E_1 & 0 \end{pmatrix}$$

$$\mathcal{L}_{EM} = -\frac{\alpha\hbar c}{16\pi} \vec{R}_{\mu\nu} \vec{R}^{\mu\nu} \rightarrow \text{electromagnetism in vacuum}$$

Gauss law counting topological charge:

$$Q_{el}(\mathcal{S}) = \frac{e_0}{4\pi} \oint_{\mathcal{S}(u,v)} du dv (\partial_u \vec{n} \times \partial_v \vec{n}) \cdot \vec{n}$$

Jacobian of closed surface $\rightarrow S^2$: $\det[\vec{n}, \vec{n}_\mu, \vec{n}_\nu] = \vec{n} \cdot (\vec{n}_\mu \times \vec{n}_\nu)$

No parton structure (!) for **electron**: point-like,

only field deformation not to exceed 511keVs, $\sim$ vacuum polarization

Faber:

$$\int_{\sim 1.4\text{fm}}^{\infty} \frac{1}{2} |E|^2 4\pi r^2 dr = 511\text{keV}$$

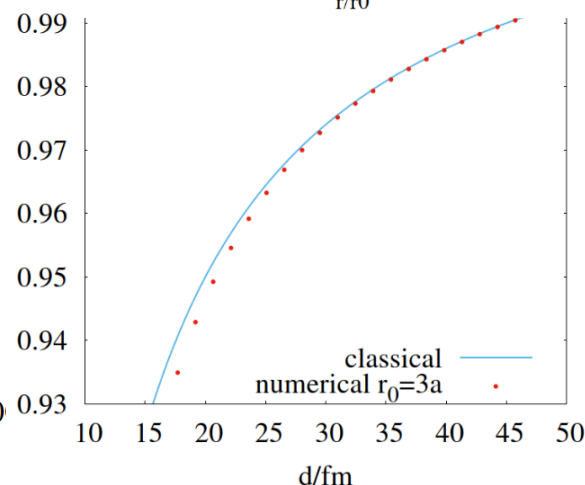
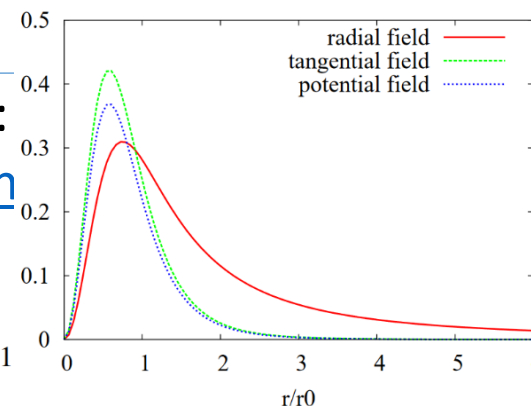
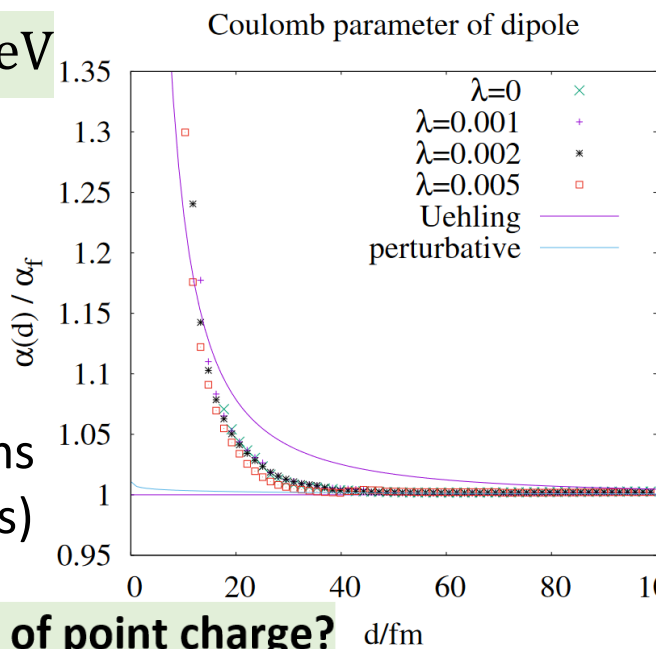
Experimental effects of finite size?

Coulomb deformation, Faber:

Running coupling:

$$\alpha \approx \frac{1}{137} \rightarrow \approx \frac{1}{127} \text{ in 90 GeVs}$$

To hide finite size in Feynman diagrams
(+ renormalization to remove infinities)



Point-like, but with deformed electric field of point charge?

Experimental boundaries for size of electron?

Point-like, but with deformed electric field of point charge?

[Dehmelt](#) 1988, Penning trap (Nobel in 1989): $R < 10^{-22}m$
 extrapolating from g-factor: “(...) electron as **three smaller**
much heavier new fermions [[Brodsky, Drell, 1980](#)] (...)”

Neutron (udd): $g \approx -3.8$ $\langle r_n^2 \rangle \approx -0.1 \text{ fm}^2$

Classically: $g = \frac{2m}{q} \frac{\mu}{L} = \frac{2m}{q} \frac{\int Adl}{\omega l} = \frac{m}{q} \frac{\int \rho_q(r) r^2 dr}{\int \rho_m(r) r^2 dr}$

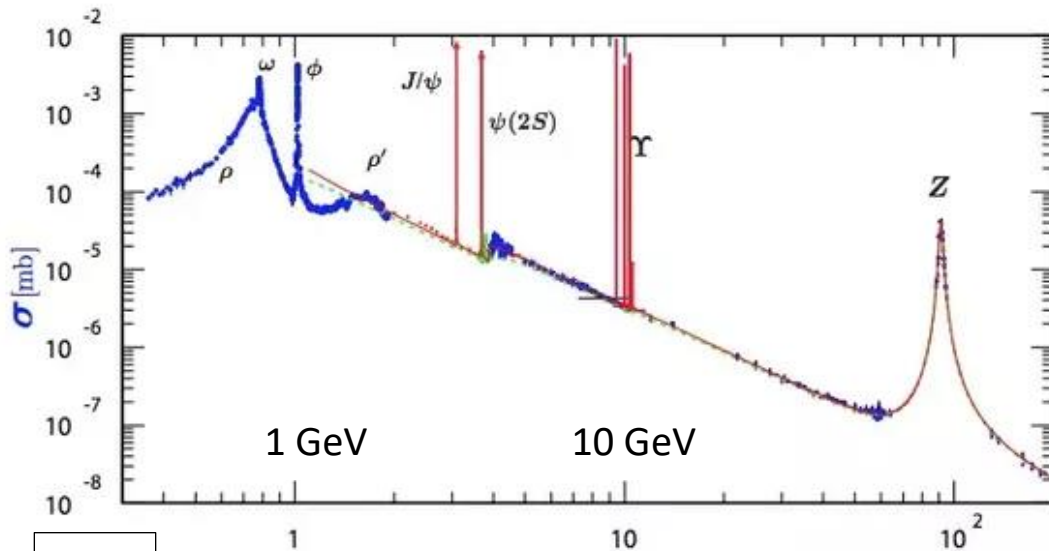
Electron-positron cross-section:

Which energy should we use??? (Lorentz contraction!),

~line in log-log, $\approx 100\text{nb}$ for 1GeV $\gamma \approx 1000$

Extrapolating to resting: $\gamma = 1$ $\sigma \propto \gamma^{-2}$

we get $\approx 100\text{mb}$: $r \approx 2 \text{ fm}$? (energy $< 511\text{keV}$!)



GeV

$$\int_{\sim 1.4\text{fm}}^{\infty} \frac{1}{2} |E|^2 4\pi r^2 dr = 511\text{keV}$$

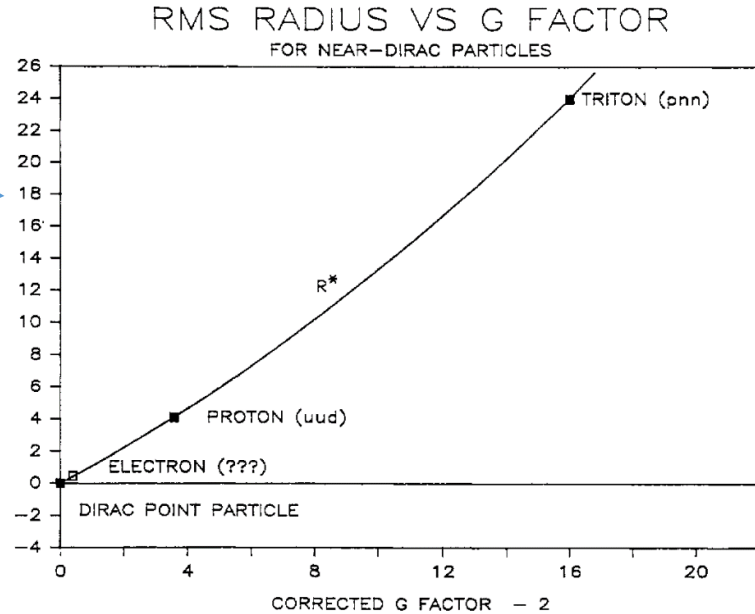
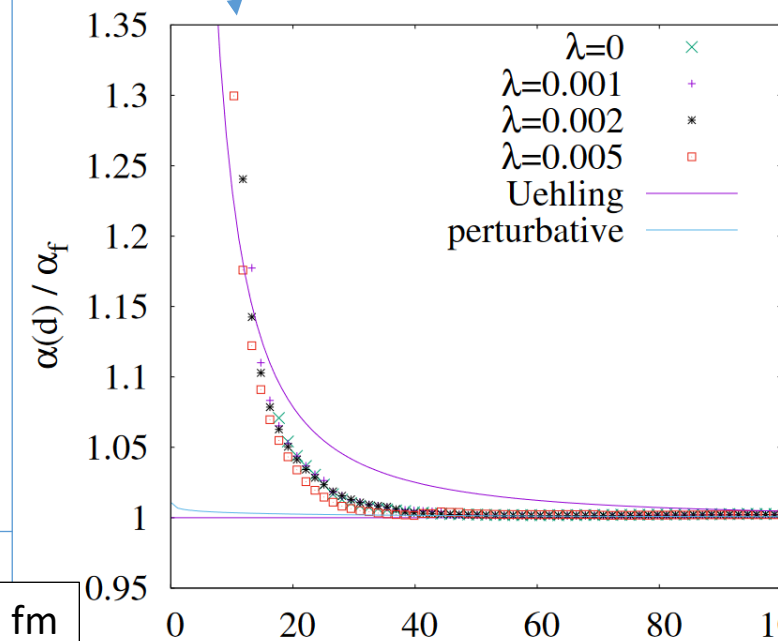


Fig. 8. Normalized RMS radius $R^* = R/\lambda_c$ vs. corrected g -factor minus 2 for near-Dirac particles [3]. A parabola has been fitted to the data points. Recent theories conjecture that the electron, similar to proton and triton, is composed of three smaller fermions. The data point at the origin represents a Dirac point particle of finite arbitrary charge and mass.

Running coupling – deformation of α (Coulomb) **proper in Faber's model**

Coulomb parameter of dipole



fm

Popular skyrmion models

Solid state, nucleus: PRL 2018

U – tensor field (of matrix)

$\Gamma_i = \partial_i U U^{-1}$ local rotation

$$\mathcal{L} = \text{Tr}[\Gamma_\mu, \Gamma_\nu][\Gamma^\mu, \Gamma^\nu] + \dots$$

Same **curvature**² ... but $V \propto \text{Tr}(\mathbf{1} - U)$ for unitary: single minimum $U = \mathbf{1}$

conserved baryon number – violated e.g. in baryogenesis, Hawking radiation

Vacuum (far from particles) filled with $U \approx \mathbf{1}$, only short-range interaction!

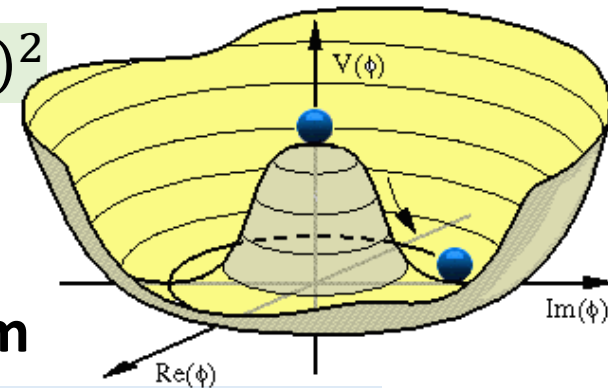
No EM, charge (**proton = neutron**), no long-range e.g. Coulomb interaction

(**proton lighter** than **neutron**, **conserved electric charge**)

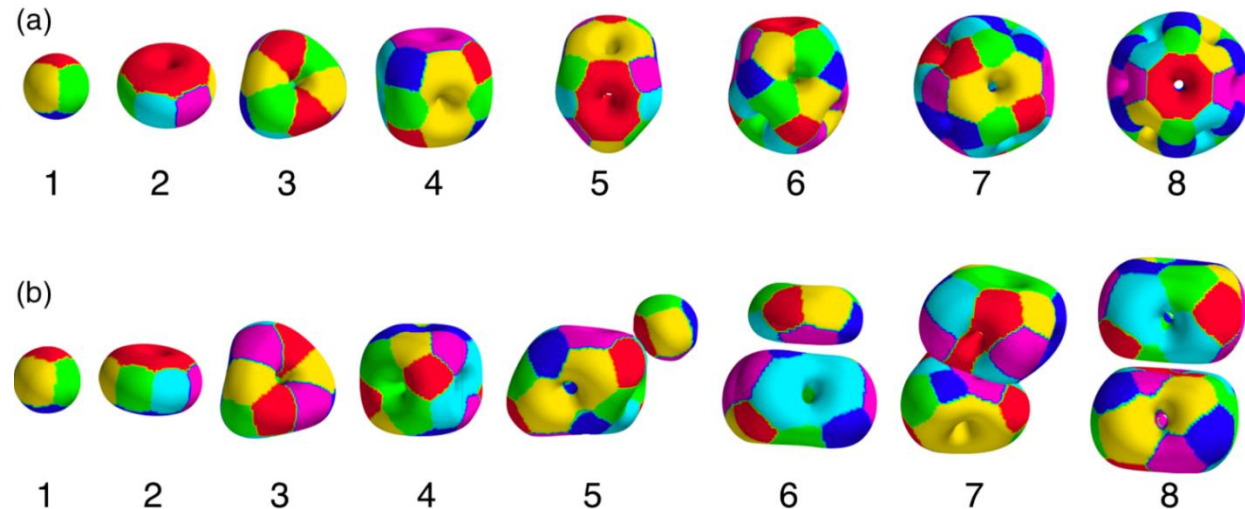
For long-range: use **topologically nontrivial vacuum** (minimum of potential)

E.g. **Higgs potential** “Mexican hat”: $V(\vec{n}) = (||\vec{n}||^2 - 1)^2$

- zero is not minimum (inflation, charge regularization)
- **Dynamics in minimum** corresponds to massless particles (Goldstone bosons), like **electromagnetism**



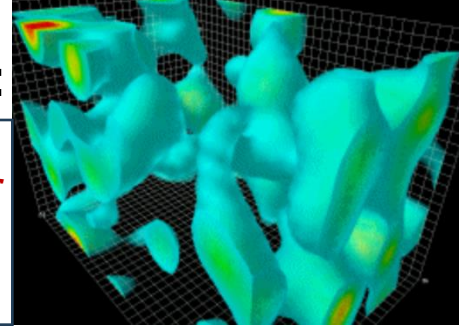
$$\vec{R}_{\mu\nu} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu, \quad \mathcal{L}_{EM} = -\frac{\alpha\hbar c}{16\pi} \vec{R}_{\mu\nu} \vec{R}^{\mu\nu} \quad \text{EM with quantized charge (Faber)}$$



Lattice QCD: non-perturbative, seen as **fundamental**??? ... lk:

Lagrangian: $\mathcal{L} = -\frac{1}{2} \text{Tr}[F_{\mu\nu} F^{\mu\nu}] + \sum_{f=1}^6 \bar{\psi}_f (iD - m_f) \psi_f$

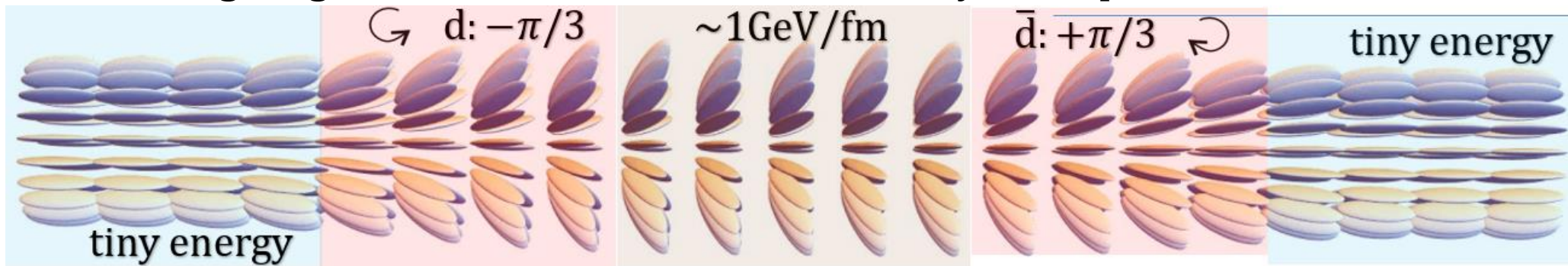
(solved Boltzmann) EM $\rightarrow$ Yang-Mills Dirac, densities



It has **Dirac term** for 6 quarks, which like Schrödinger hides Feynman **averaging**
No EM, what is EM field configuration of these charged particles $E \sim r^{-2}$?

What about **quark strings** modeled as topological vortices?

More fundamental? Still **Yang-Mills**, but **particles are topological excitations**
e.g. to get such Dirac as effective from Feynman path ensembles



Living in **3+1D**, **field dynamics** should also **happen there**, e.g.:

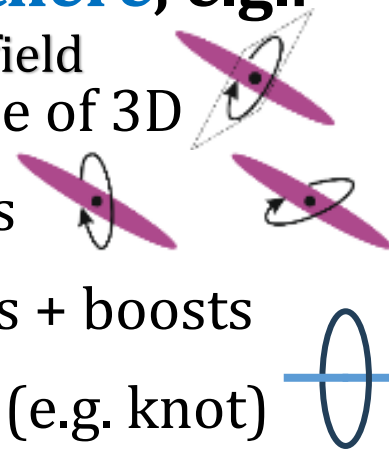
e.g. electron has **real angular momentum** – of the field

U(1) ~ SO(2) - **quantum phase**: rotations restricted to 2D plane of 3D

SU(2)_L ~ SO(3) - e.g. **neutrinos**: unrestricted 3D field rotations

SL(2, C)_L ~ SO⁺(1, 3) – Lorentz group e.g. in **Dirac**: 3D rotations + boosts

SU(3) Yang-Mills: 3D rotations with coupled 2D phase rotation (e.g. knot)



+ twists e.g. for angular momentum conservation
 Affine connection $SO(1,3) \ni O \rightarrow O(I + \epsilon \Gamma_\mu)$:
 local rotations, boosts: $\Gamma_\mu = O^T \partial_\mu O =$
 of $M = O D O^T$ $D \approx \text{diag}(g, 1, \delta, 0)$

QM+EM tilts twist ψ

$g_{\text{grav}} \gg 1 \gg \delta_{\sim \hbar} > 0$ shape

$L_{QED} = -\hbar c \bar{\psi} \gamma^\mu D_\mu \psi - mc^2 \bar{\psi} \psi - F_{\mu\nu} F^{\mu\nu} / 4$

QM: $\delta \Gamma^1 \Gamma^2, \delta \Gamma^1 \Gamma^3$, electromagnetism: $\Gamma^2 \Gamma^3$, GEM: $g \tilde{\Gamma}^1 \tilde{\Gamma}^2, g \tilde{\Gamma}^2 \tilde{\Gamma}^3, g \tilde{\Gamma}^3 \tilde{\Gamma}^1$

3D curvature EM+QM

$$\vec{R}_{\mu\nu} \equiv \vec{R}_{\mu\nu}^{ee} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu$$

0th axis boost GEM

$$\vec{R}_{\mu\nu}^{gg} = \vec{\Gamma}_\mu^g \times \vec{\Gamma}_\nu^g$$

4D EMQM-GEM interact.:

$$\Gamma^g \equiv \tilde{\Gamma} \quad \vec{R}_{\mu\nu}^{eg} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu^g$$

light bending, time dilat.

GEM equations	Maxwell's equations
$\nabla \cdot \mathbf{E}_g = -4\pi G \rho_g$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$
$\nabla \cdot \mathbf{B}_g = 0$	$\nabla \cdot \mathbf{B} = 0$
$\nabla \times \mathbf{E}_g = -\frac{\partial \mathbf{B}_g}{\partial t}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
$\nabla \times \mathbf{B}_g = -\frac{4\pi G}{c^2} \mathbf{J}_g + \frac{1}{c^2} \frac{\partial \mathbf{E}_g}{\partial t}$	$\nabla \times \mathbf{B} = \frac{1}{\epsilon_0 c^2} \mathbf{J} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$



+ short-range (to particles) axes deformations $\partial_\mu D$ $O^T M_\mu O = \bar{M}_\mu + \partial_\mu D$

$$O^T \bar{F}_{\mu\nu} O = O^T [M_\mu, M_\nu] O = [\bar{M}_\mu, \bar{M}_\nu] + [\partial_\mu D, \bar{M}_\nu] + [\bar{M}_\mu, \partial_\nu D]$$

EM + QM + GEM + Yang-Mills gluons ($L_{YM} = -G_{\mu\nu}^a G_a^{\mu\nu} / 4$) + Higgs particle

weak: $SU(2)_{LR} \sim SO(3)$ neutrino 3D rotations, strong: $SU(3)$ baryon with twist
 teleparallelism – 4 vector field Einstein's attempt to unify EM + gravity

Vacuum (long distance) dynamics:

1) EM: **quantized** electric charge

with **Coulomb** in $V(r) \propto r^{-1}$

(+ **magnetism** from **Lorentz invariance**)

S^2 : **Gauss law counts topological charge**

2) S^1 quantum phase: **Berry, pilot wave**

e.g. for Mach-Zehnder interfer.

Unify EM S^2 + QM $S^1 \rightarrow SO(3)$

"extended quantum phase"

governed by **wave equation**:

Maxwell $\square A_\mu \propto J_\mu$ + Klein-Gordon $\square \psi$

Momentum operator:

$$\hat{P} = -qA - i\hbar\nabla$$

$A \sim \Gamma$ also hides derivative

current/flow: $\vec{J} \sim \vec{v}_s m = \hbar \vec{\nabla} \varphi - q \vec{A}$

$$2\pi k = \Delta\varphi = \frac{q}{\hbar} \oint_{\partial S} A \cdot dl = \frac{q}{\hbar} \oint_S B \cdot dS$$

EM-hydro analogies: vorticity $2\vec{\Omega} = \vec{\nabla} \times \vec{U} \sim \vec{B} = \vec{\nabla} \times \vec{A}$

Aharonov-Bohm (**Berry**): electron $\sim$ hydro

Zeeman: Coriolis $\vec{F}_\Omega = -m(\vec{V} \times 2\vec{\Omega})$

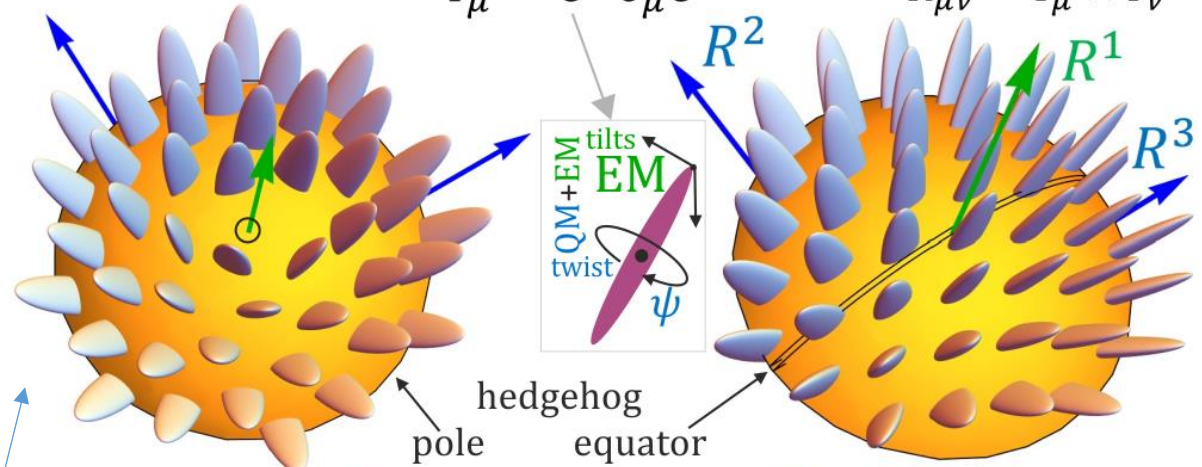
as **Lorentz force**: $\vec{F}_B = q(\vec{v} \times \vec{B})$

green: tilt-tilt of $\vec{n}$ main axis EM high energy main curvature

blue: tilt-twist QM phase low energy curvatures

$$\Lambda = (\mathbf{1}, \delta, 0) \Rightarrow \vec{A}_\mu \approx (\delta^2 \Gamma_\mu^1, \Gamma_\mu^2, \Gamma_\mu^3), \quad \vec{F}_{\mu\nu} \approx (R_{\mu\nu}^1, \delta R_{\mu\nu}^2, \delta R_{\mu\nu}^3)$$

$\approx$ fixed far from charge $\Gamma_\mu = O^T \partial_\mu O \quad \vec{R}_{\mu\nu} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu$



$$\mathcal{L}_{QED} = -\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi - F_{\mu\nu} F^{\mu\nu} / 4$$

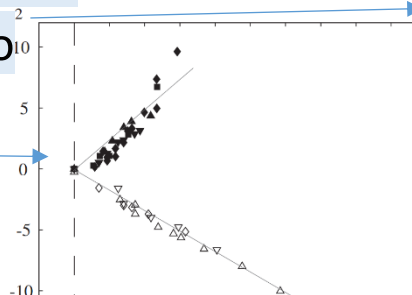
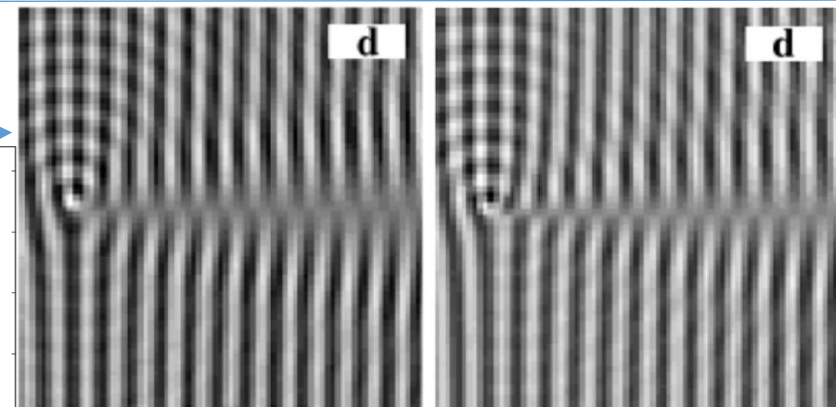
Aharonov-Bohm $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ gauge? A extended quantum phase for topological charge quantization

Maxwell: $\square A_\mu \propto J_\mu$

(dual) $A_\mu^* = [M, M_\mu] \cong$ affine connection $F^* \cong$ curvature

ψ 2D $\xrightarrow{M=ODO^T}$ 3D $\square \psi \propto -\psi \mid \hat{P} = -i\hbar \nabla - qA$

QM+EM tilts EM twist



QED $\hbar c$ tiny $\rightarrow$ unify, also gravity

$$\mathcal{L}_{QED} = -\hbar c \bar{\psi} \gamma^\mu \partial_\mu \psi - mc^2 \bar{\psi} \psi - F_{\mu\nu} F^{\mu\nu} / 4$$

Landau-de Gennes/EM-like Lorentz-invariant Lagrangian, vacuum for long-range interactions

$V=0$: $M = O D O^T$ $O \in SO(1,3)$

stress-energy tensor+Higgs
 O : 3D rotation + 4D boost

$D = \text{diagonal}(g, 1, \delta, 0)$

gravity + EM + QM
 $g \gg 1$ boosts 1 $1 \gg \delta > 0$
 $\sim 1/G$ 1 Planck $\hbar$

$$\mathcal{L} = - \sum_{\alpha\beta\mu\nu} F_{\mu\nu\alpha\beta} F^{\mu\nu\alpha\beta} - V(M)$$

EM-like Higgs-like
 $SO(1,3)$ minimum
for $F_{\mu\nu\alpha\beta} = [\partial_\mu M, \partial_\nu M]_{\alpha\beta}$
charge quantization
 F : shape D weighted curvature R

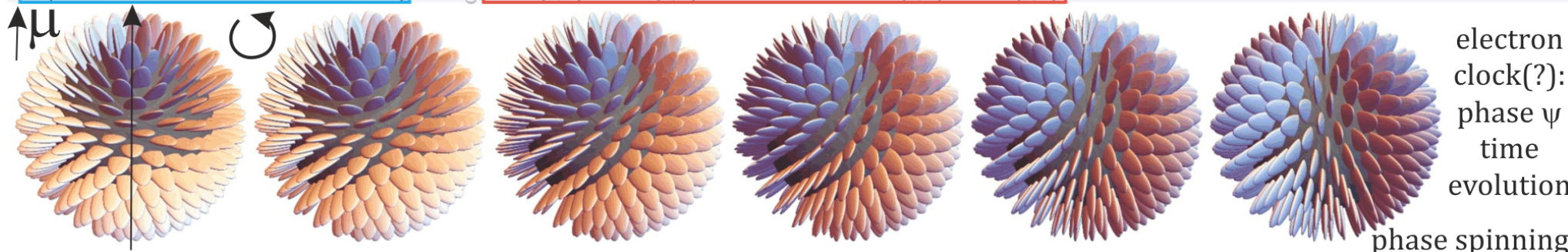
gauge? A
Aharonov-Bohm $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$
Maxwell: $\square A_\mu \propto J_\mu$
extended quantum phase for topological charge quantization
(dual) $A_\mu^* = [M, M_\mu]$
 $\cong$ affine connection
 $F^* \cong$ curvature
 ψ $2D \xrightarrow{M=ODO^T} 3D$
 $\square \psi \propto -\psi$ $\hat{P} = -i\hbar \nabla - qA$
tilts EM
QM+EM twist
 $U(1)$

$\bar{M}_\mu = \{ \{0, g \tilde{r}_\mu^1, g \tilde{r}_\mu^2, g \tilde{r}_\mu^3\}, \{-g \tilde{r}_\mu^1, 0, r_\mu^3, -r_\mu^2\}, \{-g \tilde{r}_\mu^2, r_\mu^3, 0, \delta r_\mu^1\}, \{-g \tilde{r}_\mu^3, -r_\mu^2, \delta r_\mu^1, 0\} \}; (* g \gg 1 \gg \delta > 0 \text{ approximation} *)$

$(\bar{F}_{\mu\nu} = \text{FullSimplify}[\text{coms}[\bar{M}_\mu, \bar{M}_\mu /. \mu \rightarrow \nu], \text{sub}]) // \text{MatrixForm}$

0	$g (r_\nu^3 \tilde{r}_\mu^2 - r_\nu^2 \tilde{r}_\mu^3 - r_\mu^3 \tilde{r}_\nu^2 + r_\mu^2 \tilde{r}_\nu^3)$	$g (r_\nu^3 \tilde{r}_\mu^1 + \delta r_\nu^1 \tilde{r}_\mu^3 - r_\mu^3 \tilde{r}_\nu^1 - \delta r_\mu^1 \tilde{r}_\nu^3)$	$g (-r_\nu^2 \tilde{r}_\mu^1 + \delta r_\nu^1 \tilde{r}_\mu^2 + r_\mu^2 \tilde{r}_\nu^1 - \delta r_\mu^1 \tilde{r}_\nu^2)$
0	$-\delta R_{\{\mu,\nu\}}^3 - g^2 \tilde{R}_{\{\mu,\nu\}}^3$	$\delta R_{\{\mu,\nu\}}^3 + g^2 \tilde{R}_{\{\mu,\nu\}}^3$	$\delta R_{\{\mu,\nu\}}^2 - g^2 \tilde{R}_{\{\mu,\nu\}}^2$
0	$-\delta R_{\{\mu,\nu\}}^2 + g^2 \tilde{R}_{\{\mu,\nu\}}^2$	$-\delta R_{\{\mu,\nu\}}^1 - g^2 \tilde{R}_{\{\mu,\nu\}}^1$	$R_{\{\mu,\nu\}}^1 + g^2 \tilde{R}_{\{\mu,\nu\}}^1$
0	0	0	0

EM/QM - gravity interactions
QM: tilt-twist
EM: tilt-tilt
EM-QM-GEM unifi.



[arXiv:2501.04036](https://arxiv.org/abs/2501.04036) toy model of electron **clock**/neutrino oscillation [mechanism](#)

Relativistic QM: **mass of particle propels its clock** – what is its mechanism?

$\psi = \psi_0 e^{-iEt/\hbar}$ for $E = mc^2$ – electron clock, neutrino oscillations (3 masses)

Also “[time crystal](#)” – with periodic process already in the lowest energy state

Lorentz invariant $R^2 - (\phi_0\psi_1 - \phi_1\psi_0)^2$

$$\phi_1 = \partial_1 \phi$$

shape of particle
energetically favors

$$\psi_0 = \partial_0 \psi$$

phase evolution

[codes to calculate](#)

Clock of kink from mass by curvature coupling

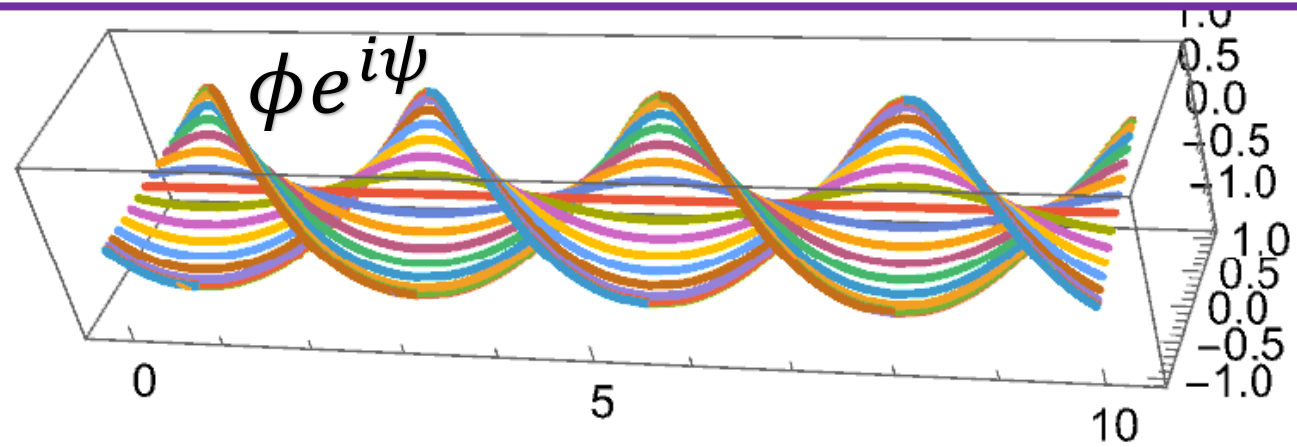
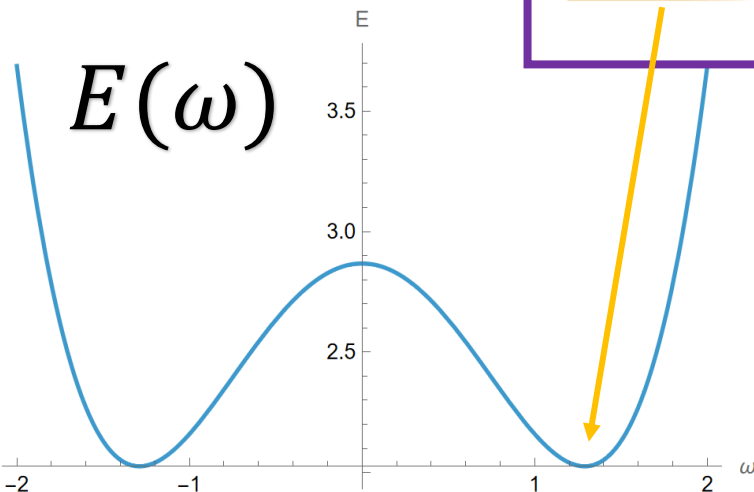
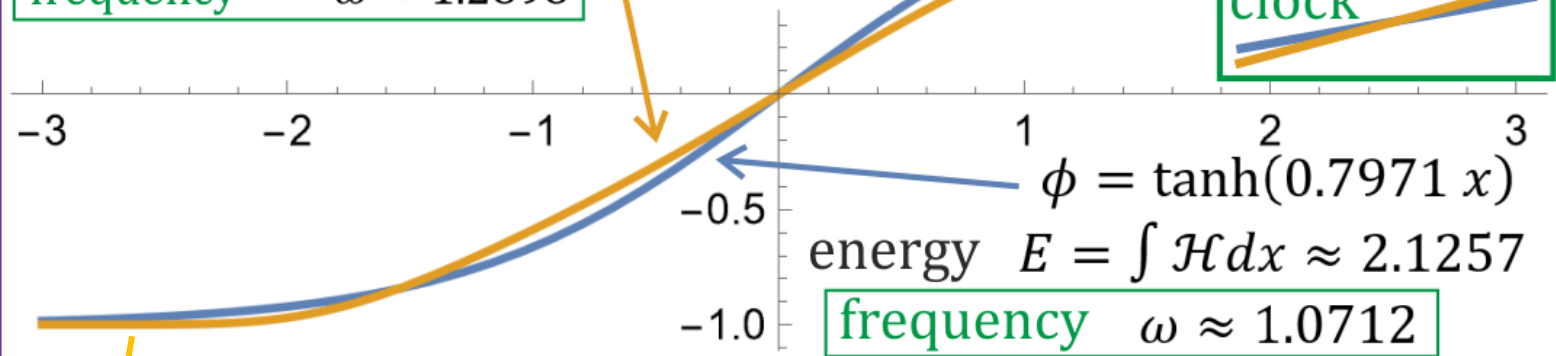
Kink minimizing energy E for Hamiltonian:

$$\mathcal{H} = \phi_0^2 + \phi_1^2 + (1 - \phi^2)^2 - (\phi_0\psi_1 - \phi_1\psi_0)^2 + (\phi_0\psi_1 - \phi_1\psi_0)^4$$

$$\phi = \tanh(0.6326x + 0.0198x^3 + 0.0203x^5)$$

$$\text{energy } E = \int \mathcal{H} dx \approx 2.0252$$

$$\text{frequency } \omega \approx 1.2898$$



electron
clock(?):
phase ψ
time
evolution

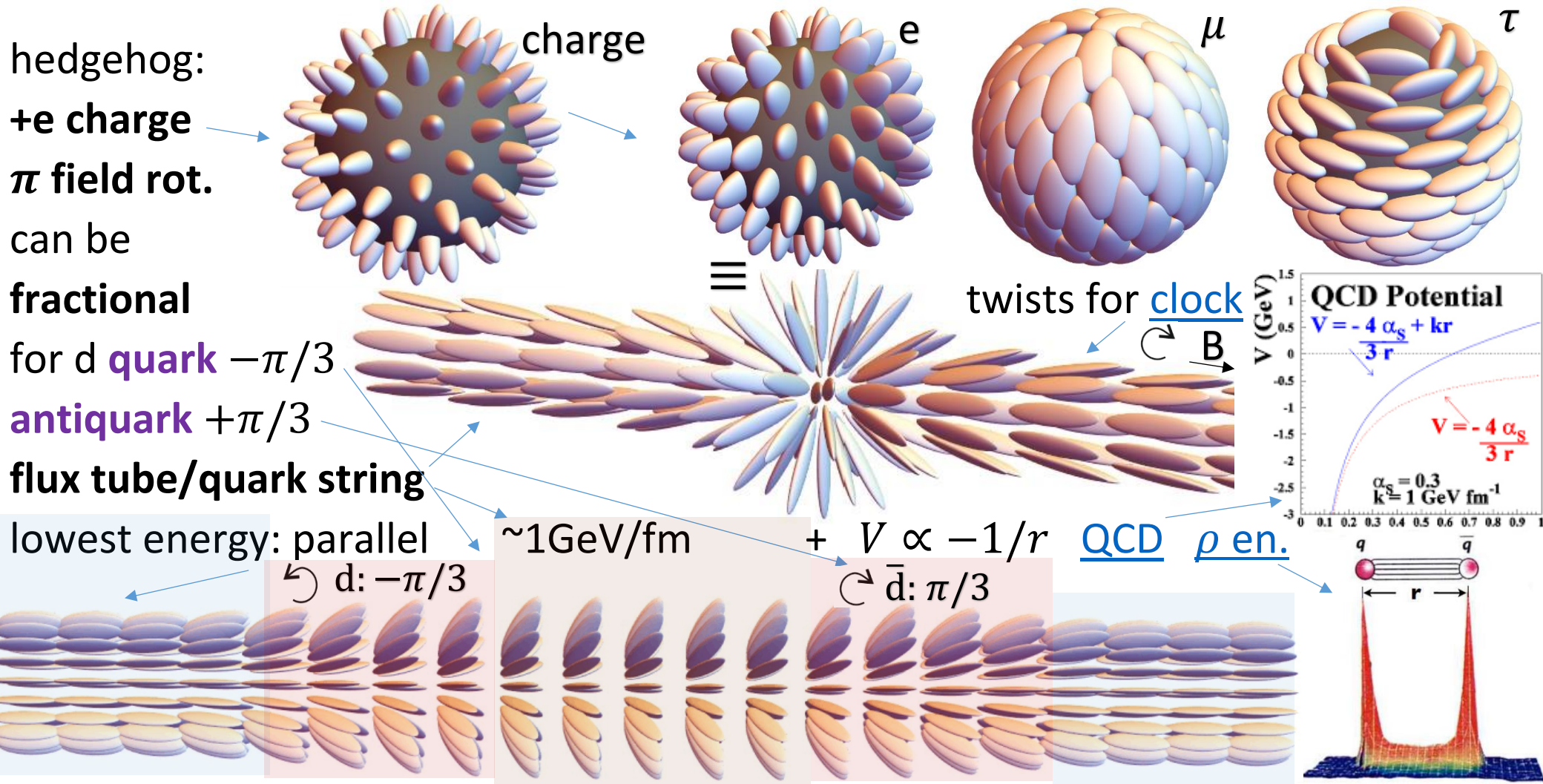
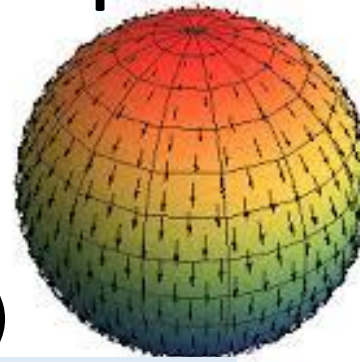
Why no naked charges in physics – leptons need magnetic dipole?

Maybe [hairy ball theorem](#)? - no continuous vector field on sphere

Uniaxial $\rightarrow$ biaxial nematic: $S^2 \rightarrow SO(3)$ vacuum: + $U(1)$ phase

- Required additional linear topological defect (magnetic?)
- Hedgehog of 1 of axes: same charge, different mass (3 leptons?)

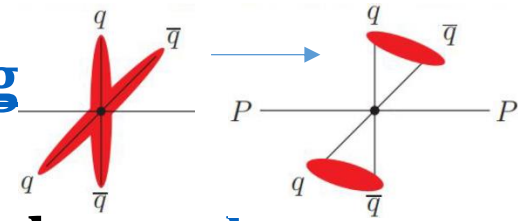
3x3 tensor field: $M = O \cdot \text{diag}(\lambda_i) \cdot O^T$, Higgs-like: $V(M) = \sum_i (\lambda_i - \lambda_i^0)^2$



Hadronization: high energy **quark string reconnecting**

→ **simple loops**: 3 neutrinos, if twisted: pions, kaons,

inward/outward **rotation**: **quark** → **hedgehog** ≡ **charge**, knots: **baryons** ... **nuclei**



Decays e.g.

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

$$\pi^0 \rightarrow \gamma + \gamma$$

$$\Sigma^0 \rightarrow \Lambda^0 + \gamma$$

+s: left/right twists?

$$n^0 \rightarrow p^+ + e^- + \bar{\nu}_e \quad 0s$$

$$\Lambda^0 \rightarrow p^+ + \pi^-, n^0 + \pi^0$$

$$\Xi^0 \rightarrow \Lambda^0 + \pi^0 \quad 123s$$

$$\Xi^- \rightarrow \Lambda^0 + \pi^-$$

$$\Omega^- \rightarrow \Xi + \pi, \Lambda^0 + K^-$$

$$K^+ \rightarrow \mu^+ + \nu_\mu, 2 \vee 3\pi$$

$$K_S^0 \rightarrow 2\pi$$

$$K_L^0 \rightarrow 3\pi, \dots$$

$$\omega \rightarrow 3\pi, \dots$$

$$\eta \rightarrow \gamma + \gamma, 3\pi^0$$

$$\phi(1020) \rightarrow 2K$$

$$\Delta^{++} \rightarrow p^+ + \pi^+ \quad (2 \text{ hedgehogs})$$

Electron, neutrino ... neutral particle oscillations

strangeness decay (?):

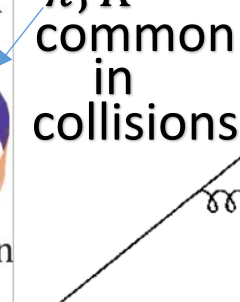
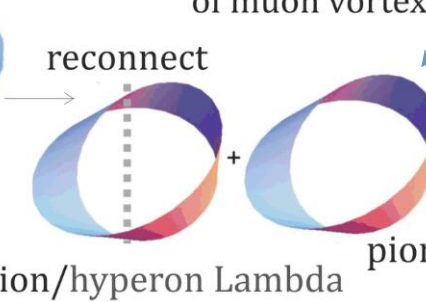
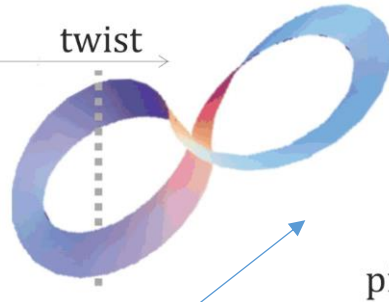
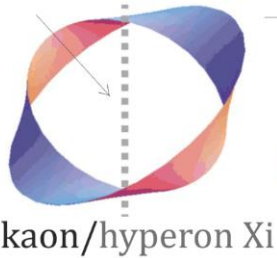
eventual vortex

twist

as internal twists (half-rotations) of muon vortex

reconnect

π, K common in collisions



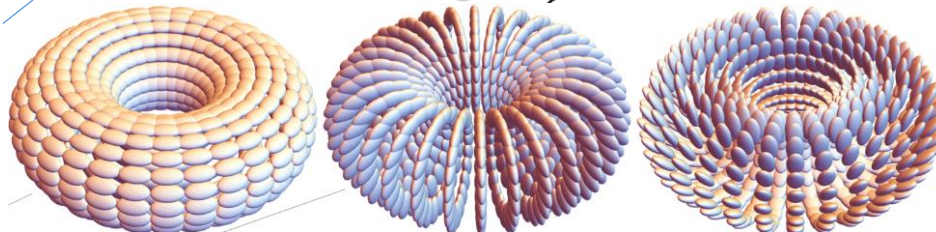
kaon/hyperon Xi

pion/hyperon Lambda

pion

3 neutrinos

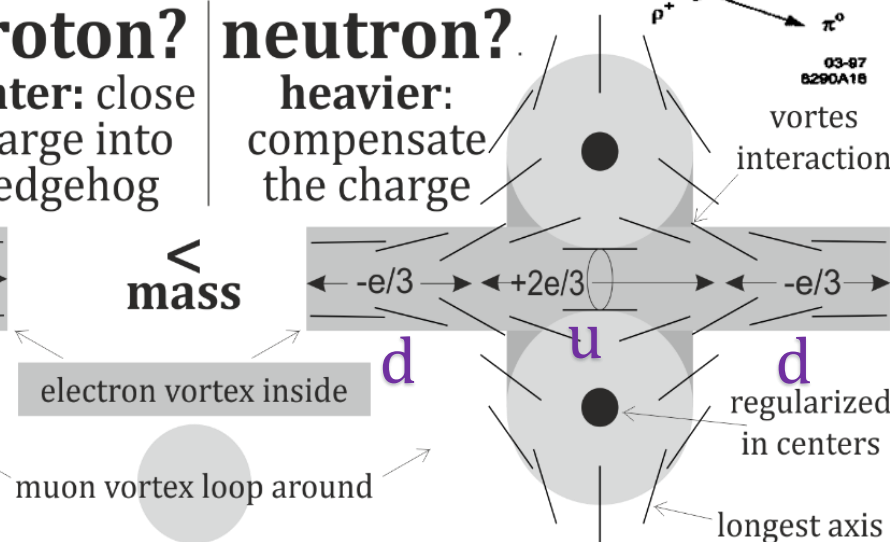
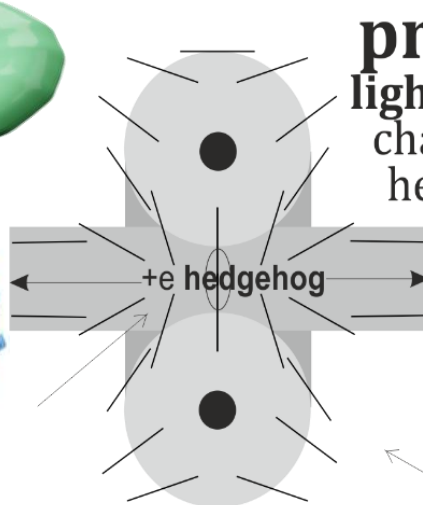
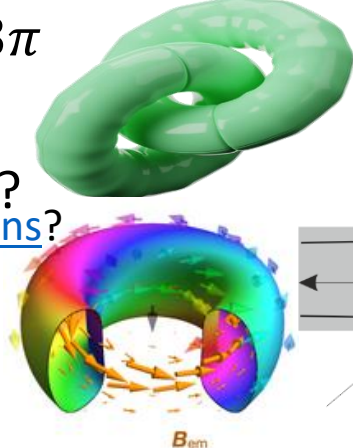
stable, oscillating ↔



proton?
lighter: close charge into hedgehog

neutron?
heavier: compensate the charge

Hopfions?



03-97
5290A18

vortices interaction

< mass

electron vortex inside

muon vortex loop around

d

u

regularized in centers

longest axis

knots?

deforms

no EDM.

proton?
lighter: close
charge into
hedgehog

neutron?
heavier:
compensate
the charge

vortex
interaction

S $\leftarrow -e/3 \rightarrow$

mass

electron vortex inside

muon vortex loop around

regularized
in centers

longest axis

neutrinos?

Wilson 1962

3 gluon flux tubes with 6 quark-antiquark pairs

$$u, c, t: +\frac{2}{3}\pi$$

due to loop
around?

udu merged in proton? No EDM

Gluons SU(3) – rotate with phase

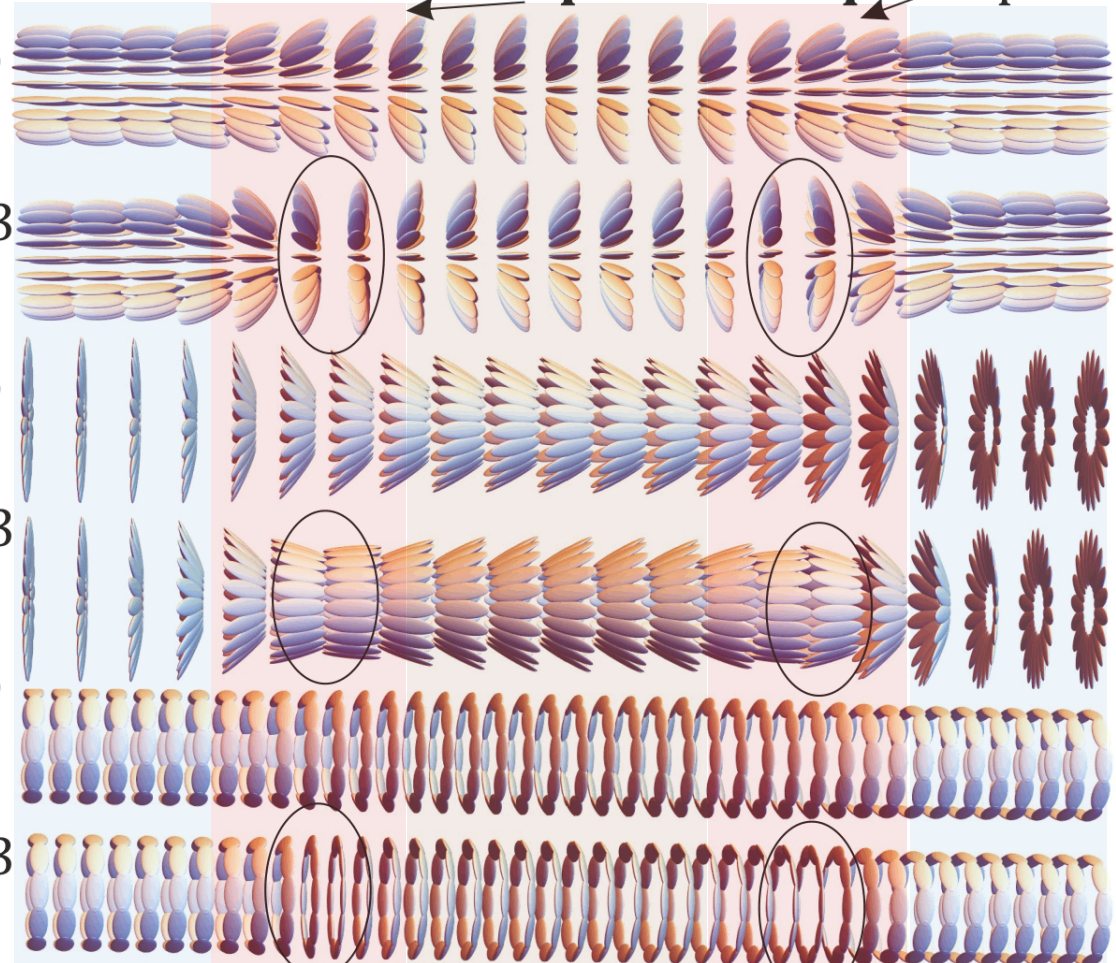
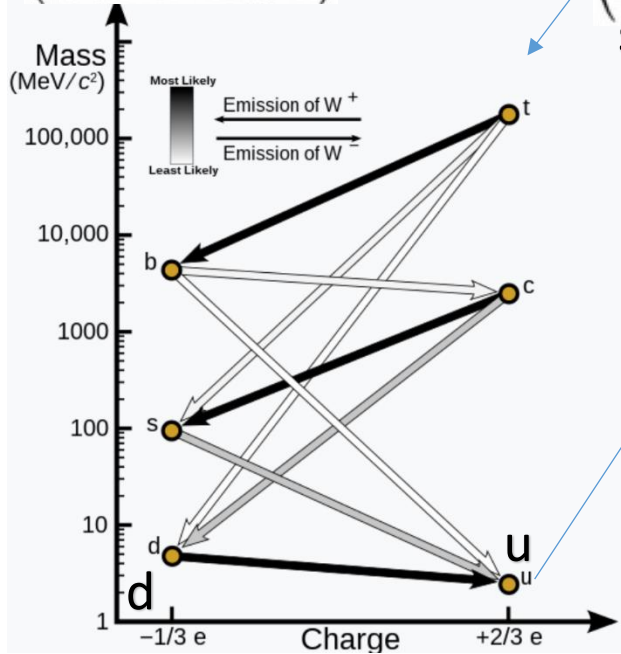
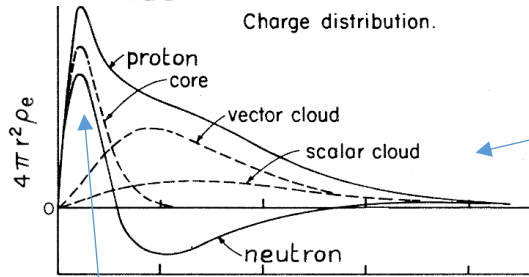
$$\begin{pmatrix} 0.8 & 0.5 & 0.2 \\ 0.4 & 0.6 & 0.7 \\ 0.4 & 0.6 & 0.7 \end{pmatrix} \xrightarrow[\text{PMNS}]{\text{CKM}} \begin{pmatrix} 1 & 0.2 & 0.001 \\ 0.2 & 1 & 0.01 \\ 0.001 & 0.01 & 1 \end{pmatrix}$$
 stabilized by loop

stabilized
by loop

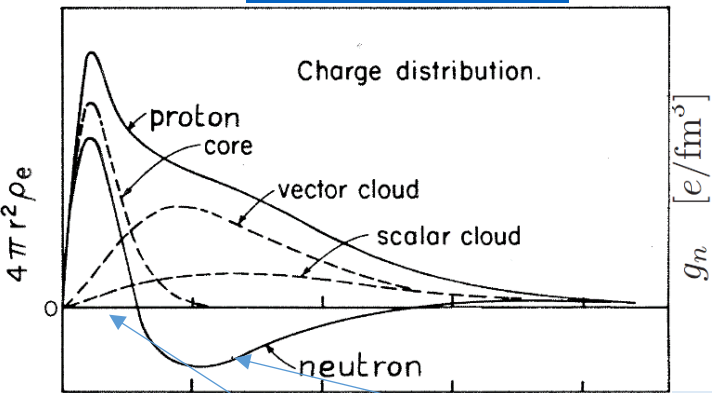
reduced
by loop
around

$$-1/3$$
$$+2/3$$

u

$$-1/3$$
$$+2/3$$
$$-1/3$$
$$+2/3$$


Baryons: [Wilson 1962](#)



Neutron: “+” core, “-” shell charge

g-factors: $g_e \approx -2$, $g_\mu \approx -2$, $g_p \approx +5.6$, $g_n \approx -3.8$

Deuteron: “two baryons share required charge”

0.2859 e·fm² - large electric quadrupole moment

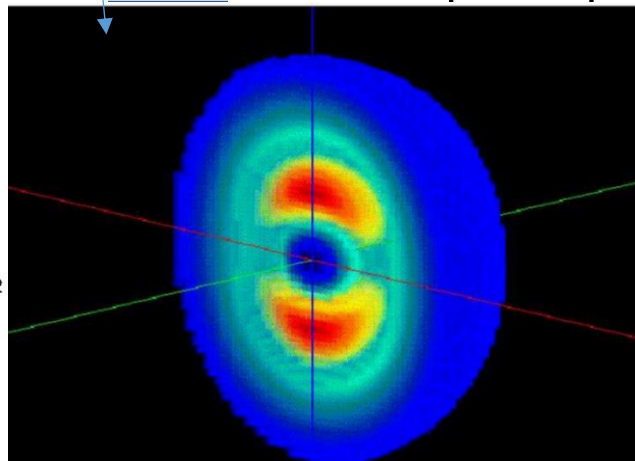
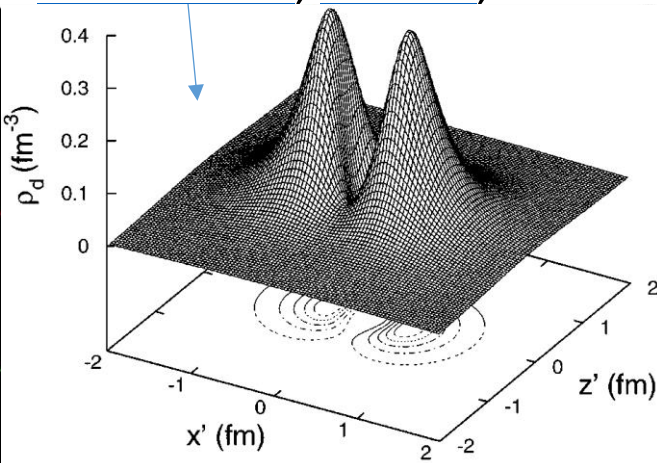
“+ - +” (EDM zero), how to get it for p+n ???

$\mu_d \approx \mu_p + \mu_n$ – aligned spins/angular mom.?

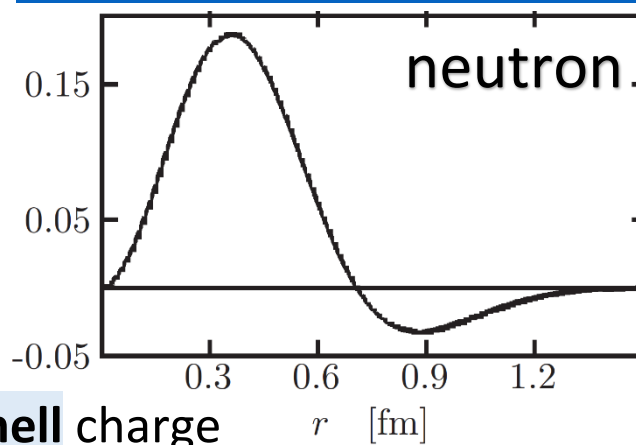
0.857 vs 0.879 μ_N magnetic dipole moments

[The deuteron: structure and form factors](#) (2001)

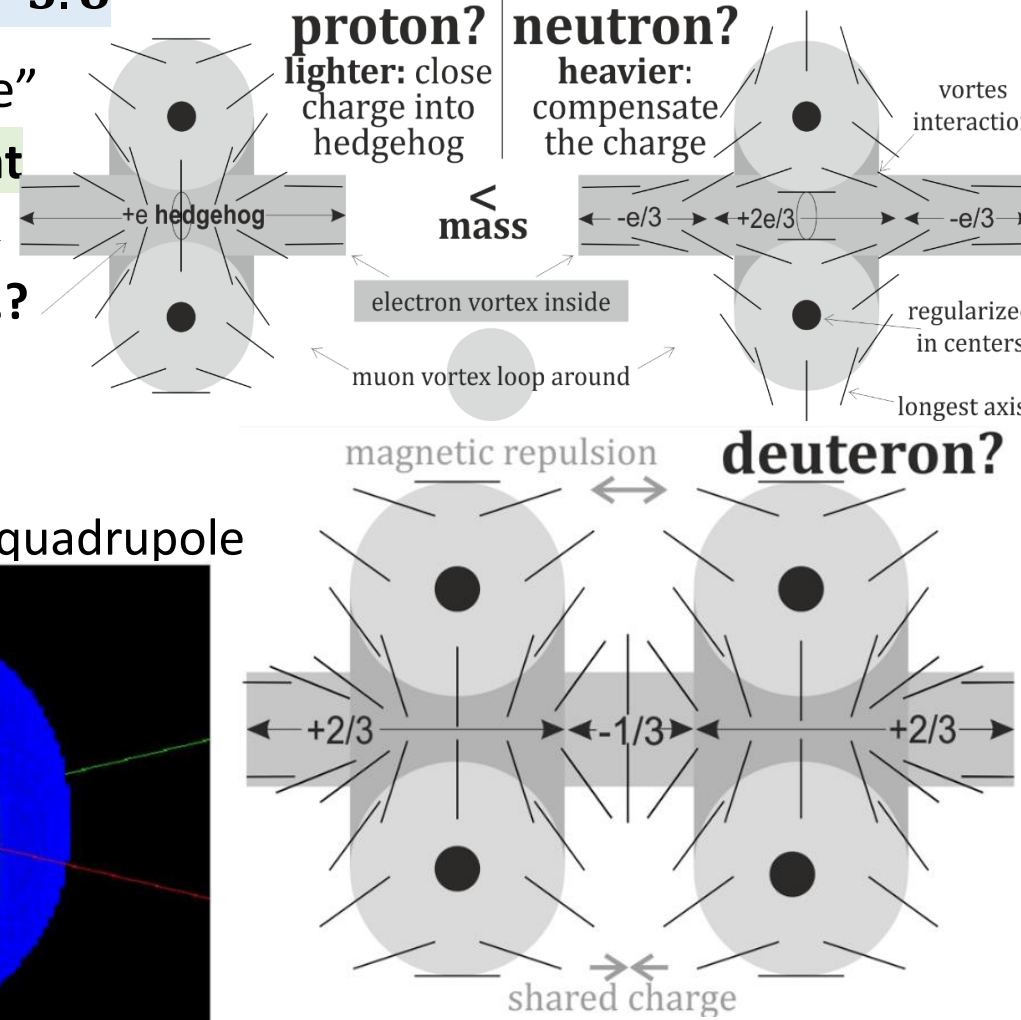
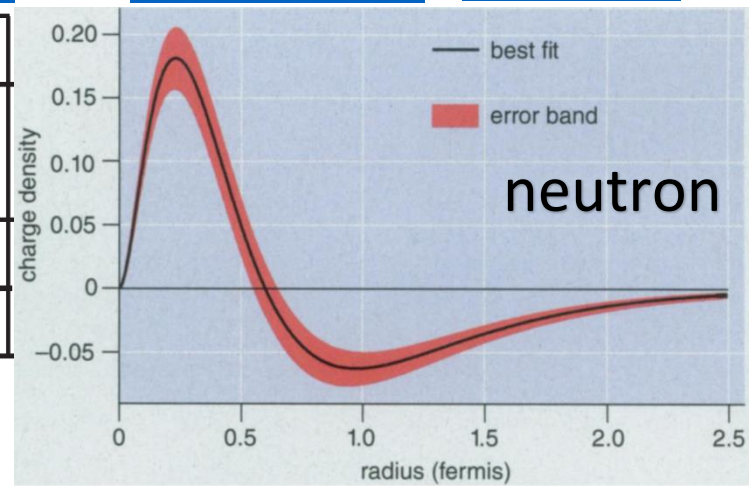
[Forest 1996](#), [Ericson](#), s+d orbitals **fitted** for seen quadrupole

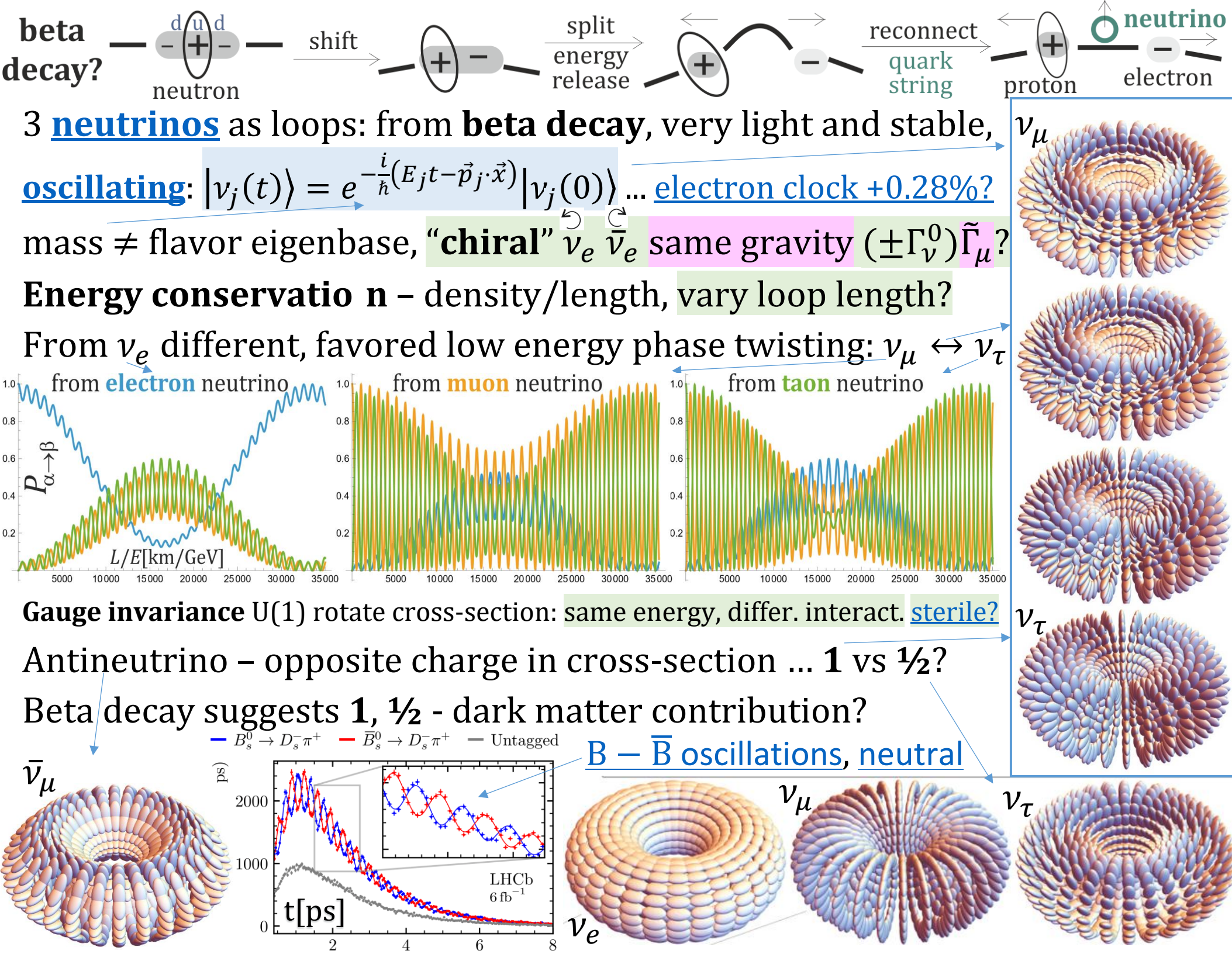


[Haddad, Suleiman 1999:](#)



[Smith 2010 \(BLAST ex.\)](#)





Basic neutrino issues (1000x larger than proton):

LSND (1993-1998) 0.8MeV p-H₂O, Cu, 30m air, 3.8 σ

MiniBooNE (2012-19): 8MeV p, 450m earth, 4.7 σ

Prod.: $p \rightarrow \pi^+ \rightarrow e^+ \bar{\nu}_\mu \nu_e$ detect: $\bar{\nu}_e + p \rightarrow e^+ + n$

Short distance π^+ excess of $\bar{\nu}_e$ in a beam of $\bar{\nu}_\mu$?

Gallium anomaly: SAGE 1998, GALLEX, BEST 2022

$^{51}\text{Cr} + e^- \rightarrow ^{51}\text{V} + \nu_e$ close source for

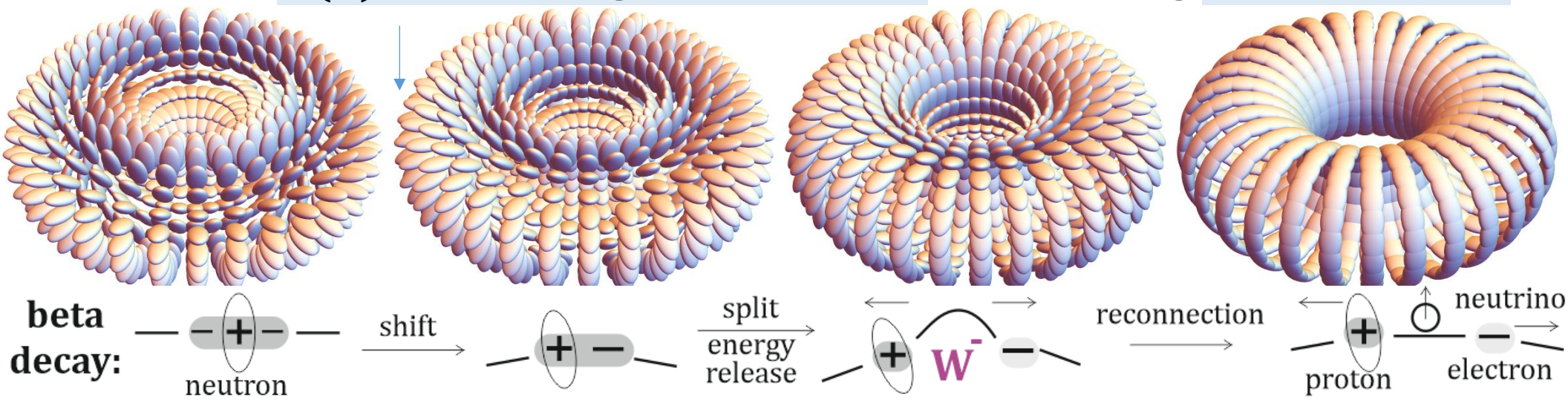
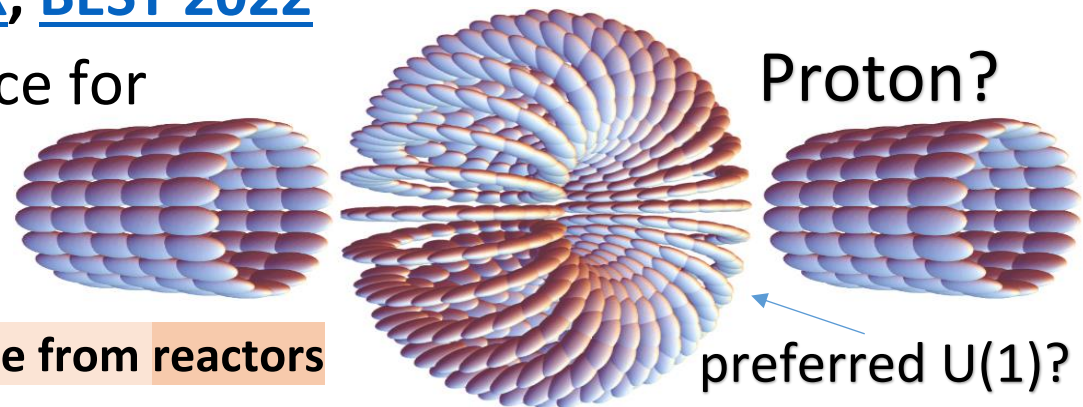
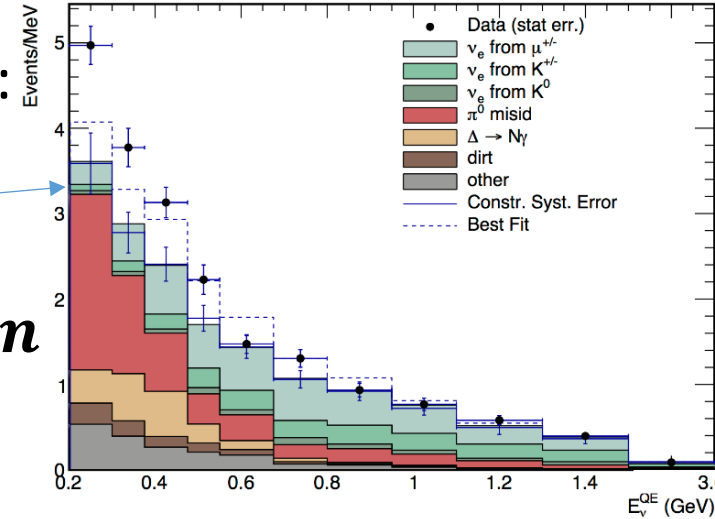
$^{71}\text{Ga} + \nu_e \rightarrow ^{71}\text{Ge} + e^-$ detector

$\nu_e \rightarrow \nu_e$ 15-20% deficiency (nucleus)

Reactor anomaly – deficit of $\bar{\nu}_e$ at short distance from reactors

Sterile neutrino? – beside flavor e.g. as axis along ... (also initial evolution?)

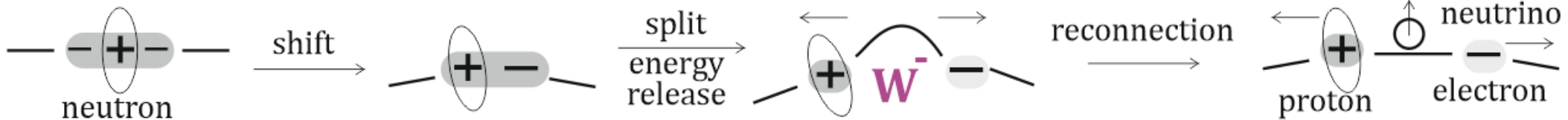
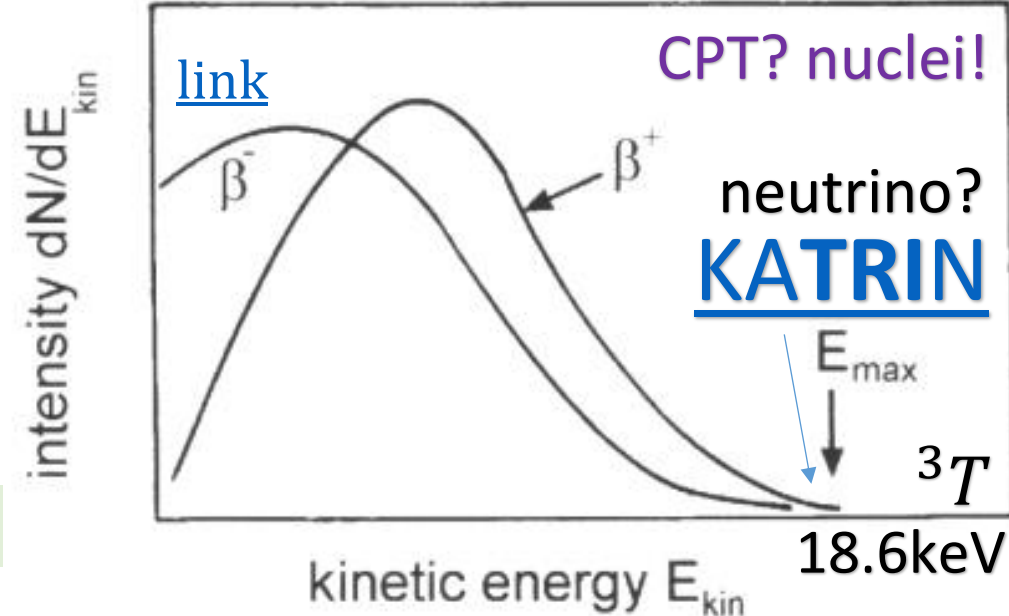
there is also $U(1)$ intrinsic degree of freedom – influencing interaction rate



β^- decay – escaping local minimum through **tunneling** – from $E = 0$

β^+ decay – escaping local minimum through **tension in nucleus** – “snaps” releasing energy as kinetic

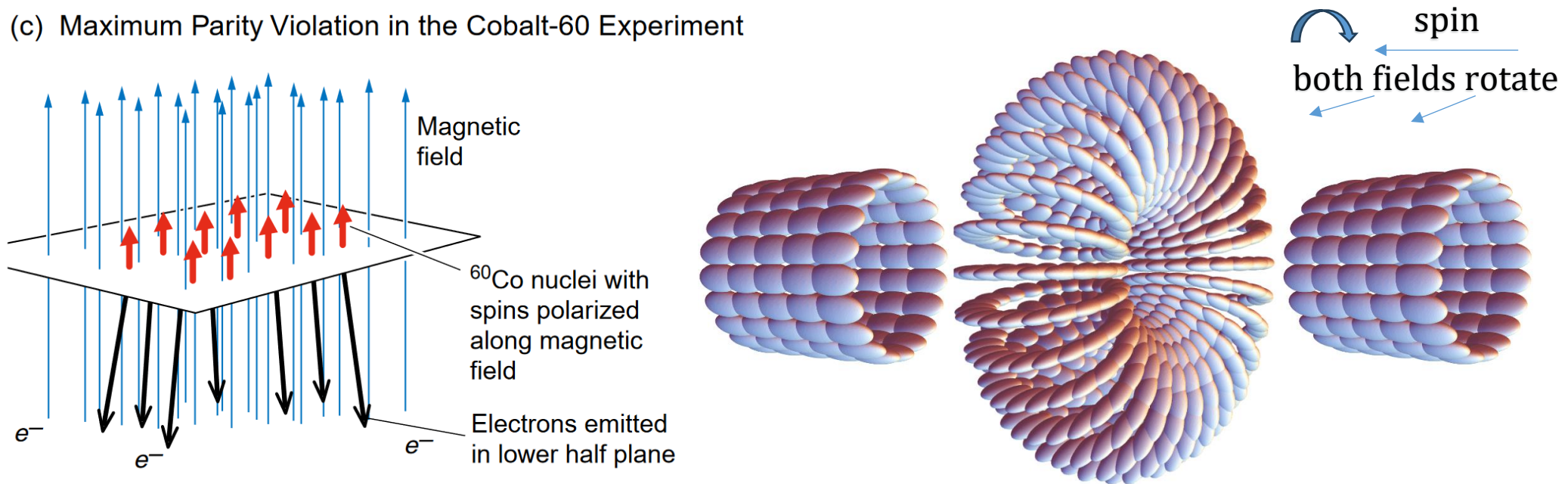
Neutrino of varying length and mass?



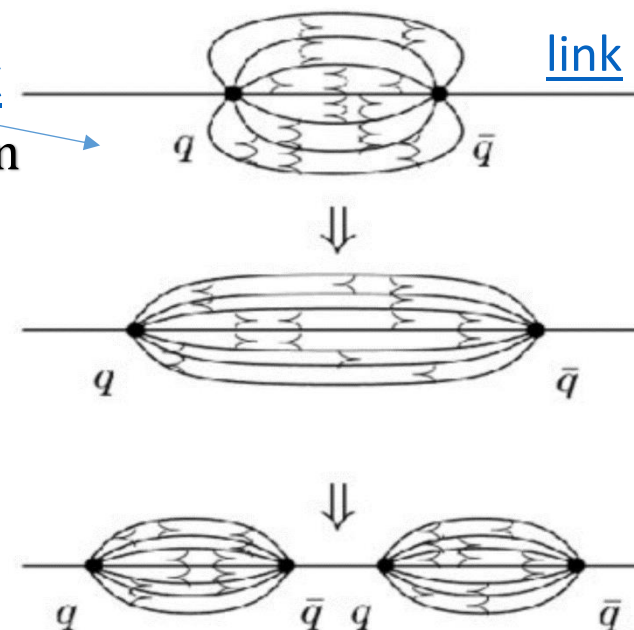
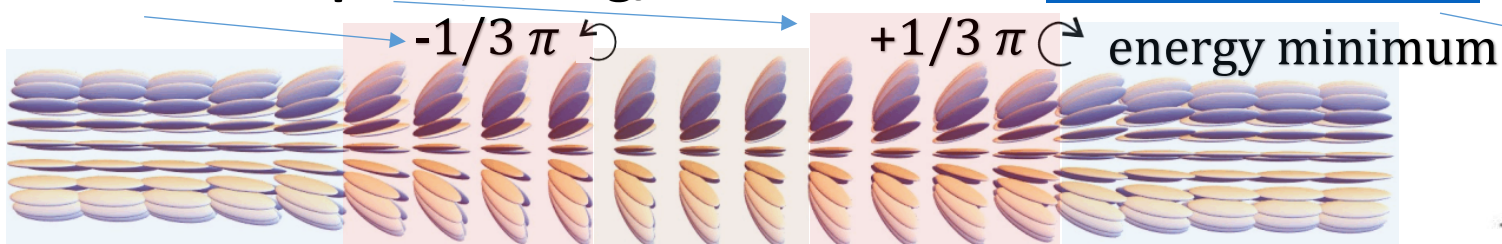
Chiral symmetry breaking – in equations (Lagrangian) or **solutions?**

[C. S. Wu experiment](#) (1956) ... because antiparticle: same $\tilde{\Gamma}$ mass, opposite Γ ?

(c) Maximum Parity Violation in the Cobalt-60 Experiment



Quark-antiquark, string/QCD flux tube: [Abrikosov vortex](#) [link](#)



Binding of [halo nuclei](#) like [Boron-8](#) with [proton](#) for 770ms

[Ne-17](#): 2 halo [protons](#) 109ms, [1p P-26](#), [2p S-27](#) 44,16ms

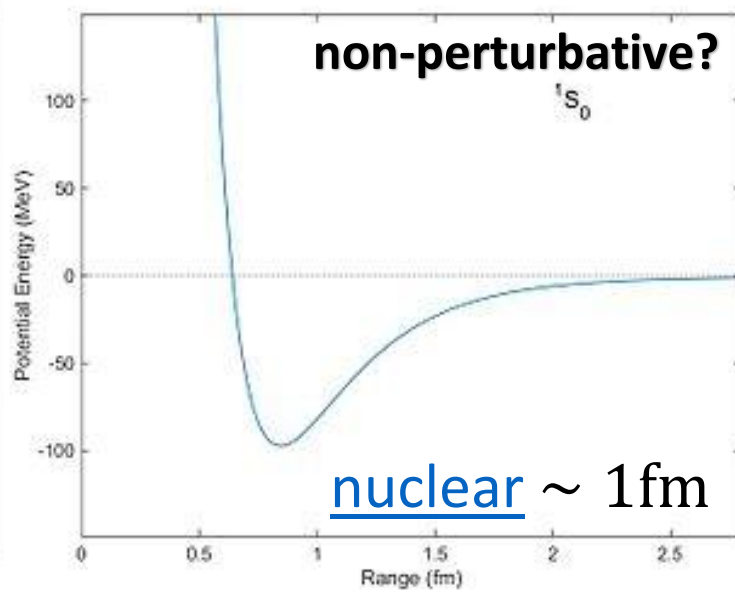
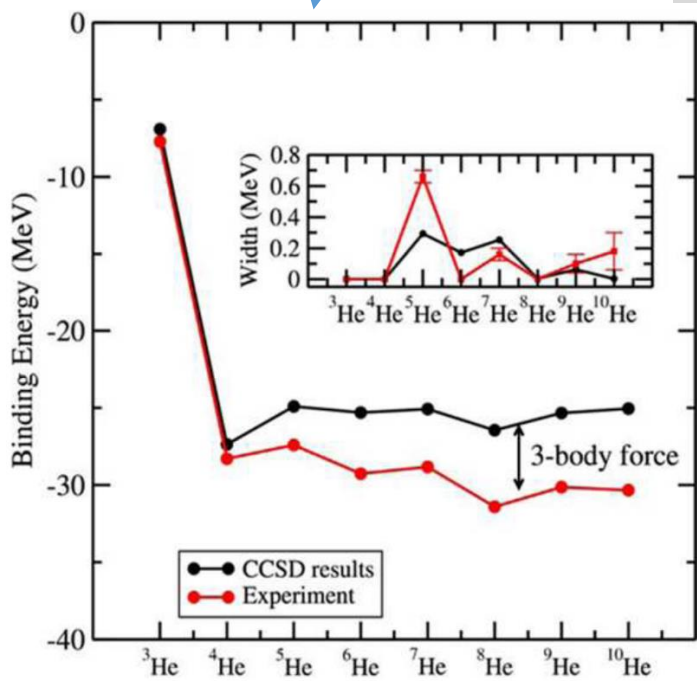
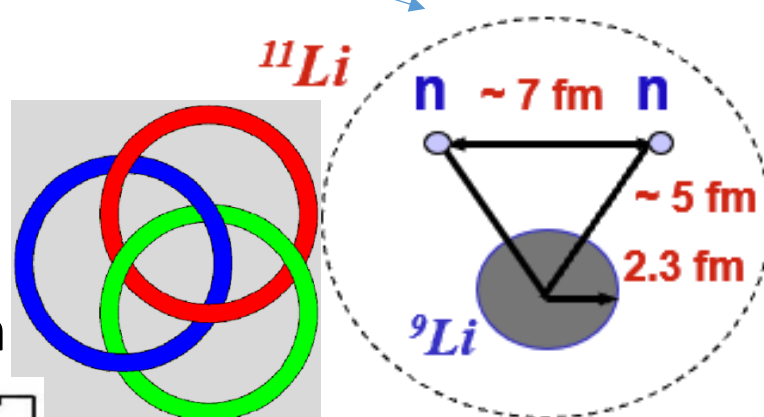
[Li-11](#): 8.6 ms lifetime in ~5 fm distance

[Efimov state](#) field conf.?

[Borromean halo nuclei](#)

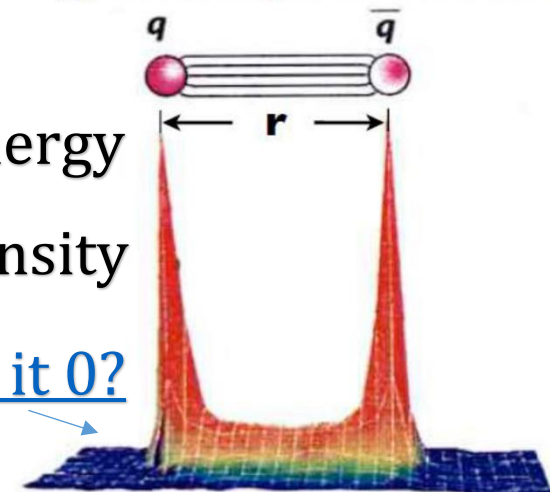
requiring [3-body forces](#)

[diagrams](#) - He6: He4+nn

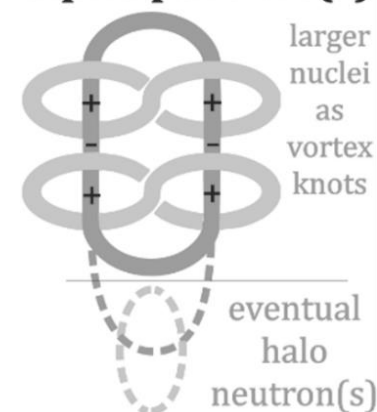


Energy density

[is it 0?](#)



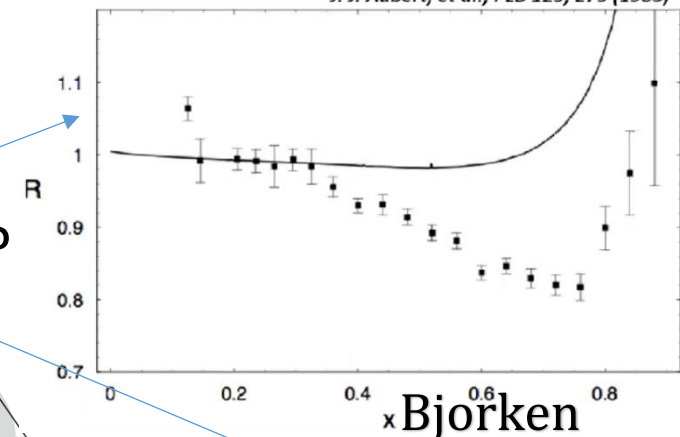
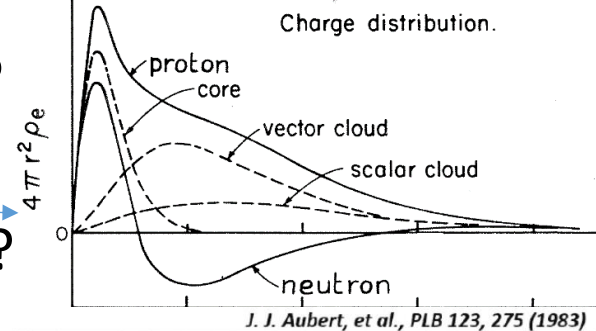
binding
larger
nuclei?



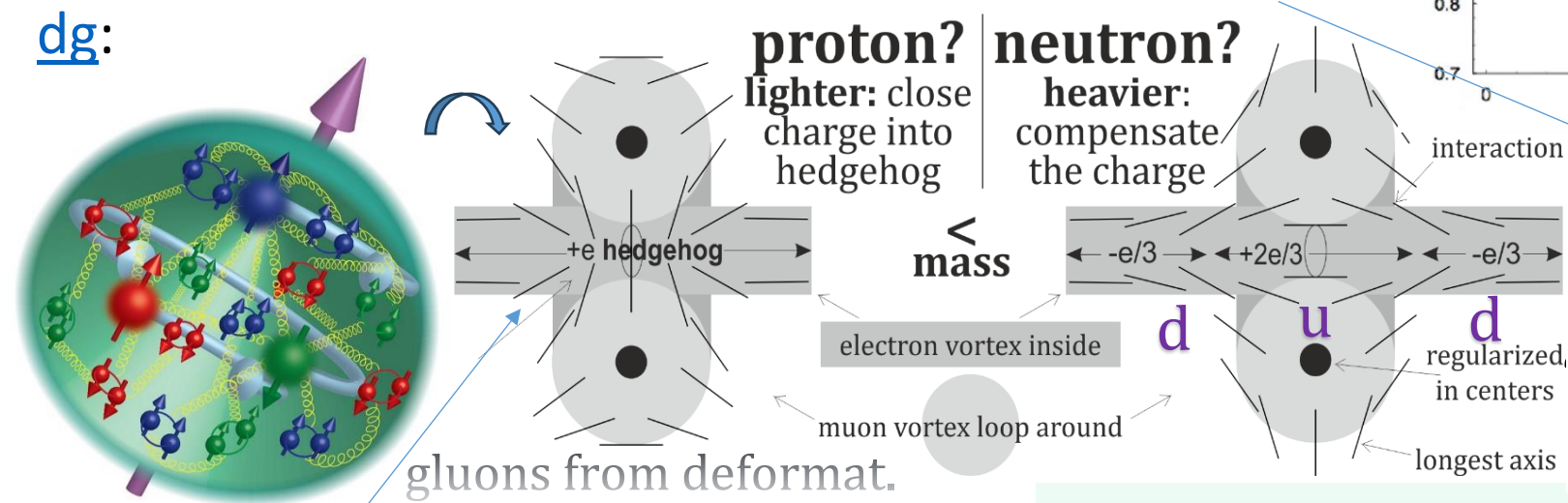
Quarks: $\sim 1\%$ of $m_p < m_n$ (?) mass. Clock like electron?

Some issues below, charge distribution more objective

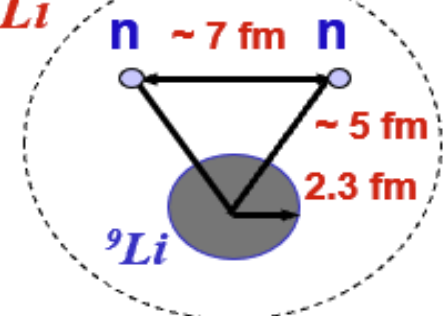
- spin crisis: $\sim 40\% + \sim 40\%$ gluons + $\sim 20\%$ angular moment.?
- proton radius puzzle: +electron: 877, +muon: 842am
- neutron lifetime puzzle 878s bottle vs 887s beam.
- EMC effect: quark distribution modified inside nuclei
- three-nucleon force? deuteron quadrupole, halo nuclei?



dg:



^{11}Li



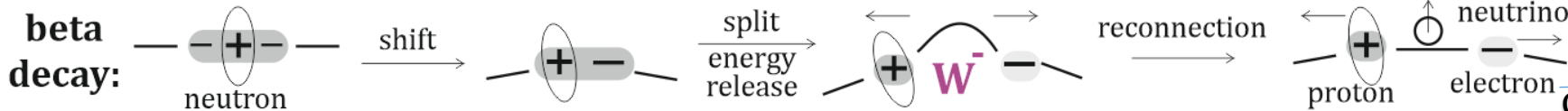
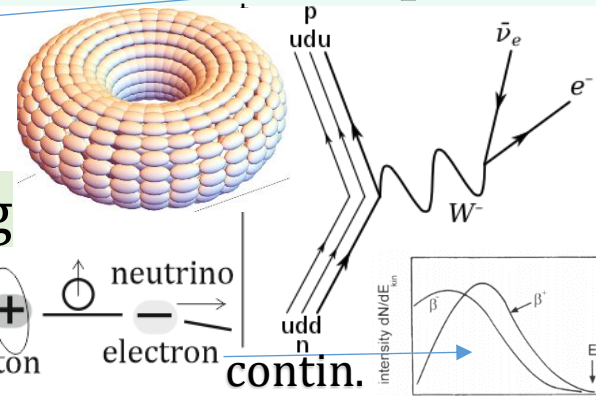
Higgs part.: deformation excitation?

$$\mathcal{L}_{\text{QCD}} = \bar{\psi}_i (i\gamma^\mu (D_\mu)_{ij} - m \delta_{ij}) \psi_j - \frac{1}{4} G_{\mu\nu}^a G_a^{\mu\nu}$$

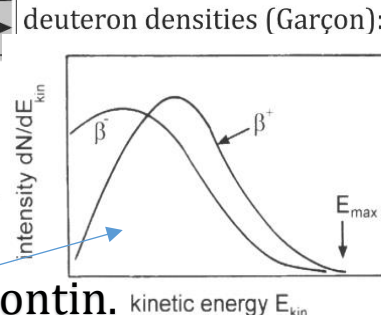
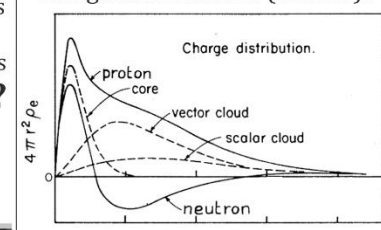
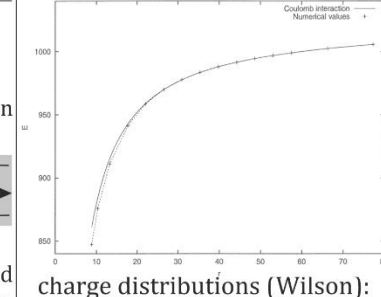
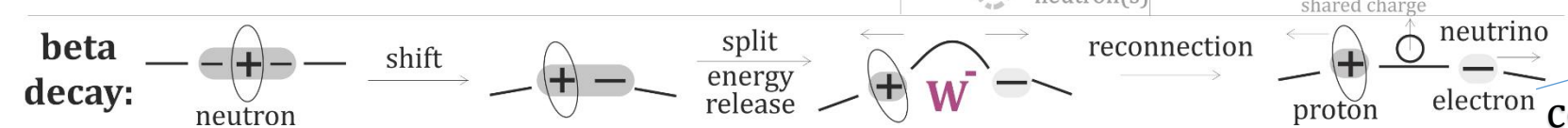
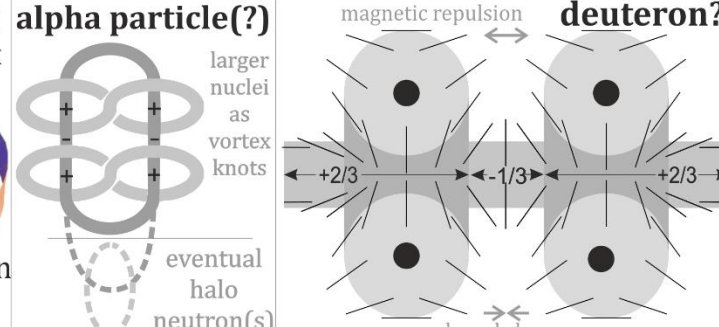
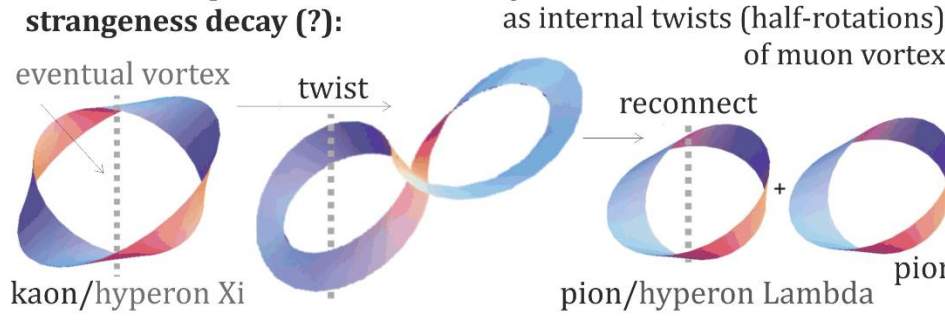
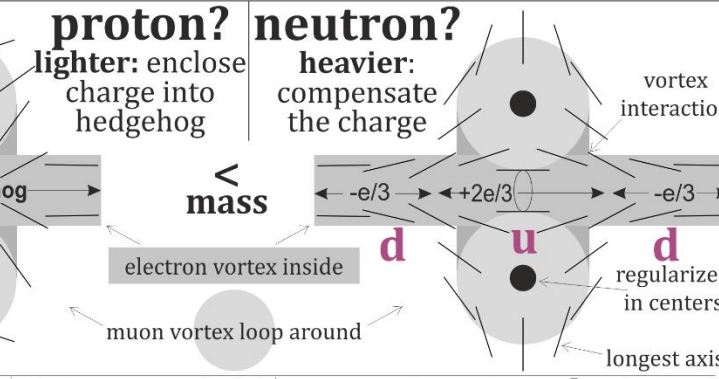
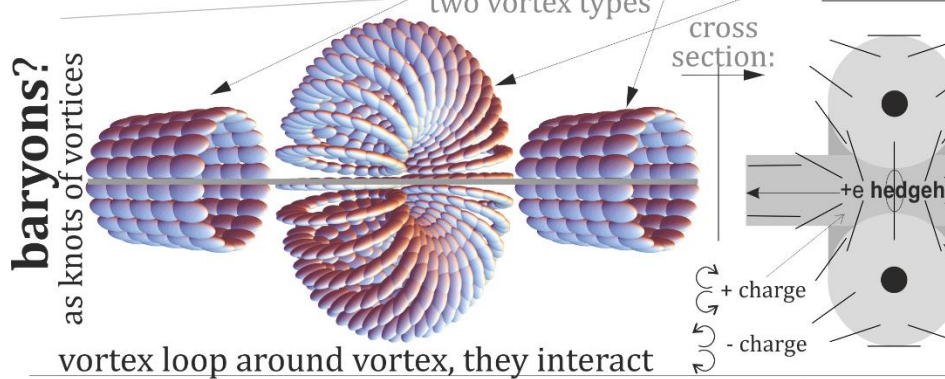
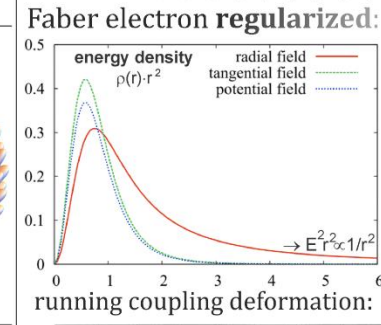
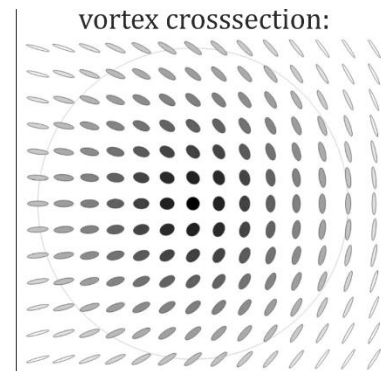
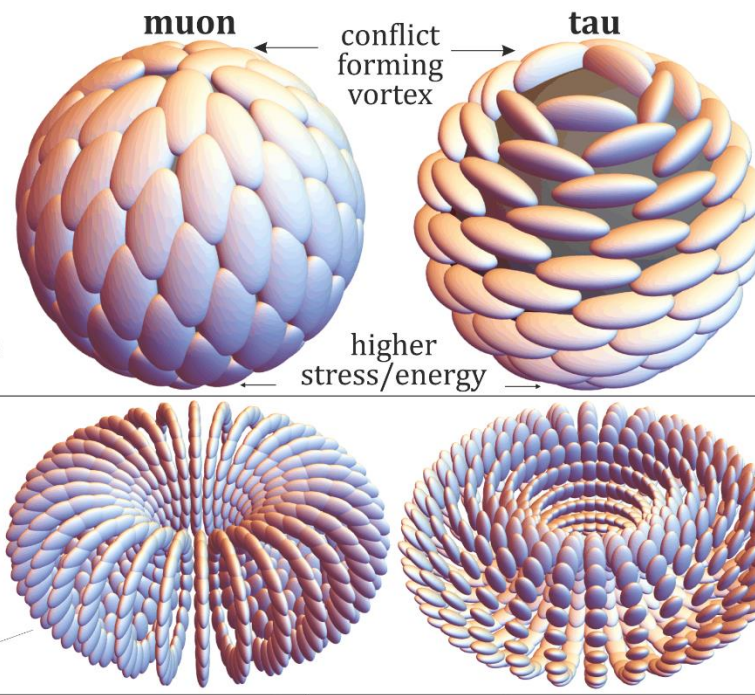
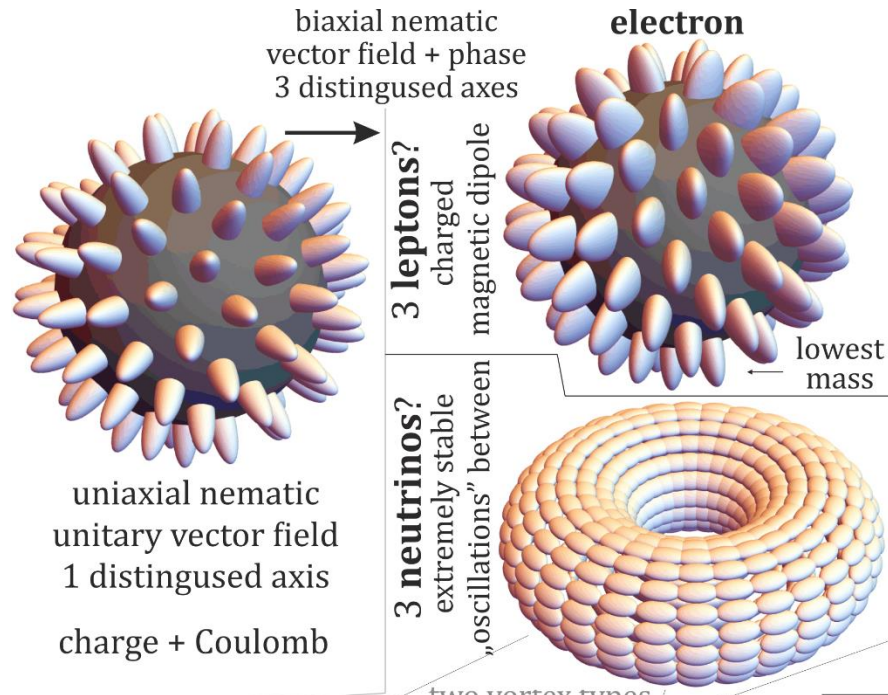
Gluons: $G = \Gamma_{\text{potential}} \times \Gamma_i$ curvatures, short distance

$W^\pm$ boson: vortex charged bubble e.g. in beta decay?

Z^0 boson: in/out rotation $e^- e^+ \rightarrow$ hadrons, quark string



gluon flux tube/color/quark string



Application(?): **stimulated** proton decay (**ultimate energy source mc^2**)

Is baryon number ultimately conserved? No Gauss law for baryon number...

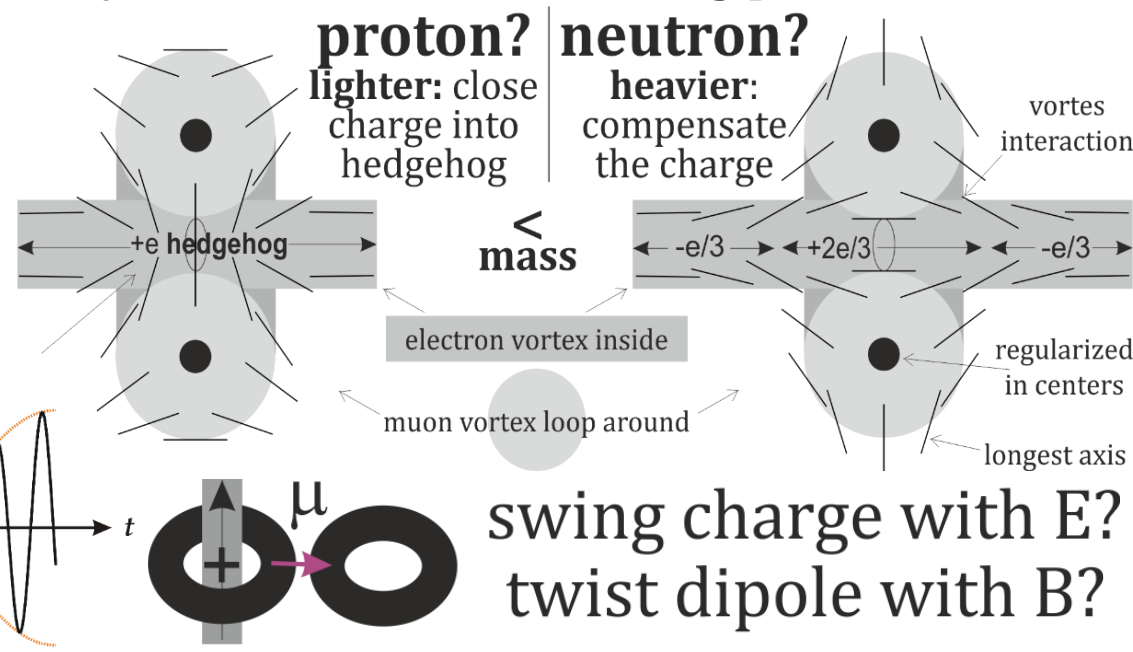
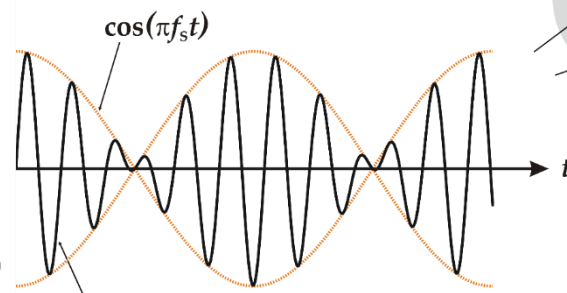
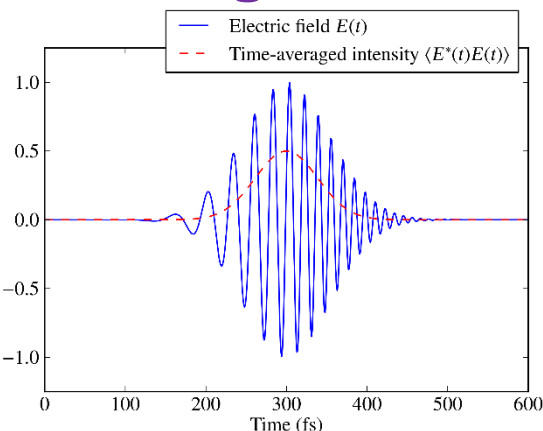
Required by baryogenesis (more matter than antimatter from Big Bang) and Hawking radiation: collapse into black hole, gather mc^2 massless evaporation

Not found in room temperature water ... maybe we should search in LHC, or centers of neutron stars: **Bizarre object 10 million times brighter than the sun defies physics, NASA says**

M82 X-2 pulsar X-ray **100-500x than allowed** by Eddington limit – where this energy comes from???

If **baryon is knot** of topological vortices, we could e.g. “swing it out to untangle”,
~1000x energy density of fission **from any matter: $mc^2 \sim 0.02\text{g/human lifetime}$**

E.g. using **EM wave of extreme frequency** ... toward fast incoming protons?
beats effect, ultrashort laser pulse
of **numerically optimized shape**
to swing out to untangle baryon



Standard liquid crystal models ([review1](#), [review2](#))

Uniaxial: $\vec{n} \in S^2$ (Higgs potential?), Oseen-Frank model (1958) :

$$E[\vec{n}] = \int_{\Omega} K_1 |\nabla \cdot \vec{n}|^2 + K_2 |\vec{n} \cdot (\nabla \times \vec{n})|^2 + K_3 |\vec{n} \times (\nabla \times \vec{n})|^2 + (|\vec{n}|^2 - 1)^2$$

Biaxial: Landau-de Gennes (1974, Nobel Prize in 1991) of preferred shape

($Q \equiv$) $M = \sum_{i=1}^3 \lambda_i \vec{n}_i \vec{n}_i^T$ preferred $(\lambda_1, \lambda_2, \lambda_3)$: $V(M)$ Higgs-like potential

$$E[M] = \int_{\Omega} |\nabla M|^2 + \frac{A}{2} \text{Tr}(M^2) - \frac{B}{3} \text{Tr}(M^3) + \frac{C}{4} (\text{Tr}(M^2))^2$$

(or $V(M) = \sum_i (\lambda_i - \Lambda_i)^2$ or $\sum_i (\text{Tr}(M^i) - c_i)^2$ tough choice!)

→ particles?: Higgs-like potential, e.g. as above, with **Coulomb, EM-like**:

$$\mathcal{L}_{EM} = -F_{\mu\nu} F^{\mu\nu} \quad \text{for} \quad F_{\mu\nu} \propto R_{\mu\nu} = \Gamma_{\mu} \times \Gamma_{\nu} \quad \Gamma_{\nu} = (\partial_{\nu} \vec{n}) \times \vec{n}$$

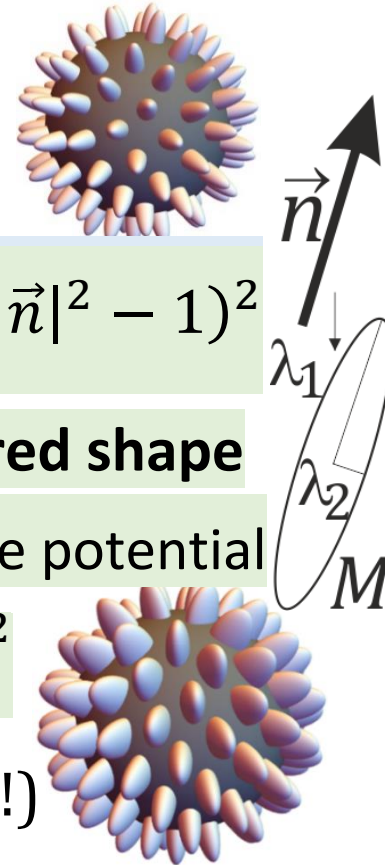
$R_{\mu\nu}$ curvature: Gauss law counts (quantized) topological charge

with kinetic: $\mathcal{L} \sim \sum_{i=1}^3 ||\partial_i \vec{n} \times \partial_0 \vec{n}||^2 - \sum_{1 \leq i < j \leq 3} ||\partial_i \vec{n} \times \partial_j \vec{n}||^2 - V(\vec{n})$

$$\text{Biaxial: } F_{\mu\nu\alpha\beta} = [\partial_{\mu} M, \partial_{\nu} M]_{\alpha\beta} \quad \mathcal{L} = -F_{\mu\nu\alpha\beta} F^{\mu\nu\alpha\beta} - V(M)$$

Lorentz-invariant Lagrangian mechanics leading to **electromagnetism**
+ ~Klein-Gordon for twists + **gravito-electromagnetism** for 0th axis boosts

Einstein-Hilbert: $\mathcal{L} \sim R \sqrt{-g}$ spacetime curvature, here field curvature ... Newton?



Intermediate step: transform unitary “**director**” field → **rotation** matrix field O

Affine **connection**: $O \rightarrow O(I + \epsilon \Gamma_\mu)$ for $\Gamma_\mu = O^T \mathbf{O}_\mu = O^T \partial_\mu O$ anti-symmetric

Denote its coordinates with 2 vectors:

EM: $\vec{\Gamma}_\mu := (\Gamma_{\mu,32}, \Gamma_{\mu,13}, \Gamma_{\mu,21})$

GEM: $\vec{\Gamma}_\mu^g := (\Gamma_{\mu,01}, \Gamma_{\mu,02}, \Gamma_{\mu,03})$ tiny boosts of 0th axis

(boosts +) $\Gamma_\mu = O^T O_\mu = \begin{pmatrix} 0 & \vec{\Gamma}_{\mu,1}^g & \vec{\Gamma}_{\mu,2}^g & \vec{\Gamma}_{\mu,3}^g \\ -\vec{\Gamma}_{\mu,1}^g & 0 & -\vec{\Gamma}_{\mu,3} & \vec{\Gamma}_{\mu,2} \\ -\vec{\Gamma}_{\mu,2}^g & \vec{\Gamma}_{\mu,3} & 0 & -\vec{\Gamma}_{\mu,1} \\ -\vec{\Gamma}_{\mu,3}^g & -\vec{\Gamma}_{\mu,2} & \vec{\Gamma}_{\mu,1} & 0 \end{pmatrix}$

3D 1st axis **curvature**: $\vec{R}_{\mu\nu} \equiv \vec{R}_{\mu\nu}^{ee} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu$ - **electromagnetism** we will focus on

4D 0th axis curvature: $\vec{R}_{\mu\nu}^{gg} = \vec{\Gamma}_\mu^g \times \vec{\Gamma}_\nu^g$ - **GEM** approximation of general relativity

4D EM-GEM interaction: $\vec{R}_{\mu\nu}^{eg} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu^g, \vec{R}_{\mu\nu}^{ge} = \vec{\Gamma}_\mu^g \times \vec{\Gamma}_\nu = -\vec{R}_{\nu\mu}^{eg}$ e.g. **GEM slowing EM**: time dilation Fermat principle: Sun's light bend

$F_{\mu\nu} = [\Gamma_\mu, \Gamma_\nu]$ in place of curvature for EM with topologically quantized charge?

$$[\Gamma_\mu, \Gamma_\nu] = \begin{pmatrix} 0 & -\vec{R}_{\mu\nu}^{eg} + \vec{R}_{\nu\mu}^{eg} \\ \vec{R}_{\mu\nu}^{eg} - \vec{R}_{\nu\mu}^{eg} & \vec{R}_{\mu\nu}^{ee} + \vec{R}_{\mu\nu}^{gg} \end{pmatrix} := \begin{pmatrix} 0 & -\vec{R}_{\mu\nu,1}^{eg} + \vec{R}_{\nu\mu,1}^{eg} & -\vec{R}_{\mu\nu,2}^{eg} + \vec{R}_{\nu\mu,2}^{eg} & -\vec{R}_{\mu\nu,3}^{eg} + \vec{R}_{\nu\mu,3}^{eg} \\ \vec{R}_{\mu\nu,1}^{eg} - \vec{R}_{\nu\mu,1}^{eg} & 0 & -\vec{R}_{\mu\nu,3}^{ee} - \vec{R}_{\mu\nu,3}^{gg} & \vec{R}_{\mu\nu,2}^{ee} + \vec{R}_{\mu\nu,2}^{gg} \\ \vec{R}_{\mu\nu,2}^{eg} - \vec{R}_{\nu\mu,2}^{eg} & \vec{R}_{\mu\nu,3}^{ee} + \vec{R}_{\mu\nu,3}^{gg} & 0 & -\vec{R}_{\mu\nu,1}^{ee} - \vec{R}_{\mu\nu,1}^{gg} \\ \vec{R}_{\mu\nu,3}^{eg} - \vec{R}_{\nu\mu,3}^{eg} & -\vec{R}_{\mu\nu,2}^{ee} - \vec{R}_{\mu\nu,2}^{gg} & \vec{R}_{\mu\nu,1}^{ee} + \vec{R}_{\mu\nu,1}^{gg} & 0 \end{pmatrix}$$

GEM – confirmed by Gravity Probe B
approximation of **general relativity**:

<https://en.wikipedia.org/wiki/Gravitoelectromagnetism>:

$[\Gamma_\mu, \Gamma_\nu]$ vanishes in flat spacetime

GEM causes spatial curvature:

$$0 = \partial_\mu \partial_\nu O - \partial_\nu \partial_\mu O = \partial_\mu (O \Gamma_\nu) - \partial_\nu (O \Gamma_\mu) = O[\Gamma_\mu, \Gamma_\nu]$$

GEM equations	Maxwell's equations
$\nabla \cdot \mathbf{E}_g = -4\pi G \rho_g$	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$
$\nabla \cdot \mathbf{B}_g = 0$	$\nabla \cdot \mathbf{B} = 0$
$\nabla \times \mathbf{E}_g = -\frac{\partial \mathbf{B}_g}{\partial t}$	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$
$\nabla \times \mathbf{B}_g = -\frac{4\pi G}{c^2} \mathbf{J}_g + \frac{1}{c^2} \frac{\partial \mathbf{E}_g}{\partial t}$	$\nabla \times \mathbf{B} = \frac{1}{\epsilon_0 c^2} \mathbf{J} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t}$

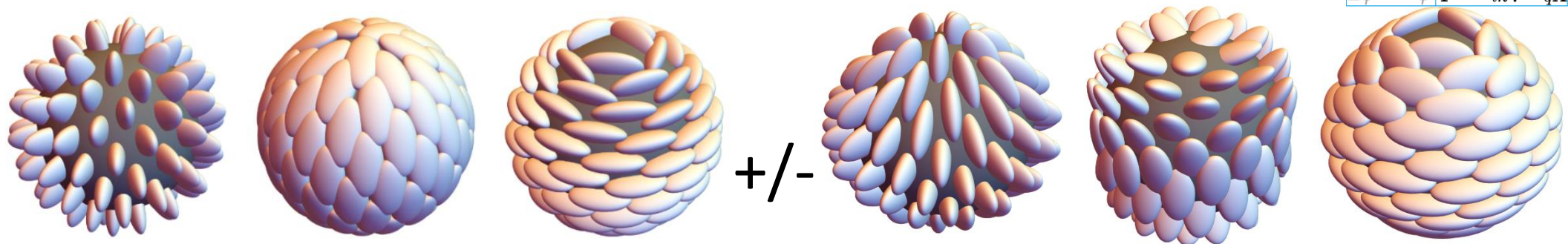
Let O matrix **rotate some objects**, e.g. **ellipsoid** in biaxial nematic

$M(x) \equiv \mathbf{M} = \mathbf{O} \mathbf{D} \mathbf{O}^T$ field for $\mathbf{O} \mathbf{O}^T = I$ rotation

$D = \text{diag}(\lambda_0, \lambda_1, \lambda_2, \lambda_3)$ with **Higgs-like** e.g. $V(\mathbf{M}) = \sum_i (\lambda_i - \Lambda_i)^2$

For $\Lambda_0 > \Lambda_1 > \Lambda_2 > \Lambda_3$ fixed preferred shape – **vacuum state**

E, B Ahar. Bohm
 $\partial_\mu A_\nu - \partial_\nu A_\mu$
 Maxwell: $\square A_\mu \propto J_\mu$ A gauge?
 extended quantum phase for topological charge quantization
 $A_\mu^* = [M, M_\mu]$ EM
 $\cong$ affine connection $Q^M + EM$
 $F^* \cong$ curvature
 ψ $M = \mathbf{O} \mathbf{D} \mathbf{O}^T$
 $\square \psi \propto -\psi$ $\hat{P} = -i\hbar \nabla - qA$



(3D) F tensor containing curvature (so Gauss law gives topological charge)

Let us postulate: $F_{\mu\nu} = [M_\mu, M_\nu] = \partial_\mu M_\nu - \partial_\nu M_\mu$, getting **vacuum**:

Curvatures:

Gauss law $\propto$
 topological charge

$\Lambda_1 \gg \Lambda_2 \approx \Lambda_3$

High energy: **EM**

Low energy: **QM**

[QED](#), ψ [Lorentz group](#)

$$O^T F_{\mu\nu} O = O^T [M_\mu, M_\nu] O \approx [\Gamma_\mu D - D \Gamma_\mu, \Gamma_\nu D - D \Gamma_\nu] =$$

$$(\Lambda_1 - \Lambda_2)(\Lambda_3 - \Lambda_1)(\Lambda_2 - \Lambda_3) \begin{pmatrix} 0 & \frac{-\vec{R}_{\mu\nu,3}}{\Lambda_1 - \Lambda_2} & \frac{\vec{R}_{\mu\nu,2}}{\Lambda_3 - \Lambda_1} \\ \frac{\vec{R}_{\mu\nu,3}}{\Lambda_1 - \Lambda_2} & 0 & \frac{-\vec{R}_{\mu\nu,1}}{\Lambda_2 - \Lambda_3} \\ \frac{-\vec{R}_{\mu\nu,2}}{\Lambda_3 - \Lambda_1} & \frac{\vec{R}_{\mu\nu,1}}{\Lambda_2 - \Lambda_3} & 0 \end{pmatrix}$$

$$\mathcal{L}_{QED} = \bar{\psi}(i\gamma^\mu \partial_\mu - e\gamma^\mu A_\mu - m)\psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

for $\vec{R}_{\mu\nu} = \vec{\Gamma}_\mu \times \vec{\Gamma}_\nu$ and $\Gamma_\mu = \mathbf{O}^T \mathbf{O}_\mu$, $\vec{\Gamma}_\mu := (\Gamma_{\mu,32}, \Gamma_{\mu,13}, \Gamma_{\mu,21})$ as previously

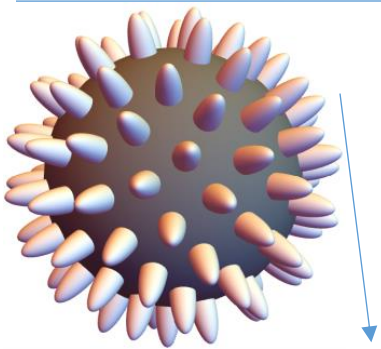
Postulate **Lagrangian** as EM:
and **four-potential** A_μ :

$$\mathcal{L} = \sum_{\mu=1}^3 \|F_{\mu 0}\|_F^2 - \sum_{1 \leq \mu < \nu \leq 3} \|F_{\mu\nu}\|_F^2 - V$$

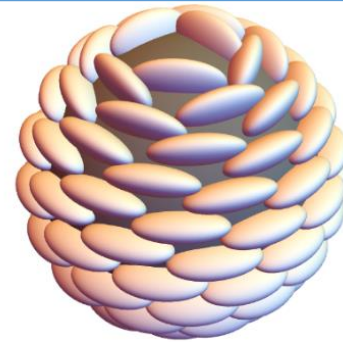
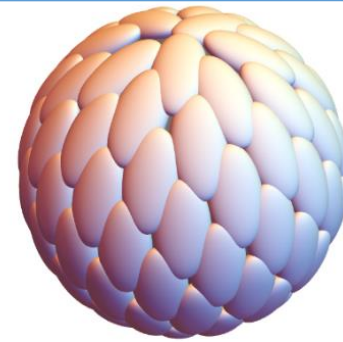
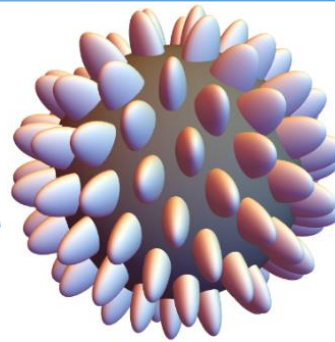
$$2F_{\mu\nu} = 2[\mathbf{M}_\mu, \mathbf{M}_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \text{for} \quad A_\mu = M M_\mu - M_\mu M \approx \quad (\text{3D vacuum})$$

$\Lambda_1 \gg \Lambda_2 \approx \Lambda_3$
High energy: **EM**
Low energy: **QM**
 $P = -i\hbar\nabla - qA$

$$\approx O \begin{pmatrix} 0 & \vec{\Gamma}_{\mu,3}(\Lambda_1 - \Lambda_2)^2 & -\vec{\Gamma}_{\mu,2}(\Lambda_1 - \Lambda_3)^2 \\ -\vec{\Gamma}_{\mu,3}(\Lambda_1 - \Lambda_2)^2 & 0 & \vec{\Gamma}_{\mu,1}(\Lambda_2 - \Lambda_3)^2 \\ \vec{\Gamma}_{\mu,2}(\Lambda_1 - \Lambda_3)^2 & -\vec{\Gamma}_{\mu,1}(\Lambda_2 - \Lambda_3)^2 & 0 \end{pmatrix} O^T$$



shape Λ
dependence
perturbation $\Lambda_2 > 0$



In **uniaxial nematic case** e.g. simplest ($\Lambda_1 = 1, \Lambda_2 = 0, \Lambda_3 = 0$): $\mathbf{M} = \vec{n} \vec{n}^T$

$$A_\mu = [M, M_\mu] = \|\vec{n}\|^2 \begin{pmatrix} 0 & (\vec{n} \times \vec{n}_\mu)_3 & -(\vec{n} \times \vec{n}_\mu)_2 \\ -(\vec{n} \times \vec{n}_\mu)_3 & 0 & (\vec{n} \times \vec{n}_\mu)_1 \\ (\vec{n} \times \vec{n}_\mu)_2 & -(\vec{n} \times \vec{n}_\mu)_3 & 0 \end{pmatrix}$$

leading to EM $F_{\mu\nu}$ curvature as Faber: $\partial_\mu(\vec{n} \times \vec{n}_\nu) - \partial_\nu(\vec{n} \times \vec{n}_\mu) = 2 \vec{n}_\mu \times \vec{n}_\nu$

General: small perturbation $\Lambda_2 > 0$, **shape Λ dependence** in $\Gamma_\mu \xrightarrow{\Lambda} A_\mu, R_{\mu\nu} \xrightarrow{\Lambda} F_{\mu\nu}$

$$\frac{\partial \|F_{\mu\nu}\|_F^2}{\partial(\partial_\alpha A_\beta)} = \frac{1}{2} \frac{\partial \|\partial_\mu A_\nu - \partial_\nu A_\mu\|_F^2}{\partial(\partial_\alpha A_\beta)} = (\delta_{\mu\alpha}\delta_{\nu\beta} + \delta_{\mu\beta}\delta_{\nu\alpha})F_{\alpha\beta}$$

Euler-Lagrange equations: extended EM with topological charge quantization

(integration by parts - last term should vanish as in [Lorentz gauge condition](#)):

[Maxwell's equations:](#)

F : E, B fields

EM: $V \sim AJ$

+ [Klein-Gordon](#):

$V \sim m^2 A^2$

+ [GEM](#): $V \sim AJ^g$

J, J^g : four-currents

$$\frac{\partial \mathcal{L}}{\partial A_\alpha} = \frac{d}{dx_0} \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\alpha)} + \sum_{i=1}^3 \frac{d}{dx_i} \frac{\partial \mathcal{L}}{\partial(\partial_i A_\alpha)}$$

$$\frac{\partial V}{\partial A_\alpha} = \partial_0 F_{0\alpha} - \sum_{i=1}^3 \partial_i F_{i\alpha} = \square A_\alpha - \partial_\alpha \left(\partial_0 A_0 - \sum_i \partial_i A_i \right)$$

All 3x3 or 4x4 anti-symmetric matrices, $\square = \partial_{00} - \partial_{11} - \partial_{22} - \partial_{33}$

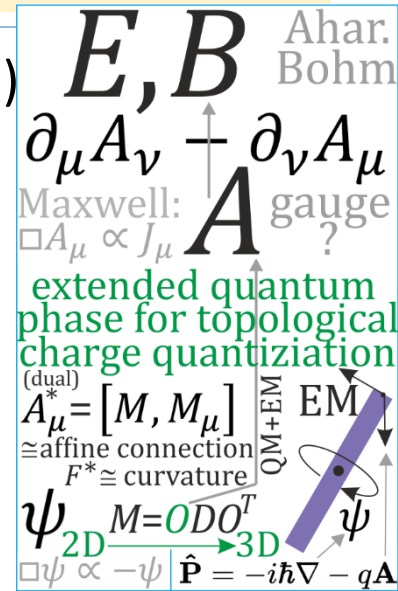
How to choose $V(M)$ or $V(A)$ $\sum_i (\lambda_i - \Lambda_i)^2$? $\sum_\mu (\|A_\mu\|_F^2 - 1)^2$? $\det M = 1$?

$A_\mu = [M, \partial_\mu M] \approx (\epsilon_{ijk}(\Lambda_i - \Lambda_j) \vec{\Gamma}_{\mu,k})_{ij}$ contains velocity – **Higgs $V(A)$ would enforce M clock**

Energy density/Hamiltonian ($\|A\|_F^2 = \text{Tr}(AA^T)$, last term should vanish)

$$\mathcal{H} = \sum_{\mu=1}^3 \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)} \partial_0 A_\mu - \mathcal{L} = \sum_{\mu=1}^3 F_{0\mu} \bullet (2F_{0\mu} + \partial_\mu A_0) - \mathcal{L}$$

$$\mathcal{H} = \sum_{0 \leq \mu < \nu \leq 3} \|F_{\mu\nu}\|_F^2 + V + \sum_{\mu=1}^3 F_{0\mu} \bullet \partial_\mu A_0$$



Model: field $M(t, x, y, z) \equiv \mathbf{M} = \mathbf{O} \mathbf{D} \mathbf{O}^T$ of real symmetric 3x3 matrices, $(V \approx 0)$ Assume **vacuum case:** $(\lambda_i) \approx (\Lambda_i) = (1, \delta, 0)$ for tiny δ related with Planck $\hbar$
 $OO^T = I$ describes **local rotation**, $D = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$ **shape** of "molecule" (to be extended to 4x4 tensor field – adding gravitoelectromagnetism)
 $(\lambda_0, \lambda_1, \lambda_2, \lambda_3)$ is **shape** preferring (Λ_i) shape fixed by model, e.g. $V = \sum_i (\lambda_i - \Lambda_i)^2$
 With **Lagrangian:** $\mathcal{L} = \sum_{\mu=1}^3 ||F_{\mu 0}||_F^2 - \sum_{1 \leq \mu < \nu \leq 3} ||F_{\mu \nu}||_F^2 - V$
 for $F_{\mu \nu} = [\partial_\mu M, \partial_\nu M] = \partial_\mu A_\nu - \partial_\nu A_\mu$ $A_\mu = [M, \partial_\mu M]$
 zeroing 3x3 M variations: **3 rotations**, and 3 axis elongations – zeroing the lowest δ :
twist: $\sim$ Klein-Gordon $X^2 \cdot \Gamma^3 = X^3 \cdot \Gamma^2$
tilt1: $X^1 \cdot \Gamma^3 = 0$
tilt2: $X^1 \cdot \Gamma^2 = 0$
 all 3 **elongations:** $|B^1| = |E^1|$ (dominated in 4D)
 for $X^i := (-\nabla \cdot B^i, \partial_0 B^i + \nabla \times E^i)$ as in **Maxwell equations**
 $\Gamma_\mu = O^T O_\mu$
 $\vec{R}_\mu = (\Gamma_{\mu,32}, \Gamma_{\mu,13}, \Gamma_{\mu,21})$
 $\vec{R}_{\mu\nu} = \vec{R}_\mu \times \vec{R}_\nu$
 $\vec{B}_i = \vec{R}_{0i}$
 $\vec{E}_{1,2,3} = (\vec{R}_{32}, \vec{R}_{13}, \vec{R}_{23})$

Variating Lagrangian in vacuum ($V = 0$): **Euler-Lagrange** evolution equation:

$$0 = \sum_{\mu\nu} d_{\mu\nu} \text{Tr} (\bar{F}_{\mu\nu} ([\Gamma_\nu, [\bar{M}_\mu, G']] - [\Gamma_\mu, [\bar{M}_\nu, G']]) + \bar{F}_{\mu\nu, \nu} [\bar{M}_\mu, G'] - \bar{F}_{\mu\nu, \mu} [\bar{M}_\nu, G'])$$

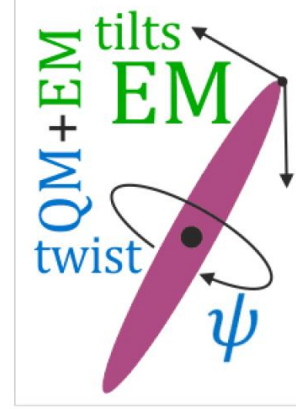
Which for **3 rotation** generators give
 $\sim$ Maxwell for 2 tilts (high energy) as:
 $\sim$ Klein-Gordon for twist (low energy) as:
 for hedgehog:

$$X^i := (-\nabla \cdot B^i, \partial_0 B^i + \nabla \times E^i)$$

$$X^1 \cdot \Gamma^3 = 0 = X^1 \cdot \Gamma^2$$

$$X^2 \cdot \Gamma^3 = X^3 \cdot \Gamma^2$$

$$2\partial_{tt}\psi = \left((\nabla - \vec{A}^{hedg})^2 + \left(\frac{\vec{A}^{hedg}}{|\vec{A}^{hedg}|} \cdot \nabla \right)^2 \right) \psi$$



```
d = DiagonalMatrix[{1, δ, 0}]; (* ellipsoid shape, δ ~ ħ *)
Gx = {{0, 0, 0}, {0, 0, -1}, {0, 1, 0}}; (* twist generator *)
Gy = {{0, 0, 1}, {0, 0, 0}, {-1, 0, 0}}; (* tilt1 generator *)
Gz = {{0, -1, 0}, {1, 0, 0}, {0, 0, 0}}; (* tilt2 generator *)
Ga = {{1, 0, 0}, {0, 0, 0}, {0, 0, 0}}; (* 3 elongation generators *)
Gb = {{0, 0, 0}, {0, 1, 0}, {0, 0, 0}};
Gc = {{0, 0, 0}, {0, 0, 0}, {0, 0, 1}}; (* Gpt of G' = Gd + dG^T *)
Gpt = Join[Table[G.d + d.Transpose[G], {G, {Gx, Gy, Gz}}], {Ga, Gb, Gc}];
com[A_, B_] := A.B - B.A; (* commutator *)
cd = {{3, 2}, {1, 3}, {2, 1}}; (* εijk *)
vect[m_] := Table[m[[cd[[i, 1]], cd[[i, 2]]], {i, 3}]; (* → rotation vector *)
Γμ = {{0, -Γμ3, Γμ2}, {Γμ3, 0, -Γμ1}, {-Γμ2, Γμ1, 0}}; (* its matrix form *)
sub = Table[Cross[{Γμ1, Γμ2, Γμ3}, {Γν1, Γν2, Γν3}] [[i]] == R{μ, ν}i, {i, 3}];
Mμ = com[Γμ, d]; Γν = Γμ /. μ → ν; Mν = com[Γν, d]; Fμν = Simplify[com[Mμ, Mν], sub];
vrip = Table[Simplify[Tr[Fμν.(com[Γν, com[Mμ, Gp]] - com[Γμ, com[Mν, Gp]]) +
(Fμν /. Table[R{μ, ν}i → R{μ, ν, ν}i, {i, 3}]).com[Mμ, Gp] - (*integrate by parts*)
(Fμν /. Table[R{μ, ν}i → R{μ, ν, μ}i, {i, 3}]).com[Mν, Gp]], {Gp, Gpt}];
vr = Simplify[Series[vrip / {2 δ^2, 2, 2, -4, 2, 2}, {δ, 0, 0}] // Normal, sub]
```

```
fin = Table[Sum[v /. μ → 0, {v, 1, 3}] - Sum[v /. {μ → cd[[i, 1]], v → cd[[i, 2]]},
{i, 3}], {v, vr}]; (* Lagrangian = Σμν ±||Fμν||^2 *)
sub1 = (* rename R curvatures as BE fields *)
Flatten[Table[{R{0, j}i → Bji, R{cd[[j, 1]], cd[[j, 2]]}i → Eji,
Table[{R{0, j, k}i → B{j, k}i, R{cd[[j, 1], cd[[j, 2], k]}i → E{j, k}i}, {k, 0, 3}]]], {i, 3}, {j, 3}]]];
Column[FullSimplify[fn = fin /. sub1], Dividers → All]
```

$(B_{1,1}^3 + B_{2,2}^3 + B_{3,3}^3) \Gamma_0^2 - (B_{1,1}^2 + B_{2,2}^2 + B_{3,3}^2) \Gamma_0^3 +$ $\Gamma_3^3 (B_{3,0}^2 + E_{1,2}^2 - E_{2,1}^2) - \Gamma_3^2 (B_{3,0}^3 + E_{1,2}^3 - E_{2,1}^3) + \Gamma_2^3 (B_{2,0}^2 - E_{1,3}^2 + E_{3,1}^2) -$ $\Gamma_2^2 (B_{2,0}^3 - E_{1,3}^3 + E_{3,1}^3) + \Gamma_1^3 (B_{1,0}^2 + E_{2,3}^2 - E_{3,2}^2) - \Gamma_1^2 (B_{1,0}^3 + E_{2,3}^3 - E_{3,2}^3)$	$\sim$Klein-Gordon
$(B_{1,1}^1 + B_{2,2}^1 + B_{3,3}^1) \Gamma_0^3 - \Gamma_3^3 (B_{3,0}^1 + E_{1,2}^1 - E_{2,1}^1) -$ $\Gamma_2^3 (B_{2,0}^1 - E_{1,3}^1 + E_{3,1}^1) - \Gamma_1^3 (B_{1,0}^1 + E_{2,3}^1 - E_{3,2}^1)$	$\sim$Maxwell1
$-((B_{1,1}^1 + B_{2,2}^1 + B_{3,3}^1) \Gamma_0^2) + \Gamma_3^3 (B_{3,0}^1 + E_{1,2}^1 - E_{2,1}^1) +$ $\Gamma_2^2 (B_{2,0}^1 - E_{1,3}^1 + E_{3,1}^1) + \Gamma_1^2 (B_{1,0}^1 + E_{2,3}^1 - E_{3,2}^1)$	$\sim$Maxwell2
$(B_1^1)^2 + (B_2^1)^2 + (B_3^1)^2 - (E_1^1)^2 - (E_2^1)^2 - (E_3^1)^2$	electric field enforces magnetic (?)
$(B_1^1)^2 + (B_2^1)^2 + (B_3^1)^2 - (E_1^1)^2 - (E_2^1)^2 - (E_3^1)^2$	-> de Broglie clock/zitterbewegung?
$(B_1^1)^2 + (B_2^1)^2 + (B_3^1)^2 - (E_1^1)^2 - (E_2^1)^2 - (E_3^1)^2$	in 4D dominated by 0-th: gravity?

$E_{\{1,3\}}^2$ - type: **1 - high energy (EM tilt-tilt)**, **2,3 low energy (QM tilt-twist)**
 $\{1,3\}$ - spatial coordinate (1,2,3), derivative (0,1,2,3)

Hedgehog ansatz for longest axis (± 1 charge), with $\psi(t, x, y, z)$ phase/twist function

In vacuum ($A = 0$) leads to **Klein-Gordon-like**: $(\hat{E} - q\phi)^2 \psi = (\hat{p} - qA)^2 \psi + m^2 \psi$

dual formulation ($E \leftrightarrow B$): $A^{hedg} = \frac{1}{r^2} (x, y, z)$ ($\hat{E} = i\hbar\partial_t$, $\hat{p} = -i\hbar\nabla$)

$\Psi = \exp(i\psi)$ $\hat{p}\Psi = -i\nabla\Psi = \Psi\nabla\psi$ here from: $X^2 \cdot \Gamma^3 = X^3 \cdot \Gamma^2$

Dirac equation? (also zitterbewegung)

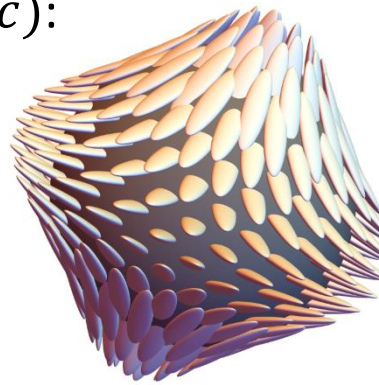
Bispinor for electron (up), positron (down)

with spin direction (a, b, c) :

$$\frac{1}{4} \begin{bmatrix} 1+c & a-ib & \pm(1+c) & \pm(a-ib) \\ a+ib & 1-c & \pm(a+ib) & \pm(1-c) \\ \pm(1+c) & \pm(a-ib) & 1+c & a-ib \\ \pm(a+ib) & \pm(1-c) & a+ib & 1-c \end{bmatrix}$$

$$S[\Lambda_{\text{boost}}] = \begin{pmatrix} e^{+\chi \cdot \sigma/2} & 0 \\ 0 & e^{-\chi \cdot \sigma/2} \end{pmatrix}$$

$$S[\Lambda_{\text{rot}}] = \begin{pmatrix} e^{+i\phi \cdot \sigma/2} & 0 \\ 0 & e^{+i\phi \cdot \sigma/2} \end{pmatrix}$$



we get wave-like: $2\partial_{tt}\psi = \left((\nabla - \vec{A}^{hedg})^2 + \left(\frac{\vec{A}^{hedg}}{|\vec{A}^{hedg}|} \cdot \nabla \right)^2 \right) \psi$

```
Q = Q0 /. {psi -> psi[t, x, y, z]}; (* assume phase dependence *)
rs = Simplify[Table[vect[Transpose[Q].D[Q, v]], {v, {t, x, y, z}}, r > 0];
BE = Simplify[Table[Cross[rs[[c[[1]]], rs[[c[[2]]]], (* find BE fields *)
{c, {{1, 2}, {1, 3}, {1, 4}, {4, 3}, {2, 4}, {3, 2}}}}]];
BEed = Simplify[Table[D[BE, v], {v, {t, x, y, z}}]]; (* BE derivatives *)
sub2 = Flatten[Join[Table[rk-1 -> rs[[k, j]], {k, 4}, {j, 3}],
Table[{B_i^j -> BE[[i, j]], E_i^j -> BE[[i+3, j]],
Table[{B_{i,k-1}^j -> BEed[[k, i, j]], E_{i,k-1}^j -> BEed[[k, i+3, j]], {k, 4}},
{i, 3}, {j, 3}]]];
(fne = FullSimplify[fne[[1 ;; 3]] /. sub2] * (x^2 + y^2 + z^2)^2) // Column (*equations:*)
- 2 z psi^{(0,0,0,1)}[t, x, y, z] + (x^2 + y^2 + 2 z^2) psi^{(0,0,0,2)}[t, x, y, z] -
2 y psi^{(0,0,1,0)}[t, x, y, z] + 2 y z psi^{(0,0,1,1)}[t, x, y, z] + x^2 psi^{(0,0,2,0)}[t, x, y, z] +
2 y^2 psi^{(0,0,2,0)}[t, x, y, z] + z^2 psi^{(0,0,2,0)}[t, x, y, z] - 2 x psi^{(0,1,0,0)}[t, x, y, z] +
2 x z psi^{(0,1,0,1)}[t, x, y, z] + 2 x y psi^{(0,1,1,0)}[t, x, y, z] + 2 x^2 psi^{(0,2,0,0)}[t, x, y, z] +
y^2 psi^{(0,2,0,0)}[t, x, y, z] + z^2 psi^{(0,2,0,0)}[t, x, y, z] - 2 (x^2 + y^2 + z^2) psi^{(2,0,0,0)}[t, x, y, z]
0,0 for 2nd, 3rd equation - they are satisfied, the first equation equalized to 0:
which turns out Klein-Gordon-like: 2\partial_{tt}\psi = \left( (\nabla - \vec{A}^{hedg})^2 + \left( \frac{\vec{A}^{hedg}}{|\vec{A}^{hedg}|} \cdot \nabla \right)^2 \right) \psi
```

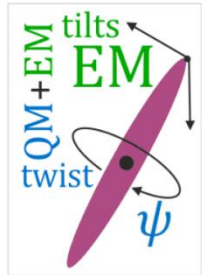
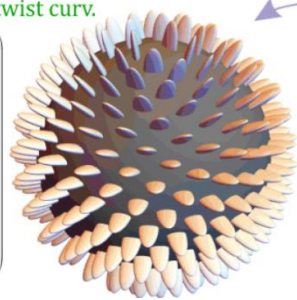
for dual: $\vec{A}^{hedg}(x, y, z) = (x, y, z)/r^2$ $\Psi = \exp(i\psi)$ $\hat{p}\Psi = -i\nabla\Psi = \Psi\nabla\psi$

```
r = Sqrt[x^2 + y^2 + z^2]; A = {x, y, z} / r^2;
gmA[f_] := Grad[f, {x, y, z}] - A * f; Adg[f_] := (A * r).Grad[f, {x, y, z}];
Simplify[fne[[1]] / r^2 - Sum[gmA[gmA[psi[t, x, y, z]]][[i, i]], {i, 3}] -
Adg[Adg[psi[t, x, y, z]]]
- 2 psi^{(2,0,0,0)}[t, x, y, z]
```

```
sph = {x -> r * Cos[theta] * Cos[phi], y -> r * Cos[theta] * Sin[phi], z -> r * Sin[theta]}; (*spherical*)
Q0 = FullSimplify[MatrixExp[phi * Gz].MatrixExp[theta * Gy].MatrixExp[psi * Gx] /.
{phi -> ArcTan[x, y], theta -> -ArcTan[Sqrt[x^2 + y^2], z]}; (* hedgehog *)
Q = Q0; tQ = Transpose[Q]; M = Simplify[Q.d.tQ];
fBE := Table[Simplify[vect[tQ.com[D[M, c[[1]]], D[M, c[[2]]]].Q]],
{c, {{t, x}, {t, y}, {t, z}, {z, y}, {x, z}, {y, x}}}}];
M0 = 0.001 * IdentityMatrix[3] + 0.05 * M /. {delta -> 0.1, psi -> 0}; (* shape to draw *)
points = SpherePoints[300];
Row[{Column[{"B_1", "B_2", "B_3", "E_1", "E_2", "E_3"}], "=", fBE /. psi -> 0 // MatrixForm,
Graphics3D[Table[Ellipsoid[p, M0 /. {x -> p[[1]], y -> p[[2]], z -> p[[3]]], {p, points}],
Gray, Sphere[{0, 0, 0}, 1]], Boxed -> False, ImageSize -> Small]]];
```

type: EM: high energy tilt-tilt curvature QM: low energy, tilt-twist curv.

$$\begin{matrix} B_1 & 1 & 0 & 2 & 0 & 3 & 0 \\ B_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ B_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ E_1 & -\frac{x(-1+\delta)}{(x^2+y^2+z^2)^{3/2}} & 0 & -\frac{xz\delta}{\sqrt{x^2+y^2}(x^2+y^2+z^2)^{3/2}} & 0 & 0 & 0 \\ E_2 & -\frac{y(-1+\delta)}{(x^2+y^2+z^2)^{3/2}} & 0 & -\frac{yz\delta}{\sqrt{x^2+y^2}(x^2+y^2+z^2)^{3/2}} & 0 & 0 & 0 \\ E_3 & -\frac{z(-1+\delta)}{(x^2+y^2+z^2)^{3/2}} & 0 & -\frac{z^2\delta}{\sqrt{x^2+y^2}(x^2+y^2+z^2)^{3/2}} & 0 & 0 & 0 \end{matrix}$$



Derivation of Maxwell-like equations for gravity (GEM), in $E = B = 0$ case

```

d = DiagonalMatrix[{g, 1, δ, 0}]; cd = {{3, 2}, {1, 3}, {2, 1}}; com[A_, B_] := A.B - B.A;
ξ = DiagonalMatrix[{-1, 1, 1, 1}]; (* signature *) coms[A_, B_] := A.ξ.B - B.ξ.A;
 $\Gamma_\mu = \{\{0, \tilde{\Gamma}_\mu^1, \tilde{\Gamma}_\mu^2, \tilde{\Gamma}_\mu^3\}, \{\tilde{\Gamma}_\mu^1, 0, -\Gamma_\mu^3, \Gamma_\mu^2\}, \{\tilde{\Gamma}_\mu^2, \Gamma_\mu^3, 0, -\Gamma_\mu^1\}, \{\tilde{\Gamma}_\mu^3, -\Gamma_\mu^2, \Gamma_\mu^1, 0\}\};$  (*3D rot + boost*)
G4 = Table[Coefficient[ $\Gamma_\mu$ , v], {v, { $\Gamma_\mu^1, \Gamma_\mu^2, \Gamma_\mu^3, \tilde{\Gamma}_\mu^1, \tilde{\Gamma}_\mu^2, \tilde{\Gamma}_\mu^3$ }}]; (* rotation + boost generators *)
dg = Table[tm = Table[0, 4, 4]; tm[[i, i]] = 1; tm, {i, 4}]; (* elongation generators *)
Gpt = Join[Table[coms[G, d], {G, G4}], dg]; (* G' size 3+3+4=10 tables *)
sub = Flatten[Table[{Cross[{ $\Gamma_\mu^1, \Gamma_\mu^2, \Gamma_\mu^3$ }, { $\Gamma_\nu^1, \Gamma_\nu^2, \Gamma_\nu^3$ }] [[i]] ==  $R_{\{\mu, \nu\}}^i$ ,
Cross[{ $\tilde{\Gamma}_\mu^1, \tilde{\Gamma}_\mu^2, \tilde{\Gamma}_\mu^3$ }, { $\tilde{\Gamma}_\nu^1, \tilde{\Gamma}_\nu^2, \tilde{\Gamma}_\nu^3$ }] [[i]] ==  $\tilde{R}_{\{\mu, \nu\}}^i$ }, {i, 3}]]]; (* EM curvatures *)
ds[a_] := Flatten[Table[{ $R_{\{\mu, \nu\}}^i \rightarrow R_{\{\mu, \nu, a\}}^i$ ,  $\tilde{R}_{\{\mu, \nu\}}^i \rightarrow \tilde{R}_{\{\mu, \nu, a\}}^i$ }, {i, 3}]]]; (* derivatives *)
cs = Flatten[Table[{ $\Gamma_\mu^i \rightarrow 0$ ,  $\Gamma_\nu^i \rightarrow 0$ }, {i, 3}]]]; (* assume EM E=B=0 here *)

```

```

Mμ = coms[ $\Gamma_\mu$ , d] /. cs;  $\Gamma_\nu = \Gamma_\mu /. \mu \rightarrow \nu$ ;  $M_\nu = coms[\Gamma_\nu, d] /. cs$ ;  $F_{\mu\nu} = Simplify[coms[M_\mu, M_\nu], sub];$ 
vr = Table[Tr[ $F_{\mu\nu} \cdot \xi \cdot (coms[\Gamma_\nu, coms[M_\mu, Gp]] - coms[\Gamma_\mu, coms[M_\nu, Gp]]) \cdot \xi +$  (* evolution equations *)
( $F_{\mu\nu} /. ds[\nu]) \cdot \xi \cdot coms[M_\mu, Gp] \cdot \xi - (F_{\mu\nu} /. ds[\mu]) \cdot \xi \cdot coms[M_\nu, Gp] \cdot \xi$ ], {Gp, Gpt}]] /. cs;
sub1 = Flatten[Table[{ $\tilde{R}_{\{0, j\}}^i \rightarrow \tilde{B}_{\{j\}}^i$ ,  $\tilde{R}_{\{cd[[j, 1], cd[[j, 2]]\}}^i \rightarrow \tilde{E}_{\{j\}}^i$ ,
Table[{ $\tilde{R}_{\{0, j, k\}}^i \rightarrow \tilde{B}_{\{j, k\}}^i$ ,  $\tilde{R}_{\{cd[[j, 1], cd[[j, 2]], k\}}^i \rightarrow \tilde{E}_{\{j, k\}}^i$ }, {k, 0, 3}]]], {i, 3}, {j, 3}]]]; (* GEM EB fields *)
fin = Simplify[Table[Sum[v /.  $\mu \rightarrow 0$ , {v, 1, 3}] - Sum[v /. { $\mu \rightarrow cd[[i, 1], \nu \rightarrow cd[[i, 2]]$ }, {i, 3}], {v, vr}]]];
Column[fnl = Limit[(fin[[4 ;; 6]] /. sub1) / 2 / g^4, g → Infinity] // FullSimplify, Dividers → All]

```

$$\begin{aligned}
& (\tilde{B}_{\{1,1\}}^3 + \tilde{B}_{\{2,2\}}^3 + \tilde{B}_{\{3,3\}}^3) \tilde{\Gamma}_0^2 - (\tilde{B}_{\{1,1\}}^2 + \tilde{B}_{\{2,2\}}^2 + \tilde{B}_{\{3,3\}}^2) \tilde{\Gamma}_0^3 + \tilde{\Gamma}_3^3 (\tilde{B}_{\{3,0\}}^2 + \tilde{E}_{\{1,2\}}^2 - \tilde{E}_{\{2,1\}}^2) - \tilde{\Gamma}_3^2 (\tilde{B}_{\{3,0\}}^3 + \tilde{E}_{\{1,2\}}^3 - \tilde{E}_{\{2,1\}}^3) + \\
& \tilde{\Gamma}_2^3 (\tilde{B}_{\{2,0\}}^2 - \tilde{E}_{\{1,3\}}^2 + \tilde{E}_{\{3,1\}}^2) - \tilde{\Gamma}_2^2 (\tilde{B}_{\{2,0\}}^3 - \tilde{E}_{\{1,3\}}^3 + \tilde{E}_{\{3,1\}}^3) + \tilde{\Gamma}_1^3 (\tilde{B}_{\{1,0\}}^2 + \tilde{E}_{\{2,3\}}^2 - \tilde{E}_{\{3,2\}}^2) - \tilde{\Gamma}_1^2 (\tilde{B}_{\{1,0\}}^3 + \tilde{E}_{\{2,3\}}^3 - \tilde{E}_{\{3,2\}}^3) \\
& - ((\tilde{B}_{\{1,1\}}^3 + \tilde{B}_{\{2,2\}}^3 + \tilde{B}_{\{3,3\}}^3) \tilde{\Gamma}_0^1) + (\tilde{B}_{\{1,1\}}^1 + \tilde{B}_{\{2,2\}}^1 + \tilde{B}_{\{3,3\}}^1) \tilde{\Gamma}_0^3 - \tilde{\Gamma}_3^3 (\tilde{B}_{\{3,0\}}^1 + \tilde{E}_{\{1,2\}}^1 - \tilde{E}_{\{2,1\}}^1) + \tilde{\Gamma}_3^1 (\tilde{B}_{\{3,0\}}^3 + \tilde{E}_{\{1,2\}}^3 - \tilde{E}_{\{2,1\}}^3) - \\
& \tilde{\Gamma}_2^3 (\tilde{B}_{\{2,0\}}^1 - \tilde{E}_{\{1,3\}}^1 + \tilde{E}_{\{3,1\}}^1) + \tilde{\Gamma}_2^1 (\tilde{B}_{\{2,0\}}^3 - \tilde{E}_{\{1,3\}}^3 + \tilde{E}_{\{3,1\}}^3) - \tilde{\Gamma}_1^3 (\tilde{B}_{\{1,0\}}^1 + \tilde{E}_{\{2,3\}}^1 - \tilde{E}_{\{3,2\}}^1) + \tilde{\Gamma}_1^1 (\tilde{B}_{\{1,0\}}^3 + \tilde{E}_{\{2,3\}}^3 - \tilde{E}_{\{3,2\}}^3) \\
& (\tilde{B}_{\{1,1\}}^2 + \tilde{B}_{\{2,2\}}^2 + \tilde{B}_{\{3,3\}}^2) \tilde{\Gamma}_0^1 - (\tilde{B}_{\{1,1\}}^1 + \tilde{B}_{\{2,2\}}^1 + \tilde{B}_{\{3,3\}}^1) \tilde{\Gamma}_0^2 + \tilde{\Gamma}_3^2 (\tilde{B}_{\{3,0\}}^1 + \tilde{E}_{\{1,2\}}^1 - \tilde{E}_{\{2,1\}}^1) - \tilde{\Gamma}_3^1 (\tilde{B}_{\{3,0\}}^2 + \tilde{E}_{\{1,2\}}^2 - \tilde{E}_{\{2,1\}}^2) + \\
& \tilde{\Gamma}_2^2 (\tilde{B}_{\{2,0\}}^1 - \tilde{E}_{\{1,3\}}^1 + \tilde{E}_{\{3,1\}}^1) - \tilde{\Gamma}_2^1 (\tilde{B}_{\{2,0\}}^2 - \tilde{E}_{\{1,3\}}^2 + \tilde{E}_{\{3,1\}}^2) + \tilde{\Gamma}_1^2 (\tilde{B}_{\{1,0\}}^1 + \tilde{E}_{\{2,3\}}^1 - \tilde{E}_{\{3,2\}}^1) - \tilde{\Gamma}_1^1 (\tilde{B}_{\{1,0\}}^2 + \tilde{E}_{\{2,3\}}^2 - \tilde{E}_{\{3,2\}}^2)
\end{aligned}$$

$$\tilde{X}^i := \left(-\nabla \cdot \tilde{B}^i, \overline{\partial_0 \tilde{B}^i + \nabla \times \tilde{E}^i} \right) \text{ Maxwell: } \tilde{X}^3 \cdot \tilde{\Gamma}^2 = \tilde{X}^2 \cdot \tilde{\Gamma}^3, \tilde{X}^3 \cdot \tilde{\Gamma}^1 = \tilde{X}^1 \cdot \tilde{\Gamma}^3, \tilde{X}^2 \cdot \tilde{\Gamma}^1 = \tilde{X}^1 \cdot \tilde{\Gamma}^2$$

Newton attraction? $\sim 10^{-36} \times$ of Coulomb???

reduced energy for closer curvature sources???

reduced distance - increased curvature² in energy

- $r \rightarrow 0$ Coulomb, $r \rightarrow \infty$ Newton – maybe $r \rightarrow r^{-1}$, **“repulsion in infinity”**?

- maybe somehow **opposite curvatures**? E.g. for protons – neutrons?

- due to **spacetime signature**? (also “negative” energy to propel the clock?)

$$\xi = \text{diag}(-1, 1, 1, 1)$$

$$AB \rightarrow A\xi B$$

$$A_{\mu}^{\nu} = \sum_{\alpha} A_{\mu\alpha} \xi^{\alpha\nu}$$

time has opposite sign

Lorentz-invariant Lagrangian?

$$\mathcal{L} = -F_{\alpha\beta\mu\nu} F^{\alpha\beta\mu\nu} - V(M)$$

$$\text{for } F_{\alpha\beta\mu\nu} = [\partial_{\mu} M, \partial_{\nu} M]_{\alpha\beta}$$

$$[A, B] \rightarrow A\xi B - B\xi A$$

$$\text{Tr}(AA^T) \rightarrow \text{Tr}(A\xi A^T \xi)$$

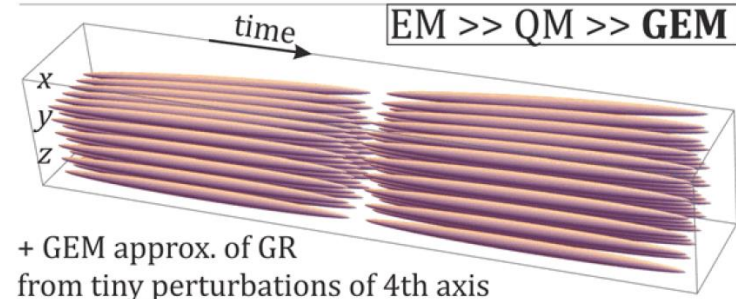
General relativity Lagrangian

([Einstein-Hilbert action](#)):

$$\mathcal{L} = R\sqrt{-g} \quad (R = R_{\alpha\beta\mu\nu} \xi^{\alpha\beta} \xi^{\mu\nu})$$

First power of (contracted) curvature,

so which is proper 1 vs 2? Mixture?



```
cos = 1 + (z - d) / Sqrt[(z - d)^2 + r^2] - (z + d) / Sqrt[(z + d)^2 + r^2]; (* Manfred Faber dipole ansatz *)
n = {Sqrt[1 - cos^2] x / r, Sqrt[1 - cos^2] y / r, cos} /. r -> Sqrt[x^2 + y^2]; (* cylindrical symmetry *)
G = Table[t = Table[0, 4, 4]; t[[1, i]] = -1; t[[i, 1]] = 1; t, {i, 2, 4}]; (* 4D rotation generators *)
M0 = {{g, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 0, 0}, {0, 0, 0, 0}}; (* rest field for {g, 1, 0, 0} axes *)
o = MatrixExp[m * {a, b, c}.G]; (* m is the mass - it is tiny, we focus on Taylor to m^1 *)
M = Series[o.M0.Transpose[o] /. {a -> n[[1]], b -> n[[2]], c -> n[[3]]}, {m, 0, 1}] // Normal // Simplify;
dM = {D[M, x], D[M, y], D[M, z]}; (* Hamiltonian taking the lowest m term: m^4 *)
H = Coefficient[Sum[Total[(dM[[i]].dM[[j]] - dM[[j]].dM[[i]])^2, 2], {i, 2}, {j, i + 1, 3}] /. y -> 0, m^4];
HH = FullSimplify[CoefficientList[H, g][[-1]]]; (* take the highest g term: g^4 *)
Es = Table[{d, NIntegrate[4 Pi * x * HH * Boole[x^2 + (z - d)^2 > 0.001], {x, 0, Infinity}, {z, 0, Infinity}]}, {d, 0.1, 3, 0.1}]; (* integrate H: total field energy *)
ft = Fit[Es, {1, 1/d}, d]; Show[ListPlot[Es], Plot[ft, {d, 0.1, 3}]] (* fit Newton potential *)
```

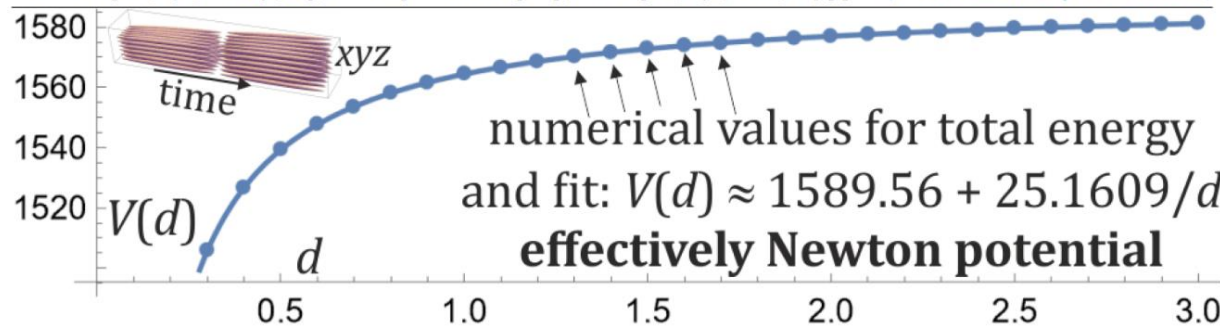


Figure 3. A simplified calculation of Newton effective potential (in GitHub - analogously as in Fig. 2, but this time instead of large spatial rotations, using tiny tilts of 0th time axis for gravity (no mass quantization). Spherically sym

by spacetime signature

$$= [M_\mu, M_\nu] = O \overline{F}_{\mu\nu} O^T \quad \text{for} \quad \overline{F}_{\mu\nu} = [\overline{M}_\mu, \overline{M}_\nu]$$

electron

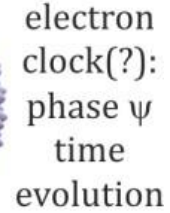
frequency $\omega \approx 1.0712$


$$= [M_\mu, M_\nu] = O \overline{F}_{\mu\nu} O^T \quad \text{for} \quad \overline{F}_{\mu\nu} = [\overline{M}_\mu, \overline{M}_\nu]$$

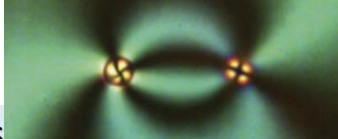
electron

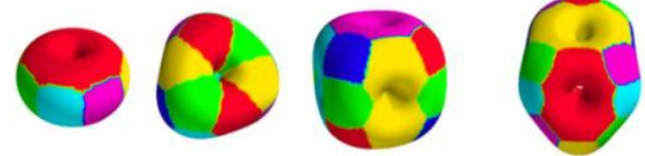


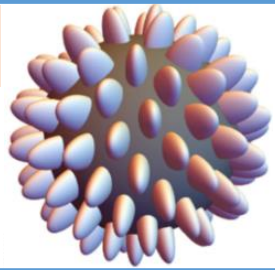
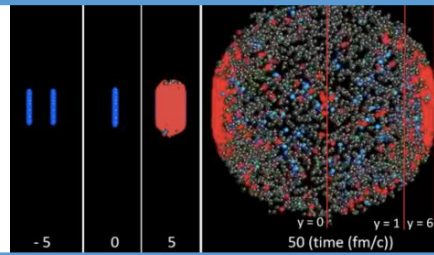
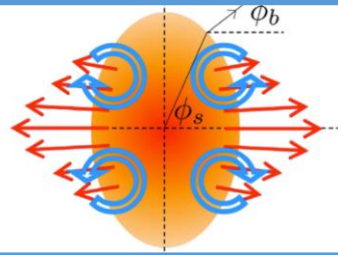
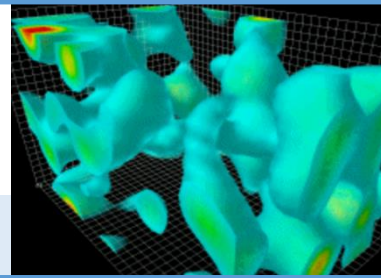
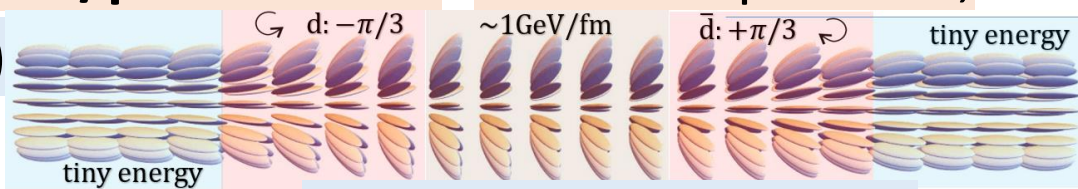
electron



Mainstream landscape ~ proposed Landau-de Gennes-Skyrme:

- Liquid crystals**: same field and potential e.g. $V(M) = \sum_i (\lambda_i - \Lambda_i)^2 / F \sim R^*$
 kinetic → Lorentz invariant: $F_{\mu\nu\alpha\beta} F^{\mu\nu\alpha\beta}$, $F_{\mu\nu\alpha\beta} = [\partial_\mu M, \partial_\nu M]_{\alpha\beta}$


- Skyrmion**: same kinetic term, potential → ~Higgs,
 topolog. charge interpretation: baryon → electric
 
- Einstein's teleparallelism**: ~4D liquid crystal, but same λ_i lengths
 → different for EM >> QM >> **GEM**, mass: **black hole** >> **electron**
 $\mathcal{L}_{QED} = -F_{\mu\nu} F^{\mu\nu} / 4 - \hbar c \bar{\psi} \gamma^\mu D_\mu \psi - mc^2 \bar{\psi} \psi$


- Spin hydrodynamics**: LHC collisions as fluid,
 but averaged: no details, charges, vortices
 Let's derive it averaging field configurations
 

- Lattice QCD**: also Yang-Mills kinetic: $\mathcal{L}_k = -\text{Tr}[F_{\mu\nu} F^{\mu\nu}]$, but
 quark probability distribution in Dirac $\sum_{f=1}^6 \bar{\psi}_f (iD - m_f) \psi_f$
 → quark field configurations, derive by **Feynman ensembles**

- Standard Model**: ~right, but effective/perturbative: creation operators,
 probability distributions (of what?)
 

Let's derive SM from nonperturbative

nonperturbatively topological vortex - quark string excitations: quark - conflict - anti-quark

Perturbative approx.: probability densities, creation operators ...

Standard model – 26+ parameters, Lagrangian →

Noncompatible with gravity (“too big infinity”)

“Taylor expansion” of some simpler model?

Where to search for it? Non-perturbative: fields

EM-hydro analogy, particles as liquid crystals?

charge quantization + Coulomb-like interaction ...

Postulate skrymion-like Lagrangian for M :

$$\mathcal{L} = -F_{\mu\nu\alpha\beta}F^{\mu\nu\alpha\beta} - V(M) \quad (\text{Higgs})$$

($R \sim$) $F_{\mu\nu\alpha\beta} = [\partial_\mu M, \partial_\nu M]_{\alpha\beta}$ (SO(1,3) vacuum) field curvature ($\hbar, G$ weighted)

EM >> quantum phase >> GEM vacuum dynamics

EM with built-in charge quantization, regularization

3 leptons: same charge – different mass, $\mu \neq 0$

3 neutrinos: very stable, oscillations, $\uparrow$ beta decay

3 families of **baryons**: $m_p < m_n, m_d < m_p + m_n$

deuteron with quadrupole moment, $\mu_d \approx \mu_p + \mu_n$

nuclei as knots – also halo neutrons in e.g. ~ 5 fm

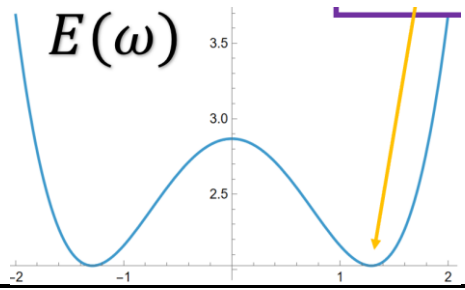
string hadronization reconnections ... effective? →

Feynman ensemble of topological defects → QFT

epicycles...?

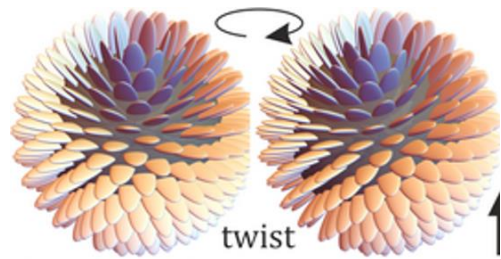
$$\begin{aligned} & -\frac{1}{2}\partial_\nu g_\mu^a \partial_\nu g_\mu^a - g_s f^{abc} \partial_\mu g_\nu^a g_\mu^b g_\nu^c - \frac{1}{4}g_s^2 f^{abc} f^{ade} g_\mu^b g_\nu^c g_\mu^d g_\nu^e + \\ & \frac{1}{2}ig_s^2(\bar{q}_i^\sigma \gamma^\mu q_j^\sigma)g_\mu^a + \bar{G}^a \partial^2 G^a + g_s f^{abc} \partial_\mu \bar{G}^a G^b g_\mu^c - \partial_\nu W_\mu^+ \partial_\nu W_\mu^- - \\ 2 \quad & M^2 W_\mu^+ W_\mu^- - \frac{1}{2}\partial_\nu Z_\mu^0 \partial_\nu Z_\mu^0 - \frac{1}{2c_w^2}M^2 Z_\mu^0 Z_\mu^0 - \frac{1}{2}\partial_\mu A_\nu \partial_\mu A_\nu - \frac{1}{2}\partial_\mu H \partial_\mu H - \\ & \frac{1}{2}m_h^2 H^2 - \partial_\mu \phi^+ \partial_\mu \phi^- - M^2 \phi^+ \phi^- - \frac{1}{2}\partial_\mu \phi^0 \partial_\mu \phi^0 - \frac{1}{2c_w^2}M \phi^0 \phi^0 - \beta_h[\frac{2M^2}{g^2} + \\ & \frac{2M}{g}H + \frac{1}{2}(H^2 + \phi^0 \phi^0 + 2\phi^+ \phi^-)] + \frac{2M^4}{g^2}\alpha_h - igc_w[\partial_\nu Z_\mu^0(W_\mu^+ W_\nu^- - \\ & W_\nu^+ W_\mu^-) - Z_\nu^0(W_\mu^+ \partial_\nu W_\mu^- - W_\mu^+ \partial_\nu W_\mu^-) + Z_\mu^0(W_\nu^+ \partial_\nu W_\mu^- - \\ & W_\nu^- \partial_\nu W_\mu^+) - igs_w[\partial_\nu A_\mu(W_\mu^+ W_\nu^- - W_\nu^+ W_\mu^-) - A_\nu(W_\mu^+ \partial_\nu W_\mu^- - \\ & W_\mu^- \partial_\nu W_\mu^+) + A_\mu(W_\nu^+ \partial_\nu W_\mu^- - W_\nu^- \partial_\nu W_\mu^+)] - \frac{1}{2}g^2 W_\mu^+ W_\mu^- W_\nu^+ W_\nu^- + \\ & \frac{1}{2}g^2 W_\mu^+ W_\nu^- W_\mu^- W_\nu^- + g^2 c_w^2(Z_\mu^0 W_\mu^+ Z_\nu^0 W_\nu^- - Z_\mu^0 Z_\nu^0 W_\mu^+ W_\nu^-) + \\ & g^2 s_w^2(A_\mu W_\mu^+ A_\nu W_\nu^- - A_\mu A_\nu W_\mu^+ W_\nu^-) + g^2 s_w c_w[A_\mu Z_\nu^0(W_\mu^+ W_\nu^- - \\ & W_\nu^+ W_\mu^-) - 2A_\mu Z_\mu^0 W_\nu^+ W_\nu^-] - g\alpha[H^3 + H\phi^0 \phi^0 + 2H\phi^+ \phi^-] - \\ & \frac{1}{8}g^2 \alpha_h[H^4 + (\phi^0)^4 + 4(\phi^+ \phi^-)^2 + 4(\phi^0)^2 \phi^+ \phi^- + 4H^2 \phi^+ \phi^- + 2(\phi^0)^2 H^2] - \\ & gMW_\mu^+ W_\mu^- H - \frac{1}{2}g\frac{M}{c_w^2}Z_\mu^0 Z_\mu^0 H - \frac{1}{2}ig[W_\mu^+(\phi^0 \partial_\mu \phi^- - \phi^- \partial_\mu \phi^0) - \\ & W_\mu^-(\phi^0 \partial_\mu \phi^+ - \phi^+ \partial_\mu \phi^0)] + \frac{1}{2}g[W_\mu^+(H \partial_\mu \phi^- - \phi^- \partial_\mu H) - W_\mu^-(H \partial_\mu \phi^+ - \\ & \phi^+ \partial_\mu H)] + \frac{1}{2}g\frac{1}{c_w}(Z_\mu^0(H \partial_\mu \phi^0 - \phi^0 \partial_\mu H) - ig\frac{s_w^2}{c_w}M Z_\mu^0(W_\mu^+ \phi^- - W_\mu^- \phi^+) + \\ & igs_w M A_\mu(W_\mu^+ \phi^- - W_\mu^- \phi^+) - ig\frac{1-2c_w^2}{2c_w}Z_\mu^0(\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) + \\ & igs_w A_\mu(\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) - \frac{1}{4}g^2 W_\mu^+ W_\mu^- [H^2 + (\phi^0)^2 + 2\phi^+ \phi^-] - \\ & \frac{1}{4}g^2 \frac{1}{c_w^2}Z_\mu^0 Z_\mu^0 [H^2 + (\phi^0)^2 + 2(2s_w^2 - 1)^2 \phi^+ \phi^-] - \frac{1}{2}g^2 \frac{s_w^2}{c_w}Z_\mu^0 \phi^0(W_\mu^+ \phi^- + \\ & W_\mu^- \phi^+) - \frac{1}{2}ig^2 \frac{s_w^2}{c_w}Z_\mu^0 H(W_\mu^+ \phi^- - W_\mu^- \phi^+) + \frac{1}{2}g^2 s_w A_\mu \phi^0(W_\mu^+ \phi^- + \\ & W_\mu^- \phi^+) + \frac{1}{2}ig^2 s_w A_\mu H(W_\mu^+ \phi^- - W_\mu^- \phi^+) - g^2 \frac{s_w}{c_w}(2c_w^2 - 1)Z_\mu^0 A_\mu \phi^+ \phi^- - \\ & g^1 s_w^2 A_\mu A_\mu \phi^+ \phi^- - \bar{e}^\lambda(\gamma \partial + m_e^\lambda)e^\lambda - \bar{\nu}^\lambda \gamma \partial \nu^\lambda - \bar{u}_j^\lambda(\gamma \partial + m_u^\lambda)u_j^\lambda - \\ 3 \quad & \bar{d}_j^\lambda(\gamma \partial + m_d^\lambda)d_j^\lambda + igs_w A_\mu[-(\bar{e}^\lambda \gamma^\mu e^\lambda) + \frac{2}{3}(\bar{u}_j^\lambda \gamma^\mu u_j^\lambda) - \frac{1}{3}(\bar{d}_j^\lambda \gamma^\mu d_j^\lambda)] + \\ & \frac{ig}{4c_w}Z_\mu^0[(\bar{\nu}^\lambda \gamma^\mu(1 + \gamma^5)\nu^\lambda) + (\bar{e}^\lambda \gamma^\mu(4s_w^2 - 1 - \gamma^5)e^\lambda) + (\bar{u}_j^\lambda \gamma^\mu(\frac{4}{3}s_w^2 - \\ & 1 - \gamma^5)u_j^\lambda) + (\bar{d}_j^\lambda \gamma^\mu(1 - \frac{8}{3}s_w^2 - \gamma^5)d_j^\lambda)] + \frac{ig}{2\sqrt{2}}W_\mu^+[(\bar{\nu}^\lambda \gamma^\mu(1 + \gamma^5)e^\lambda) + \\ & (\bar{u}_j^\lambda \gamma^\mu(1 + \gamma^5)C_{\lambda\kappa}d_j^\kappa)] + \frac{ig}{2\sqrt{2}}W_\mu^-[(\bar{e}^\lambda \gamma^\mu(1 + \gamma^5)\nu^\lambda) + (\bar{d}_j^\kappa C_{\lambda\kappa}^\dagger \gamma^\mu(1 + \\ & \gamma^5)u_j^\lambda)] + \frac{ig}{2\sqrt{2}}\frac{m_e^\lambda}{M}[-\phi^+(\bar{\nu}^\lambda(1 - \gamma^5)e^\lambda) + \phi^-(\bar{e}^\lambda(1 + \gamma^5)\nu^\lambda)] - \\ 4 \quad & \frac{g}{2}\frac{m_e^\lambda}{M}[H(\bar{e}^\lambda e^\lambda) + i\phi^0(\bar{e}^\lambda \gamma^5 e^\lambda)] + \frac{ig}{2M\sqrt{2}}\phi^+[-m_d^\kappa(\bar{u}_j^\lambda C_{\lambda\kappa}(1 - \gamma^5)d_j^\kappa) + \\ & m_u^\lambda(\bar{u}_j^\lambda C_{\lambda\kappa}(1 + \gamma^5)d_j^\kappa) + \frac{ig}{2M\sqrt{2}}\phi^-[m_d^\lambda(\bar{d}_j^\lambda C_{\lambda\kappa}^\dagger(1 + \gamma^5)u_j^\kappa) - m_u^\kappa(\bar{d}_j^\lambda C_{\lambda\kappa}^\dagger(1 - \\ & \gamma^5)u_j^\kappa) - \frac{g}{2}\frac{m_u^\lambda}{M}H(\bar{u}_j^\lambda u_j^\lambda) - \frac{g}{2}\frac{m_d^\lambda}{M}H(\bar{d}_j^\lambda d_j^\lambda) + \frac{ig}{2}\frac{m_u^\lambda}{M}\phi^0(\bar{u}_j^\lambda \gamma^5 u_j^\lambda) - \\ & \frac{ig}{2}\frac{m_d^\lambda}{M}\phi^0(\bar{d}_j^\lambda \gamma^5 d_j^\lambda) + \bar{X}^+(\partial^2 - M^2)X^+ + \bar{X}^-(\partial^2 - M^2)X^- + \bar{X}^0(\partial^2 - \\ 5 \quad & \frac{M^2}{c_w^2})X^0 + \bar{Y}\partial^2 Y + igc_w W_\mu^+(\partial_\mu \bar{X}^0 X^- - \partial_\mu \bar{X}^+ X^0) + igs_w W_\mu^+(\partial_\mu \bar{Y} X^- - \\ & \partial_\mu \bar{X}^+ Y) + igc_w W_\mu^-(\partial_\mu \bar{X}^- X^0 - \partial_\mu \bar{X}^0 X^+) + igs_w W_\mu^-(\partial_\mu \bar{X}^- Y - \\ & \partial_\mu \bar{Y} X^+) + igc_w Z_\mu^0(\partial_\mu \bar{X}^+ X^+ - \partial_\mu \bar{X}^- X^-) + igs_w A_\mu(\partial_\mu \bar{X}^+ X^+ - \\ & \partial_\mu \bar{X}^- X^-) - \frac{1}{2}gM[\bar{X}^+ X^+ H + \bar{X}^- X^- H + \frac{1}{c_w^2}\bar{X}^0 X^0 H] + \\ & \frac{1-2c_w^2}{2c_w}igM[\bar{X}^+ X^0 \phi^+ - \bar{X}^- X^0 \phi^-] + \frac{1}{2c_w}igM[\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \\ & igMs_w[\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \frac{1}{2}igM[\bar{X}^+ X^+ \phi^0 - \bar{X}^- X^- \phi^0] \end{aligned}$$

Some calculation directions, main difficulty: regularization

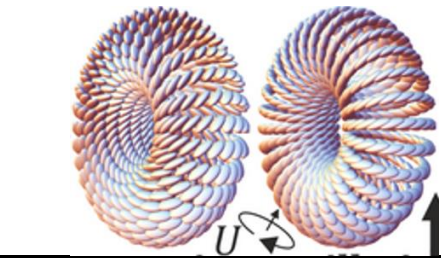


1D kinks with clock propulsion [arXiv:2501.04036](https://arxiv.org/abs/2501.04036)

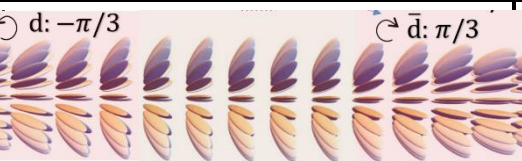
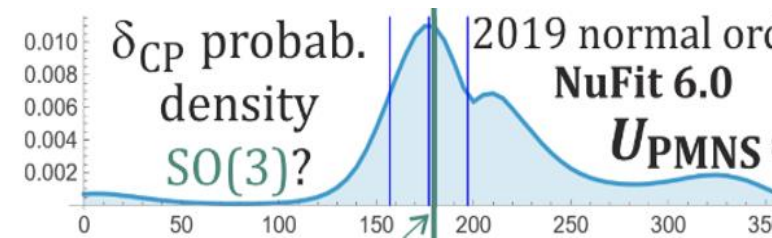
$H = \phi^2 + \psi^2 + (1 - \phi^2)^2 - R^2 + R^4$ for curvature (Lorentz-invariant): $R = \partial_0 \phi \partial_1 \psi - \partial_1 \phi \partial_0 \psi$



Electron: quantized topological charge e.g. hedgehog+clock: de Broglie/ $SO(2)$ field angular momentum + gravitational mass: local boosts preferred energy.



Neutrino oscillations by $SO(3)$ field rotation
[time crystals](#): osc. in min. E



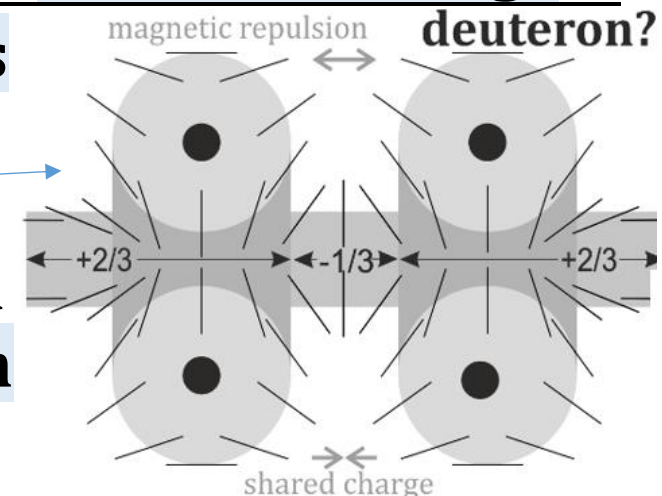
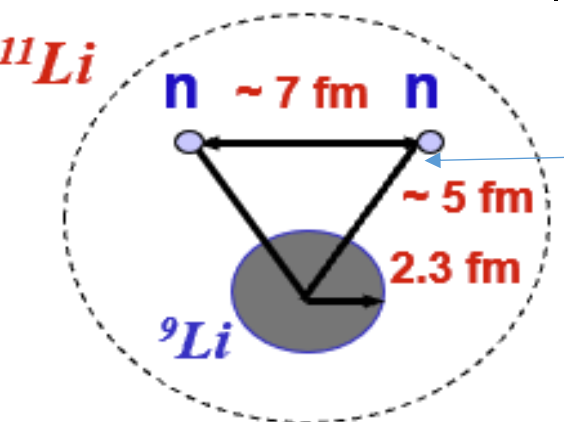
[Cornell potential](#): quark $\sim 1\text{GeV}/\text{fm}$ addition to Coulomb: by conflict between fractional charge

Nuclei as quark strings knots

Crucial e.g. for [halo nuclei](#)

for moments e.g. quadrupole

[EMC effect](#): quark distribution $\rightarrow$ fission/fusion optimization



+ gravity adding boosts: $SO(3) \rightarrow SO(1,3)$ vacuum